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The World-Model Calculus

A Sprout from the Formalization of PLN

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Introduction

1.1 The Problem

Any system that reasons under uncertainty must decide what to store. One family of answers—the one used by most PLN implementations to date—stores *truth values*: a strength $s \in [0, 1]$ and a confidence $c \in [0, 1]$ attached to each proposition. Inference rules combine these pairs, producing new truth values from old ones.

This works until you ask what happens when evidence arrives from two sources that share a common cause, or when you need to retract a faulty sensor's contribution, or when you want to explain *why* the system believes what it believes. At that point, the truth value is silent: it has already thrown away the structure needed to answer.

The underlying issue is a conflation of three layers that should be separate:

1. A **probabilistic world-model state**—the complete semantic object.

2. An **evidence algebra** for compositional aggregation—the carrier in which evidence lives.

3. **Operational truth-value views**—strength, confidence, intervals—which are derived summaries of the evidence, not the evidence itself.

When these layers are collapsed into a single (s, c) pair, the result is what we call *semantic drift*: algebraic operators are treated as if they were the logic itself, and link-level inference propagates correlation-sensitive quantities without carrying the information required to do so correctly.

A one-line example makes the cost visible. Consider two evidence states: $e_1 = (1, 0)$ and $e_2 = (1, 1)$. The second is more *informed* than the first—it contains strictly more data—but less *supportive* of the positive claim, because the negative channel has increased. A system that stores only strength cannot distinguish "strongly supported on little evidence" from "weakly supported on much evidence." That distinction is the subject of this book.

1.2 The Move

Replace truth values with **carried evidence state**. The system stores sufficient statistics—not summaries—and derives truth values as lossy projections (*views*) only when a decision must be made.

This is not a new idea. It is what Bayesian statistics has always done. The contribution of this book is to make the move precise for PLN-style uncertain reasoning, with eight normative requirements (desiderata D1–D8, Chapter 3), a formal algebra, and machine-verified proofs.

Four verbs organize the calculus: *revise* (incorporate new evidence), *query* (extract answers from the current state), *forget* (retract a source's contribution while preserving the rest), and *explain* (trace which evidence supports each conclusion). The next chapter presents the full calculus in miniature.

1.3 Four Distinctions

Four distinctions recur throughout the book. They are worth stating here so the reader can watch for them:

State vs. view. The state is what the system carries (sufficient statistics, provenance, scope labels). A view (strength, confidence, interval) is a lossy projection from the state. Views are useful for display and thresholding; they are not proof objects.

Semantics vs. compiled shortcut. The world model is the truthmaker. A compiled rule (deduction, induction, abduction) is a certified shortcut that is exact under explicit side conditions. When the conditions fail, the shortcut is not licensed.

More information vs. more support. Evidence can improve in two senses that are not the same. $(1, 1)$ is more informed than $(1, 0)$ but less supportive. The book formalizes both orders and proves they are distinct (Chapter 7).

Claim taxonomy. Every formal result in this book carries one of four labels: Core (theorem-backed, no assumptions), Conditional (proved under declared assumptions), Boundary (the claim is *false* without its stated condition), or Operational (heuristic or governance layer). Boundary results are among the most valuable: they tell you where not to waste effort.

1.4 PLN Heritage

This book grows out of the formalization of Probabilistic Logic Networks [10]. The derived link calculus and the compiled Bayesian-network constructions recover, sharpen, and delimit classical PLN-style deduction, induction, and abduction under explicit structural assumptions.

Where the original PLN rule family is correct, the present calculus proves it correct and states the conditions. Where it is incomplete or semantically unstable, the present calculus states the boundary and provides a typed, side-conditioned replacement. The eight desiderata (Chapter 3) are requirements *on* the calculus, not axioms *of* it. They come from lessons learned across PLN implementations—what should have been true, and what went wrong when it was not.

1.5 Maple Court

Maple Court: Maple Court Community Commons is the book's running example. Each apartment carries a private evidence state over questions such as whether a pipe is leaking (binary: (n^+, n^-) alarm counts), whether the elevator is healthy (categorical: Dirichlet counts over normal/slow/-faulty), and what the hallway temperature is (continuous: Normal-Gamma sufficient statistics). Shared spaces carry a building-level state. Throughout the book, Maple Court instantiates abstract concepts in teal-boxed asides like this one. All eight desiderata, all three primary carriers, and the forgetting, provenance, and conservation theorems have Maple Court examples.

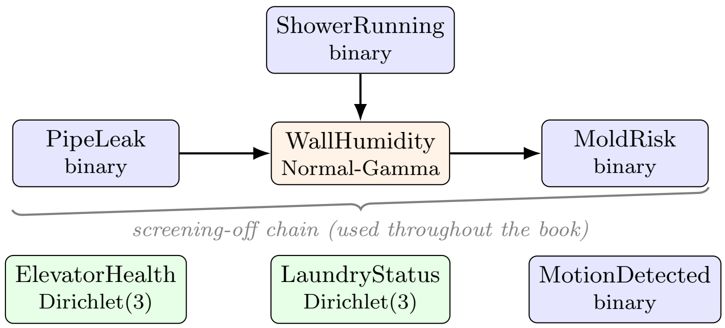


Figure 1.1: The Maple Court building model. Blue: binary evidence (alarm counts). Orange: continuous evidence (Normal-Gamma sufficient statistics). Green: categorical evidence (Dirichlet counts). WallHumidity is a collider: both PipeLeak and ShowerRunning cause it. The chain PipeLeak → WallHumidity → MoldRisk is the screening-off example used throughout.

1.6 Formal Verification

The entire development is formalized in Lean 4 (version 4.28.0) with Mathlib v4.28.0. The book-referenced files contain hundreds of kernel-checked theorems with 0 sorry and 0 `native_decide` (Appendix 27). Six application domains—gas sensor drift, steel fault classification, NAB anomaly detection, GJP geopolitical forecasting, Bayesian-network inference, and genomic hypothesis generation—have kernel-checked Lean fixtures.

This is not a decorative appendix. It is the backbone. Every "Formalized in . . . " footnote in the text points to a specific Lean file; a reader can build the file and verify the claim in minutes.

1.7 How to Read This Book

The next chapter (Chapter 2) presents the entire calculus in miniature. A reader who studies that chapter will see how the whole book fits together before entering the deeper technical development. The remaining parts are:

Part II develops the calculus in depth: evidence carriers, state operations, views, and natively typed models.

Part III covers compiled inference, Bayesian-network exactness, the no-go theorem, and large-scale inference control.

Part IV extends to temporal, causal, and normative reasoning.

Part V bridges to seven neighboring frameworks and states the limitations honestly.

Part VI validates the framework on real benchmarks across six domains.

This book is not a PLN tutorial or a replacement for the PLN book. It is a mathematical refoundation with formal verification, aimed at researchers who want to understand PLN's semantic commitments and extend the calculus to new domains.

Chapter 2

The World-Model Calculus: An Overview

This chapter presents the world-model calculus in miniature. It states the main semantic objects, the main exact regime, the main boundary, and the role of typed structure and compiled inference. A reader who studies this chapter should see how the whole book fits together before entering the deeper technical developments.

2.1 The Core Picture

At the highest level, the calculus has three typed parameters:

$$\mathit{State}, \qquad \mathit{Query}, \qquad V.$$

A state is what the system currently carries. A query is what the system may ask. The value type V is the carrier in which extracted answers live. In many applications V will be a sufficient-statistic carrier such as binary counts, Dirichlet counts, or Normal–Gamma statistics; in future work it may also be probability-valued, interval-valued, or amplitude-valued.

The most general world-model interface records only three operations:

```
class WorldModel (State Query V : Type*) where
  revise  : State -> State -> State
  empty   : State
  extract : State -> Query -> V
```

This interface is intentionally minimal. It says only that a world model is a revisable state space equipped with query extraction. No additivity, commutativity, overlap law, or probabilistic interpretation is assumed at this top level. Those laws belong to later subclasses and later chapters.

The book works mostly in one especially important specialization: the *additive regime*. There, state revision is additive, the value carrier is an additive commutative monoid, and extraction commutes with additive revision. This is the regime in which the profile-hom equivalence, Bayesian-network exactness theorems, provenance-tracked revision results, and most of the current Lean development live. But it is not presented as the whole theory. It is the main exact specialization of a broader family.

Maple Court: Maple Court Community Commons is our running example. Each apartment carries a private posterior state over questions such as whether a pipe is leaking, whether the shower is running, or whether motion indicates occupancy. Shared spaces carry a building-level state over hallway humidity, laundry-room availability, and elevator health. The same architecture appears at both scales: revise the state as reports and sensor updates arrive, query the state for evidence, and project that evidence into views only when a decision must be made.

2.2 Revision Regimes and the Additive Powerhouse

The calculus does not assume that all revision behaves the same way. It is useful to distinguish several regimes.

Additive revision. Independent or screening-off updates compose by addition. This is the main tractable regime of the book.

Overlap-aware revision. Two states may share information, so naive addition would double-count. A correction or explicit overlap structure is required.

Trust-gated revision. The merge operation depends on source trust or authority, so revision need not be symmetric or additive.

Coalition or semitopological revision. Consensus is constrained by which coalitions are actionable; not every merge is valid.

Only the first of these is developed theoremically in full generality in the present book. The others already appear in the typed family and in the formalization landscape, but they are treated as frontier directions rather than as fully developed kernels. Chapter 4, §4.3 develops the *non-additive perimeter* in more detail: the overlap extraction law, semitopological coalitions, and the recovery theorem showing that additivity is regained when overlap is small or non-actionable ($\sim$60 theorems, 0 sorry).

The additive regime is captured by the following shape:

```
class AdditiveWorldModel (State Query V : Type*)
    [AddCommMonoid State] [AddCommMonoid V]
    extends WorldModel State Query V where
  revise_eq_add : forall W1 W2, revise W1 W2 = W1 + W2
  empty_eq_zero : empty = 0
  extract_add   : forall W1 W2 q,
    extract (W1 + W2) q = extract W1 q + extract W2 q
  extract_zero  : forall q, extract 0 q = 0
```

The decisive law is `extract_add`. It says that in the additive regime, revising states and then extracting is the same as extracting first and then combining answers in the carrier V. This is the exact algebraic condition behind the cleanest part of the calculus.

Theorem 2.1 (Additive answer-profile form)**.** *For additive world models, currying the extractor yields an additive monoid homomorphism*

$$\widehat{\text{extract}} : \mathit{State} \to^{+} (\mathit{Query} \to V), \qquad W \mapsto (q \mapsto \text{extract}(W, q)).$$

Conversely, every additive monoid homomorphism $\mathit{State} \to^{+} (\mathit{Query} \to V)$ determines an additive world model. Thus additive world models are exactly answer-profile generators.

This theorem is the categorical and algebraic center of the additive regime. It says that, once additivity is assumed, a world model is one structure-preserving arrow into an answer-profile object. The extensional model $\mathit{Query} \to V$ is terminal among additive world models over the same query space: every additive world model maps canonically to its answer profile.

This does *not* trivialize the subject. The hard work is in choosing expressive state spaces, proving when compiled shortcuts are exact, and showing where local summaries fail. But it does make the kernel visible.

2.3 Evidence Carriers

Evidence is not one fixed datatype. The general pattern is that a carrier stores sufficient statistics for a conjugate family or other compositional value system, and revision in the additive regime is addition of those sufficient statistics.

In the current development three carriers are central.

Binary evidence. A two-channel carrier recording positive and negative support. This is the historical PLN carrier, now treated as one instance among others.

Dirichlet evidence. A multi-valued carrier recording category counts. This is the right instance for questions with more than two alternatives.

Normal–Gamma evidence. A continuous-domain carrier storing the sufficient statistics needed for unknown mean and variance.

In Maple Court these arise naturally. Leak alarms and motion detections are binary. Elevator health and laundry status are multi-valued. Hallway humidity and temperature are continuous. The calculus does not change across these cases; only the carrier does.

The additive regime explains why revision is so simple in these families: counts add, sufficient statistics add, and the extractor commutes with that addition. Some additional algebraic structure can then be studied on top of particular carriers. For binary evidence, the current development includes a quantale tensor and a Heyting structure. Those are mathematically real, but they were originally developed for binary evidence but have now been extended to the full k-ary Dirichlet family: coordinatewise multiplication gives a commutative monoid tensor, and the lattice gives a complete Heyting algebra (Frame) for all k, not just $k = 2$.[1]

Evidence items also carry an *epistemic kind* label (empirical, expert-elicited, model-derived, text-interpreted, logically derived) that records where the evidence came from. The same pseudo-count means different things depending on whether it arose from sensor data or expert judgment (Chapter 6). When conjugacy holds, sufficient statistics are lossless compressions of the raw observations; the raw data can optionally be retained for model switching, auditing, or distributional queries that exceed the carrier's resolution.

[1] Formalized in `Mettapedia/Logic/EvidenceDirichletQuantale.lean` (13 theorems, 0 sorry). The round-trip at $k = 2$ recovers `BinaryEvidence` exactly.

2.4 States Are Primary: Revise, Query, Forget, Explain

The book revolves around four verbs: *revise*, *query*, *forget*, and *explain*.

Revision and querying are the minimal kernel operations. Forgetting and explanation show why the kernel stores states rather than floating truth values. A state can be updated, partially retracted, audited, replayed, and explained in a way that a naked scalar view cannot.

Maple Court: Suppose three hallway humidity alerts arrive, then one sensor is found to be faulty. If the system carries only a strength estimate such as 0.83, it is unclear how to retract just the faulty source. If the system carries a revisable state with provenance, forgetting can remove the appropriate contribution while preserving answers outside the forgotten scope. The same state can also explain why the building robot now believes the hallway needs mopping: not because the confidence number crossed an arbitrary threshold in the abstract, but because specific reports and sensors revised the building state in a tracked way.

This is why provenance is not "mere metadata." It is part of the structure of stateful reasoning. Likewise, forgetting is not an optional convenience. It is a semantic operation that clarifies what kind of proof object a world-model state really is.

2.5 Information, Support, and Views

One of the deepest distinctions in the book is that evidence can improve in more than one sense. It can become *more informed*, and it can become *more supportive* of a query. These are not the same order.

For binary evidence, compare

$$e_1 = (1, 0), \qquad e_2 = (1, 1).$$

The second state is more informative than the first in the ordinary information-order sense: it contains at least as much positive and at least as much negative evidence. But it is less supportive of the positive claim, because the negative channel has increased as well. A strength view therefore decreases even as total evidence increases.

This distinction drives the architecture.

Proposition 2.2 (Views are not additive semantics)**.** *In the current development, total evidence is additive, but central operational views such as strength are not. More generally, useful readouts from evidence may be informative, calibrated, or decision-relevant without preserving the revision law of the underlying carrier.*

So a view is not a proof object. It is a projection. This is why the book keeps views after evidence and state operations, not before them. The semantics lives in state and extraction; views are derived interfaces for display, thresholding, and action.

2.6 Natively Typed World Models

The calculus is typed from the beginning. Its primary sorts are already visible in the kernel:

$$\mathit{State}, \qquad \mathit{Query}, \qquad V.$$

What later chapters develop in more detail is not a second semantics but a full native typed realization of the same calculus: a rewrite family whose terms, rules, and native types expose the same mathematical structure from inside.

In the additive regime, the core rewrite system is organized by a 2×2 calculus square: raw versus guarded, and plain versus congruence-aware. Later chapters expand this to a richer typed family, prove the encoding is biconditionally adequate (the typed and algebraic presentations agree exactly), and show that for any type T, multiset ensembles give a canonical world model—suggesting the framework may be universal for typed rewrite systems (Chapter 8).

Logical structure appears wherever the query space carries it. Quantified and higher-order queries are not presented as "extensions" to a lower theory. They are simply instances of the general *Query* parameter when one wants logical structure rich enough to express those questions.

2.7 Compiled Inference and the PLN Connection

The calculus does not treat deduction, induction, and abduction as primitive truth-value algebra. It treats them as *compiled rewrites*: certified shortcuts that are exact under explicit side conditions within declared model classes.

This is where the book most visibly overlaps with the PLN tradition. The derived link calculus and the compiled Bayesian-network rules formalize, sharpen, and delimit the classical PLN rule family. The point is not to deny that those rules can be useful. The point is to state exactly when they are correct, what they compute, and what they leave out.

Maple Court: Suppose the building model contains the chain *PipeLeak* $\to$ *WallHumidity* $\to$ *MoldRisk*. A maintenance agent wants to answer a leak-related query quickly. Rather than recomputing the full posterior from scratch, the system may use a compiled chain rule justified by the BN factorization and screening-off assumptions attached to that model class. When those side conditions hold, the compiled answer agrees with the full extractor on the covered query family. When they do not hold, the rewrite is not licensed.

That is the modernized meaning of a PLN rule inside WM-PLN: not a free-standing law of truth values, but a typed and side-conditioned shortcut to a result already determined by the underlying world model. The most exact form of chaining propagates the *full posterior distribution* through the chain, not just scalar strength and confidence. Five certified-chaining limitation theorems (Chapter 12) make the cost of the scalar approximation precise. Demand-directed backward chaining (Chapter 13) complements forward supply: demand propagation identifies which premises are most worth investigating, and the demand contracts after supply fills the information gap.

2.8 The Boundary: Why Local Chaining Cannot Be Complete

The book's main negative result explains why compiled inference must be handled so carefully.

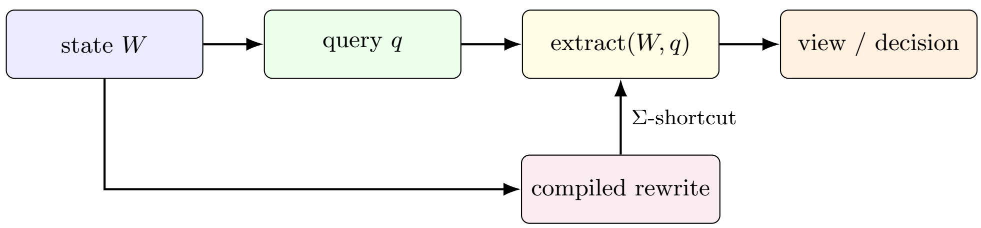
Theorem 2.3 (No-go for unrestricted local link completeness)**.** *In general, complete link evidence cannot be recovered from local per-link surrogates alone. Two world-model states can agree on all chosen local link summaries while differing on the complete evidence needed for the target query.*

This boundary theorem is not an embarrassment. It is a design theorem. It says that unrestricted local chaining cannot replace a complete semantics. It also explains why structural side conditions matter so much: compiled shards become exact only inside model classes that carry enough global structure for those shortcuts to commute with full extraction.

The no-go theorem is also one reason the non-additive frontier matters. When sources overlap, correlations matter, or trust gates revision, naive local summary propagation is even less likely to preserve the required structure. The additive regime is powerful precisely because it identifies a large, tractable class in which the profile-hom story, carrier algebra, and compiled rules all fit cleanly together.

2.9 A Compact End-to-End Picture

The overview can now be summarized in one pipeline:



The state is primary. Queries interrogate the state. Extraction returns values in a carrier. Views project from those values. Compiled rewrites accelerate families of queries under explicit side conditions. Boundaries tell us where such shortcuts cannot be complete.

The computational story is layered: revision is $O(1)$ (add sufficient statistics); single-query extraction is $O(1)$; but full joint inference over many correlated queries is as hard as the underlying model class (#P-hard for general Bayesian networks). The calculus does not make this cheaper. What it does is make the cost *visible*: the Σ-guard tells you exactly which queries admit cheap compiled shortcuts, the no-go theorem tells you which do not, and the demand layer tells you where to spend your computational budget.

That is the whole book in compressed form.

2.10 What the Calculus Does in Practice

The applications (Part VI) test the calculus on real data: gas sensor drift across 36 months (governance question: when to retrain?), geopolitical forecast aggregation (forgetting stale reliability weights improves prediction), Bayesian-network benchmarks (distributional inference reduces MAE by

1.5–3.8×), and eight PPLBench lanes (matching Stan's posteriors at $O(1)$ per update). The empirical contribution is not raw accuracy but inference-control quality: exact where licensed, bounded where partially licensed, abstaining where not.

2.11 How the Rest of the Book Unfolds

The remaining chapters slow this compact picture down. Part II develops the algebra and its instances in full. Part III formalizes compiled inference, BN exactness, and large-scale inference control. Parts IV–V extend to temporal, causal, and normative reasoning, bridge to seven neighboring frameworks, and state the limitations honestly. Part VI validates the framework empirically across six domains.

The intended reading experience is two-layered. This overview shows the crystal. The rest of the book turns the crystal in the light, proves its faces, and follows each face into deeper mathematics and larger applications.

PLN heritage. The relationship of this material to the PLN tradition—what it recovers, what it corrects, what it delimits—is discussed in the Introduction (Chapter 1).

Chapter 3

Desiderata

Before building the calculus, we state what it must achieve. This chapter lists eight normative requirements—*desiderata*—that the design is held accountable to. Each desideratum is stated as a named, numbered property so that later chapters can cite it explicitly when discharging it. The desiderata are not axioms of the calculus (those come in Chapter 4). They are requirements *on* the calculus: properties that a well-designed framework for uncertain reasoning ought to satisfy, and that we claim this one does satisfy, under conditions that will be made precise.

Why desiderata, not axioms? An axiom is a starting point for derivation. A desideratum is a target for design. The distinction matters because several of the properties below are not simultaneously achievable in full generality: the no-go theorem (Chapter 9) shows that local chaining cannot be universally complete, and the five chaining-limitation theorems (Chapter 12) show that scalar propagation inevitably loses information. The desiderata say what we *want*; the theorems say what we *get* and where we *stop*. The interplay between the two is what gives the book its shape.

Origin. These desiderata crystallized during the formalization of Probabilistic Logic Networks [11]. The original PLN design implicitly assumed most of them—evidence-based reasoning, the strength-confidence decomposition, revision as evidence combination, the idea that inference rules are derived rather than primitive. What PLN v0.9 did not do was state these requirements explicitly, prove them under typed conditions, or delineate where they fail. The present list makes the implicit contract explicit.

Independence. The eight desiderata are logically independent: for each one, there exists a system satisfying the other seven while violating it. These independence witnesses are noted after each statement. The independence claim is not a theorem of the formal development—it is a design observation that justifies each desideratum's inclusion as non-redundant.

The desiderata are organized in three layers. The first two (**D1**–**D2**) define the semantic kernel. The next two (**D3**–**D4**) constrain the evidence architecture. The final four (**D5**–**D8**) constrain inference, retraction, conservation, and self-knowledge.

Maple Court: Each desideratum will be illustrated with Maple Court's building model. The illustrations are not proofs—they are concrete instances showing that the desiderata arise naturally in a realistic evidence-management scenario, not as abstract stipulations.

3.1 D1: State Primacy

Definition 3.1 (Desideratum D1 — State Primacy)**.** A reasoning system should carry a *revisable state* that can be updated, queried, partially retracted, and explained. Truth-value summaries are derived from states, never stored as the primary representation.

The philosophical motivation is elementary: a bare number has no parts. You cannot ask *where* a strength of 0.83 came from, which observations contributed to it, or what would happen if one source were retracted. A state has structure. It records what was observed, by whom, and when. The four verbs of the calculus—revise, query, forget, explain—all require state; the first two require it minimally, the last two require it essentially.

What D1 rules out. Any system that stores only a scalar summary (a probability, a strength-confidence pair, a log-odds value) as its primary representation violates D1. Classical probability calculus violates D1 when probabilities are treated as fundamental rather than derived from evidence. PLN v0.9 satisfies D1 in spirit (it carries evidence counts) but does not formalize the state operations that D1 enables (forgetting, explanation).

Discharge. D1 is discharged by the `WorldModel` typeclass (Chapter 4), which takes State as a first-class typed parameter with `revise`, `empty`, and `extract` as operations.

Independence witness. A lookup-table system that maps queries directly to real-valued answers (no state, no revision) satisfies D2–D8 trivially but violates D1: there is nothing to revise, retract, or explain.

Maple Court: Each Maple Court apartment carries a private state over questions such as "Is there a pipe leak?" and "Is the shower running?" The state is not the strength 0.83—it is the evidence pair $(5, 1)$: five leak alarms and one no-alarm reading. The state can be revised (a new alarm arrives), queried (what is the current leak evidence?), forgotten (retract the reading from the miscalibrated sensor), and explained (the evidence came from sensors 3B-north and 3B-south, installed on 2025-01-15).

3.2 D2: Compositional Extraction

Definition 3.2 (Desideratum D2 — Compositional Extraction)**.** In the primary regime, querying a revised state should decompose into queries of the components:

$$\mathrm{extract}(\mathrm{revise}(W_1, W_2),\, q) = \mathrm{combine}(\mathrm{extract}(W_1, q),\ \mathrm{extract}(W_2, q)).$$

That is, extraction should be a homomorphism from the state monoid to the value monoid.

This is the algebraic heart of the additive regime. It says that evidence from independent sources can be combined without interference: the result of revising and then querying equals the result of querying each source separately and then combining the answers.

Why this matters. Compositional extraction is what makes the profile-hom equivalence (Theorem 4.8) possible: an additive world model *is* a monoid homomorphism from states to answer profiles. Without compositionality, there is no universal property, no clean categorical picture, and no guarantee that compiled inference shortcuts agree with full extraction.

The foundational justification. D2 is not an arbitrary algebraic choice. De Finetti's representation theorem for exchangeable sequences shows that, under exchangeability, sufficient statistics add: $(n_1^+, n_1^-) + (n_2^+, n_2^-) = (n_1^+ + n_2^+, n_1^- + n_2^-)$. The Solomonoff–de Finetti foundation (Chapter 4, §"The Solomonoff–De Finetti Foundation") establishes the categorical necessity chain:

$$\begin{gathered}\text{exchangeability} \to \text{de Finetti mixture} \to \text{Kyburg flattening} \\ \to \text{Giry bind} \to \text{evidence sufficiency.}\end{gathered}$$

D2 is the algebraic shadow of this necessity.

What D2 rules out. Any system whose query answers depend on the *order* of evidence arrival (in the independent-sources case) violates D2. Classical Bayesian updating satisfies D2 at the level of sufficient statistics but violates it at the level of posterior probabilities (normalization destroys additivity). This is exactly why the calculus operates on carriers (sufficient statistics), not on probabilities.

What D2 does not require. D2 is stated for the "primary regime" (the additive regime). It is *not* required for all revision modes. The overlap-aware, trust-gated, and coalition regimes of Chapter 4 explicitly violate D2, and this is expected: when sources share information, or when trust gates revision, naive composition is wrong. D2 identifies the regime where composition is exact; the other regimes are frontier directions where D2's algebraic convenience is traded for richer structure.

Discharge. D2 is discharged by the `AdditiveWorldModel` axiom `extract_add` (Chapter 4, §4.4) and the profile-hom equivalence (Theorem 4.8).

Independence witness. A system that carries states and supports forgetting (D1, D6) but uses a nonlinear extraction—e.g., $\text{extract}(W_1 + W_2, q) = \max(\text{extract}(W_1, q), \text{extract}(W_2, q))$—satisfies D1, D3–D8 but violates D2.

Maple Court: Apartment 3B receives two independent sensor reports about the pipe: $(3, 1)$ from the north sensor and $(2, 0)$ from the south sensor. D2 says the combined evidence is $(5, 1)$, the same whether we combine first and then query, or query each and combine the answers. The order of arrival does not matter; the combination is commutative and associative. This is why the apartment can buffer overnight readings and batch-update at dawn (the sleep-consolidation theorem, Theorem 5.1).

3.3 D3: Generic Evidence Carriers

Definition 3.3 (Desideratum D3 — Generic Carriers)**.** The value type V in which extracted answers live should be a *parameter*, not a fixed datatype. Different domains (binary yes/no, multi-valued categories, continuous measurements) should instantiate the same algebraic discipline with different carriers, and the calculus should not change across them.

The motivation is domain coverage. A framework that handles only binary queries is too narrow for practical applications. Elevator health is three-valued (normal, slow, faulty). Hallway humidity is continuous. Gene expression levels are real-valued vectors. Each domain has its own natural sufficient statistics, and a good calculus should accommodate all of them under one roof.

The algebraic discipline. D3 does not ask for V to be completely arbitrary. It asks for V to be an `AddCommMonoid`: a commutative monoid under addition with a zero element. This is the minimal structure needed for D2 (compositional extraction) to make sense. Each carrier instantiates this structure with its own sufficient statistics:

Carrier	V	**AddCommMonoid operation**
Binary	$(n^+, n^-) \in \mathbb{R}^2_{\geq 0}$	componentwise addition
Dirichlet	$(\alpha_1, \ldots, \alpha_k) \in \mathbb{R}^k_{\geq 0}$	componentwise addition
Normal-Gamma	$(n, \sum x_i, \sum x_i^2) \in \mathbb{R}^3_{\geq 0}$	componentwise addition
Amplitude	$z \in \mathbb{C}$	complex addition

The same five algebraic rules (Chapter 4, §4.6) hold for every row of this table. The calculus changes only in the choice of carrier; the algebra stays the same.

What D3 rules out. Any framework that hard-codes a single evidence type (e.g., binary strength-confidence pairs) violates D3. PLN v0.9 uses binary evidence exclusively; the world-model calculus treats binary evidence as one instance among several (Chapter 5).

Discharge. D3 is discharged by the parametric definition `WorldModel (State Query V : Type*)` with V constrained only by `[AddCommMonoid V]` in the additive regime. The three formalized carriers (binary, Dirichlet, Normal-Gamma) are concrete instances (Chapter 5).

Independence witness. PLN v0.9 (with binary evidence only) satisfies D1, D2, D4–D8 but violates D3: the carrier is not generic.

Maple Court: Maple Court uses all three primary carriers simultaneously: binary (n^+, n^-) for leak alarms and occupancy; Dirichlet $(\alpha_1, \alpha_2, \alpha_3)$ for the three-valued elevator health and laundry status; Normal-Gamma $(n, \sum x_i, \sum x_i^2)$ for continuous humidity readings. D3 ensures that the same revision law, the same extraction interface, and the same forgetting mechanism work across all three—only the carrier changes.

3.4 D4: Evidence over Views

Definition 3.4 (Desideratum D4 — Evidence over Views)**.** Operational readouts—strength, confidence, interval bounds, posterior means—are *derived projections* from richer evidence carriers. They must never be treated as the primary semantic objects. Evidence is what you carry; views are what you display.

This is the desideratum that most sharply distinguishes the world-model calculus from its predecessors. Classical probability theory operates on probabilities. PLN v0.9 operates on strength-confidence pairs. NARS operates on frequency-confidence pairs. All of these treat the projected value as the thing being stored and manipulated. D4 says: store the evidence, not the projection.

Why views are not proof objects. A strength value $s = 0.83$ tells you nothing about how much evidence produced it. The evidence pairs $(5, 1)$ and $(500, 100)$ both have strength ≈ 0.83, but the second is far more informative. More importantly, views are not additive (Proposition 2.2): the strength of combined evidence is not the sum of individual strengths. Any system that propagates views through inference chains accumulates artificial precision—the five chaining-limitation theorems of Chapter 12 quantify this loss precisely.

The categorical reading. In categorical terms, evidence carriers are objects in the category of additive commutative monoids; views are morphisms from carriers to simpler monoids (or non-monoids). The key fact is that most views are *not* monoid homomorphisms: they do not preserve the additive structure. This is why views cannot serve as the semantic layer—they break the algebraic structure that D2 provides.

What D4 rules out. Any system that stores only s and c (strength and confidence), discarding the underlying counts, violates D4. Such a system cannot support forgetting (you cannot retract part of a scalar), cannot distinguish high-confidence low-evidence from high-confidence high-evidence, and cannot propagate full posteriors through inference chains.

Discharge. D4 is discharged by the architecture of the calculus: extraction returns values in the carrier V (the evidence layer), and views are defined as separate functions on V (Chapter 7). The strength non-monotonicity theorem (Theorem 7.1) proves that views are not order-preserving, making the separation essential rather than conventional.

Independence witness. A system that stores full evidence carriers but has no extraction homomorphism (violating D2) still satisfies D4. Conversely, a system with compositional extraction into a carrier isomorphic to its views (e.g., $V = \mathbb{R}$ with extraction returning strength directly) satisfies D2 but violates D4.

Maple Court: Two Maple Court humidity sensors report binary evidence for "pipe leak in 3B": sensor north gives $(5, 1)$ and sensor south gives $(2, 0)$. Combined evidence is $(7, 1)$, yielding strength $s = 7/8 = 0.875$ and confidence $c = 8/10 = 0.8$. A system satisfying D4 stores $(7, 1)$—not the scalar pair $(0.875, 0.8)$. Why? Because if sensor north is later found to be miscalibrated (D6), the system can retract $(5, 1)$ to recover $(2, 0)$; from the scalars alone, the retraction is impossible. The evidence pair is the proof object; the strength–confidence view is the projection.

3.5 D5: Explicit Side Conditions

Definition 3.5 (Desideratum D5 — Explicit Side Conditions)**.** Every compiled inference shortcut must carry its structural assumptions as a typed, machine-checkable predicate Σ. When Σ holds, the shortcut is *exact*: it agrees with full extraction. When Σ fails, the shortcut is not licensed. There are no hidden assumptions.

This desideratum addresses a deep problem in practical uncertain reasoning: rules that work under unstated assumptions. The PLN deduction formula is correct—but only under screening-off and positivity ($0 < s_B < 1$). Naive Bayes classification is exact—but only under the conditional independence assumption encoded in a fork BN. D5 says: state the assumption. Prove the rule. Check the assumption at runtime.

The Σ-discipline. In the calculus, every compiled rewrite has the form:

$$\frac{\Sigma \vdash \text{sideCondition} \quad \Sigma; \Gamma \vdash \text{premises}}{\Sigma; \Gamma \vdash \text{conclusion}}$$

The side condition Σ appears above the line. It is not optional. It is a typed proposition that can be proved, checked by a decision procedure, or required as a hypothesis.

What D5 rules out. Any system that applies inference rules unconditionally—without checking or even stating their side conditions—violates D5. PLN v0.9 states rules as universal formulas ("deduction always gives $s_{AC} = \ldots$"); the world-model calculus states them as conditional theorems ("if screening-off and positivity, then $s_{AC} = \ldots$"). The no-go theorem (Chapter 9) shows that unconditional application is unsound in general: two world-model states can agree on all local per-link summaries while differing on the target query.

Discharge. D5 is discharged by the Σ-guarded rewrite architecture of Chapter 10 and the BN exactness theorems of Chapter 11. Each compiled rule in the catalog (Chapter 10, §"The Compiled Rule Catalog") has an explicit Σ column. The bridges chapter (Chapter 17) shows that the same Σ-discipline applies across all seven neighboring frameworks.

Independence witness. A purely extensional world model that stores full joint distributions and answers queries by marginalization satisfies D1–D4, D6–D8 but has no compiled rules and therefore trivially satisfies (or vacuously satisfies) D5. The interesting violation is a system *with* compiled rules but *without* explicit side conditions—this is essentially PLN v0.9, which violates D5 while approximately satisfying the others.

Maple Court: The compiled chain rule for PipeLeak $\to$ WallHumidity $\to$ MoldRisk carries the side condition $\Sigma = \{\text{ScreenOff}(\text{PipeLeak}, \text{WallHumidity}, \text{MoldRisk}),\ 0 < s_{\text{Humidity}} < 1\}$. When this Σ holds—which it does for Maple Court's BN structure, where the DAG guarantees d-separation—the compiled formula gives the exact posterior. When an additional direct leak-to-mold pathway is added (breaking screening-off), Σ fails and the rule is not licensed. D5 ensures that this failure is detected, not silently ignored.

3.6 D6: Retractability

Definition 3.6 (Desideratum D6 — Retractability)**.** It must be possible to *retract* evidence within a declared scope while provably preserving all answers outside that scope. Formally: for every scope S and state W, there exists a forgetting operation forget(S, W) such that

$$\forall q \notin S, \quad \text{extract}(\text{forget}(S, W), q) = \text{extract}(W, q).$$

Retractability is what separates stateful reasoning from stateless truth-value manipulation. If a sensor is found to be miscalibrated, its readings must be retractable. If a privacy constraint requires apartment-level data to expire after 24 hours, the system must forget that data without corrupting building-level answers. If a correction arrives ("observation 3 was a false alarm"), the system must remove that observation's contribution while preserving everything else.

Why retractability is not deletion. Deleting evidence is easy: set the state to zero. Retractability is hard because it requires *surgical* removal: forget the evidence within scope S while proving that answers outside S are unchanged. This requires the state to carry enough structure to distinguish what came from where—which is precisely what provenance provides (Chapter 6, §6.4).

What D6 rules out. Any system that cannot undo a revision—either because it stores only aggregate summaries (violating D1) or because it lacks provenance tracking—violates D6. Classical Bayesian updating can, in principle, undo a conjugate update by subtracting the likelihood contribution, but this requires knowing which data point to subtract. Without provenance, the data point is lost in the aggregate.

Discharge. D6 is discharged by the `ForgettingLayer` structure and the outside-invariant law of Chapter 6, §6.2. Its practical value is demonstrated in Chapter 21, §21.5: forgetting stale forecaster reliability weights improves prediction on regime-change questions.

Independence witness. An append-only evidence store (states grow monotonically, no retraction) satisfies D1–D5, D7–D8 but violates D6.

Maple Court: Three hallway humidity alerts arrive, then sensor 3B-north is found to be miscalibrated. If the system stores only the aggregate $(5, 1)$, it cannot retract sensor 3B-north's contribution without knowing how much of the $(5, 1)$ came from that sensor. With provenance, the state records $\{(\text{3B-north}, (3, 1)), (\text{3B-south}, (2, 0))\}$, and forgetting scope $S = \text{3B-north}$ yields $(2, 0)$. The building-level humidity answer (which does not depend on 3B's pipe sensor) is unchanged.

3.7 D7: Evidence Conservation

Definition 3.7 (Desideratum D7 — Evidence Conservation)**.** Evidence can be neither *fabricated* (inference cannot create support that was not present in the premises) nor *overcounted* (the same observation cannot contribute twice to the same query). Formally, three conservation laws hold under exact-inverse forgetting:

1. **Anti-hallucination**: a revision Δ within scope S contributes zero evidence to queries outside S.

2. **Noether conservation**: the revised state agrees with the base state on queries outside the revision's scope.

3. **Zero leakage**: the observation count of Δ at queries outside its scope is zero.

D7 is the evidence-theoretic analogue of conservation laws in physics. Just as energy cannot be created or destroyed, evidence cannot be fabricated or lost. The "anti-hallucination" property is particularly important in the context of large language models and generative AI: a system satisfying D7 cannot output a confident assertion unless the evidence to support it actually exists in the state.

Two faces of conservation. The calculus establishes conservation from two perspectives:

- **State-level** (Chapter 6): the conservation theorems (Theorem 6.1) show that scoped forgetting preserves evidence outside the scope.
- **Inference-path level** (Goertzel [7]): a discrete quantale Noether theorem showing that reinforcement is constant along geodesic inference paths, a hallucination bound on conclusion strength, and an evidence monotonicity theorem (a quantale data-processing inequality).

Together, the two perspectives show that evidence integrity holds whether you look at state retraction or at inference chaining.

What D7 rules out. Any system where chaining inference rules can amplify evidence beyond what the premises contain violates D7. This includes systems where deduction applied twice produces stronger conclusions than the original evidence warrants. The stamp-disjointness condition in the revision rule (Chapter 10) is one mechanism that enforces D7: stamps prevent the same observation from being counted twice.

Discharge. D7 is discharged by the evidence conservation pack (`EvidenceConservationPack` in Chapter 6, §"Conservation Theorems") and the exact-inverse no-go theorem (if evidence leaked, forgetting cannot exactly undo it).

Independence witness. A system with scoped forgetting (D6) but no conservation guarantee (e.g., a sloppy retraction that removes too much or too little) satisfies D6 but violates D7. Conversely, a system that never forgets (violating D6) trivially satisfies D7 by never modifying the state.

Maple Court: Suppose the building model infers mold risk from pipe-leak evidence via a chain of compiled rules. D7 guarantees that the mold-risk evidence in the conclusion cannot exceed the pipe-leak evidence in the premises. If only $(5, 1)$ of leak evidence exists, the mold-risk conclusion has at most $(5, 1)$ worth of evidential support (modulated by the chain's conditional probabilities). The anti-hallucination property says: if the pipe-leak evidence is entirely within scope "apartment 3B," then the mold-risk answer for apartment 5C is unaffected.

3.8 D8: Honest Boundaries

Definition 3.8 (Desideratum D8 — Honest Boundaries)**.** The calculus must state what it *cannot* do as precisely as what it can do. Impossibility results—no-go theorems, coverage collapse bounds, non-uniqueness demonstrations—are first-class contributions, not embarrassments. Every boundary theorem has the same formal status as every positive theorem.

D8 is a meta-desideratum: it constrains how the calculus presents itself. A framework that claims universality without stating its limits is either trivial or dishonest. The world-model calculus earns its credibility by proving both what works and what does not.

The boundary catalog. The book contains a substantial family of boundary results:

- **No-go for local link completeness** (Chapter 9): per-link summaries cannot reconstruct the joint.
- **Five chaining limitations** (Chapter 12): variance accumulation, sensitivity amplification, coverage collapse, topology-CPT gap, and no universal error bound.
- **Classical inclusion-exclusion gap** (Chapter 13): the first two inclusion-exclusion terms do not determine the union cardinality—a classical fact, formalized as a guard condition.
- **Trail-free non-stabilization** (Chapter 13): re-deriving conclusions without trails can oscillate.

- **Pooling non-uniqueness** (Chapter 13): the four pooling axioms do not pin down additive fusion.
- **No binary mixed policy** (Chapter 14): the classical two-channel intensional mixture is refuted.
- **Exact-inverse no-go** (Chapter 6): if evidence leaked, forgetting cannot exactly undo it.
- **Strength non-monotonicity** (Chapter 7): more evidence can yield lower strength.

Each boundary theorem carries a BOUNDARY label, making it visually and taxonomically distinct from CORE and CONDITIONAL results.

What D8 rules out. Any framework that presents only positive results—"our system can do X, Y, Z"—without stating where it fails violates D8. This includes marketing-style presentations of AI systems ("universal reasoning engine") that omit computational complexity, soundness gaps, or regime limitations.

Discharge. D8 is discharged by the boundary catalog above and by the claim taxonomy that labels every theorem as CORE, CONDITIONAL, or BOUNDARY.

Independence witness. A framework with correct positive results but no proved impossibility theorems violates D8 while potentially satisfying D1–D7.

3.9 The Desiderata as a Whole

The eight desiderata form a layered specification:

ID	Name	Layer	Primary discharge
D1	State Primacy	Semantic kernel	Ch 4: `WorldModel`
D2	Compositional Extraction	Semantic kernel	Ch 4: profile-hom
D3	Generic Carriers	Evidence architecture	Ch 5: three carriers
D4	Evidence over Views	Evidence architecture	Ch 7: lossy projections
D5	Explicit Side Conditions	Inference discipline	Ch 10: Σ-guards
D6	Retractability	State management	Ch 6: forgetting layer
D7	Evidence Conservation	State management	Ch 6: conservation pack
D8	Honest Boundaries	Self-knowledge	Ch 9: no-go theorems

Necessity. Are all eight necessary? The independence witnesses show that dropping any one produces a system that, while coherent, fails in a specific and important way: without D1, you cannot retract; without D2, you cannot compose; without D3, you are limited to one domain; without D4, you confuse projections with semantics; without D5, your shortcuts are unsound; without D6, your state is append-only; without D7, your inference can hallucinate; without D8, you do not know your own limits.

Sufficiency. Are the eight jointly sufficient for "responsible uncertain reasoning"? Not in full generality. At least three additional properties are needed for a complete system but are not yet fully developed in this book:

- **Computational tractability**: the calculus says *what* the correct answer is but does not bound the *cost* of computing it.
- **Non-additive regimes**: D2 holds only in the additive regime; overlap-aware and trust-gated revision need their own algebraic discipline.
- **Measure-theoretic grounding**: the evidence carriers operate over the extended reals ($\mathbb{R}^{\infty}_{\geq 0}$ for binary evidence, $\mathbb{R}$ for Normal-Gamma), not over finite types. The terminal measure-valued world model connecting these carriers to formal probability measures is an open program.

These are recorded as frontier directions (Chapter 19), not as current desiderata. The eight listed here are the contract that the present book fulfills.

The PLN heritage. The relationship to PLN is discussed in Chapter 1. In brief: PLN v0.9 satisfies D1, D2 in spirit but lacks D3 (single carrier), D5 (no Σ), D6 (no formal forgetting), D8 (no no-go theorems).

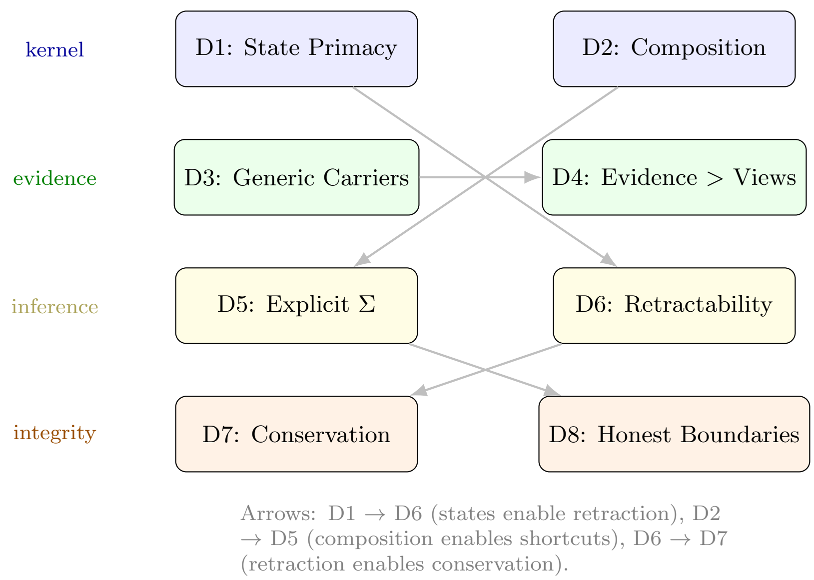


Figure 3.1: The eight desiderata organized by layer. Arrows show enablement relationships: D1 (state primacy) enables D6 (retractability); D2 (compositional extraction) enables D5 (explicit side conditions for compiled shortcuts); D6 (retractability) enables D7 (conservation).

Maple Court: Maple Court's building model satisfies all eight desiderata. It carries states (D1), combines sensor reports additively (D2), handles binary, Dirichlet, and Normal-Gamma evidence (D3), derives strength and confidence as views (D4), gates compiled chain rules on BN screening-off (D5), retracts miscalibrated sensor data (D6), conserves evidence under retraction (D7), and reports where local chaining fails (D8). The desiderata are not abstract stipulations imposed from outside—they are the properties that make a building automation system trustworthy.

Part II

The Calculus in Depth

Chapter 4

The World-Model Calculus

What does it mean for a system to carry uncertain knowledge? It means the system has a *state* that can be revised as new information arrives, and that can be *queried* to extract answers about the world. The world-model calculus makes this precise.

4.1 The General Interface

At its most abstract, a world model has three things: something it knows (a *state*), questions it can ask (*queries*), and a way to extract answers from what it knows. In Lean, this is a typeclass with three typed parameters and three operations:

```
class WorldModel (State Query V : Type*) where
  revise  : State -> State -> State
  empty   : State
  extract : State -> Query -> V
```

A `State` is what the system carries. A `Query` is what the system may ask. V is the type in which extracted answers live. `revise` combines two states. `empty` is the state of knowing nothing. `extract` answers a query from a state.

No laws are assumed at this level. How $\mathrm{extract}(\mathrm{revise}(W_1, W_2), q)$ relates to the individual extractions depends on the *revision regime*—a choice that determines the character of the calculus.

The value type V is generic. Choosing V determines the domain:

Value type V	**Domain**
(n^+, n^-)	Binary observations
$(\alpha_1, \ldots, \alpha_k)$	Multi-valued observations
$(n, \sum x_i, \sum x_i^2)$	Continuous observations
$\mathbb{C}$	Quantum amplitudes

Each of these carries more information than a bare probability: the sufficient statistics retain what is needed for further Bayesian updating, not just a point estimate. Classical probabilities are always available as *views*—lossy projections from richer carriers (Chapter 7).[1]

The query type is equally generic. It may be a simple propositional language, a first-order or higher-order logic, a temporal query (Chapter 15), an intensional-sort query (Chapter 14), or a deontic modality (Chapter 16). Logical structure appears wherever the query space carries it—quantified and higher-order queries are instances of the generic `Query` parameter, not extensions to a lower theory.

The generality of this interface is genuine. Seven neighboring frameworks—Markov Logic Networks, ProbLog, PLN v0.9, classical FOL, the Giry monad, and the ρ-calculus—are shown to be `WorldModel` instances with proved extraction agreements; NARS has a proved Galois connection to PLN evidence (Chapter 17). Pearl's structural causal models [27] are also `WorldModel` instances: the state is a system of structural equations $X_i = f_i(\mathrm{pa}(X_i), U_i)$ plus a joint distribution over exogenous variables U_i. Revision updates the equations or incorporates observational data. Queries span Pearl's three-level causal hierarchy: observational ($P(Y \mid X = x)$), interventional ($P(Y \mid \mathrm{do}(X = x))$), and counterfactual. The do-calculus rules become compiled rewrites under causal-DAG side conditions—exactly the pattern developed in Part III. Revision of SCMs with conflicting equations is *not* additive; it requires the overlap-aware or trust-gated regimes described in the next section.[2]

Maple Court: Each Maple Court apartment carries a private state over questions such as whether a pipe is leaking or whether the shower is running. Querying "Is there a pipe leak?" extracts binary evidence $(1, 4)$—one leak alarm and four no-alarm readings. Shared spaces carry a building-level state

[1] The first three carriers are fully formalized in Lean (Chapter 5). The amplitude carrier $\mathbb{C}$ connects to Meredith's framework [25] and is a natural future instance.

[2] Formalized in `Mettapedia/Logic/WorldModelBase.lean`.

over hallway humidity and elevator health. The same three operations—revise, empty, extract—appear at both scales.

4.2 Revision Regimes

Chapter 2 introduced four revision regimes. Here we develop the additive regime in depth and situate the others concretely.

Additive. Independent updates compose by addition: $\text{extract}(W_1+W_2, q) = \text{extract}(W_1, q)+\text{extract}(W_2, q)$. This is the main tractable regime and the subject of §4.4–§4.11.

Overlap-aware. When two sensor networks share a source—say, two apartments' humidity sensors both pick up the same hallway vent—naive addition double-counts. An explicit overlap correction removes the shared component.

Trust-gated. When a tenant's verbal report contradicts an automated sensor, the system must weigh source authority. Revision is gated by trust scores: not symmetric, not additive.

Coalition. When multiple apartments must agree before updating the building model—e.g., a majority vote on whether the elevator needs maintenance—only coalitions satisfying an actionability predicate may revise.

The additive regime is developed in full generality in this book. The others are formalized in Lean and treated here in less depth, but they are *not* future work—the key theorems are proved, and this section presents them.

4.3 The Non-Additive Perimeter

The non-additive regimes are not afterthoughts. They formalize the situations where naive evidence addition fails: correlated sources, trust-weighted merges, and quorum-based consensus. The Lean formalization contains $\sim$60 theorems across these regimes, all kernel-checked with 0 sorry.

4.3.1 The Overlap Extraction Law

When two evidence sources share information, addition double-counts the shared component. The `OverlapWorldModel` replaces the additive extraction law with an overlap-corrected version:

$$\mathrm{extract}(\mathrm{revise}(W_1, W_2), q) = \mathrm{combine}\big(\mathrm{extract}(W_1, q),\ \mathrm{extract}(W_2, q),\ \mathrm{overlap}(W_1, W_2, q)\big).$$

When the overlap function returns zero (independent sources), this reduces to the additive law. When overlap is nonzero, the correction prevents double-counting.

Theorem 4.1 (Additivity recovery under independence (Core))**.** *If* $\mathrm{overlap}(W_1, W_2, q)$ *is trivial for all* q*, then the overlap extraction law reduces to the additive extraction law:* $\mathrm{extract}(\mathrm{revise}(W_1, W_2), q) = \mathrm{extract}(W_1, q) + \mathrm{extract}(W_2, q)$.

This theorem is the formal boundary between the additive and overlap regimes: additivity is not assumed—it is *recovered* when the independence condition holds.[3]

The `SubtractiveOverlapLayer` makes the correction explicit via inclusion-exclusion:

$$\mathrm{extract}(\mathrm{merge}(W_1, W_2), q) = \mathrm{extract}(W_1, q) + \mathrm{extract}(W_2, q) - \mathrm{overlap}(W_1, W_2, q).$$

When overlap is zero (independent sources), the subtraction vanishes and additive extraction is recovered exactly (`inclusionExclusion_of_zero_overlap`). This is the formal analogue of the Dempster-Shafer subadditivity correction.

In the probability hypercube (Knuth–Skilling [20]), the overlap regime corresponds to the Dempster-Shafer vertex (subadditive belief functions), where $\mathrm{bel}(A \cup B) \geq \mathrm{bel}(A) + \mathrm{bel}(B) - \mathrm{bel}(A \cap B)$:

WM regime	**Logic**	**Repr.**	**Hypercube vertex**
Additive	Boolean	point	classical Bayesian
Overlap-aware	Boolean	bounds	imprecise probability
Evidence-tracking	Heyting	2D bounds	PLN evidence
Trust-gated	Heyting	point	weighted trust

Each regime is determined by two independent structural decisions: does the event algebra satisfy excluded middle (Boolean vs. Heyting), and does

[3] Formalized in `Mettapedia/Logic/WorldModelOverlap.lean` (7 theorems, 0 sorry).

the representation track per-channel evidence or summarize to a point? The KS representation theorem gives the unique algebra for each vertex.[4]

The carrier axis is orthogonal: binary, Dirichlet, and Normal-Gamma carriers are all formalized and compose with any merge-law vertex. The quantum/classical boundary is whether the carrier has a lattice ordering: $\mathbb{C}$ does not, placing quantum inference at a distinct vertex (`quantum_not_classical`).[5]

4.3.2 Coalition Revision and Semitopology

The `CoalitionWorldModel` requires a proof of actionable(W_1, W_2) before revision can fire. The actionable sets form a *semitopology*: a family closed under union but *not* under intersection—strictly weaker than a topology.

Example 4.2 (Quorum semitopology)**.** Three agents $\{0,1,2\}$. Coalitions $\{0,1\}$ and $\{0,2\}$ are both actionable (two-agent quorum). But their intersection $\{0\}$ is *not* actionable—a single agent cannot force revision. This is the formal content of "a majority must agree."

Theorem 4.3 (Additivity recovery under semitopological independence (Core))**.** *If both states' supports are actionable and disjoint, then the coalition extraction law reduces to the additive extraction law.*

The deepest result connects overlap to coalitions:

Theorem 4.4 (Additivity recovery after forgetting non-actionable overlap (Core))**.** *When the overlap footprint between two states is non-actionable (the shared evidence does not form a legitimate coalition), forgetting the overlap recovers additivity on the remainders.*

This is the formal answer to "when can we safely treat correlated sources as independent?"—when the correlation structure is below the actionability threshold.[6]

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[4] Classification kernel-checked in `Mettapedia/Logic/WMHypercubeClassification.lean` (17 items, 0 sorry).

[5] Hypercube positions are kernel-checked in `Mettapedia/ProbabilityTheory/KnuthSkilling/Examples/PreciseVsImpreciseGrounded.lean` (1071 lines, 26 theorems, 0 sorry).

[6] Formalized in `Mettapedia/Logic/PLNSemitopology.lean` (241 lines, 23 theorems) and `Mettapedia/Logic/PLNSemitopologyProvenanceBridge.lean` (294 lines, 22 theorems). Both 0 sorry.

Maple Court: Two apartments share a hallway humidity vent. Their sensor readings are correlated—naive addition double-counts the shared signal. The overlap layer explicitly tracks this correlation. When the building steward forgets the shared vent's contribution, the remaining per-apartment evidence is independent, and additivity recovers. The quorum structure matters too: if the building requires two apartments to agree before revising the hallway humidity state, a single apartment's reading cannot force an update—that is the semitopological actionability constraint.

7

4.3.3 The Additive Regime as the Main Powerhouse

The non-additive perimeter is formalized because it is *real*—correlated sensors exist, quorum requirements exist, and the book would be dishonest to pretend otherwise. But the additive regime remains the main powerhouse for two reasons: (1) it is where the profile-hom equivalence, BN exactness, provenance tracking, and compiled inference all live, and (2) the recovery theorems above show that the additive regime is *approached* whenever overlap is small or non-actionable. The rest of this chapter develops the additive regime in depth; the non-additive theorems appear again when needed (multideduction overlap in Chapter 13; forgetting in Chapter 6).

To be explicit: the book's strongest exact theorems—profile-hom equivalence, BN chain/fork/collider exactness, evidence conservation, sleep consolidation—are additive-regime theorems. The non-additive regimes are real and formalized ($\sim$90 theorems, 0 sorry), but they are not yet developed to comparable breadth. This asymmetry is not weakness; it is intellectual hygiene about what has been proved and what remains open.

4.4 The Additive Regime

The additive regime assumes that state revision is additive, that the value carrier is an additive commutative monoid, and that extraction commutes with addition:

```
class AdditiveWorldModel (State Query V : Type*)
    [AddCommMonoid State] [AddCommMonoid V]
```

[7]Both examples kernel-checked: `Mettapedia/Logic/PLNMapleCourtOverlapDemo.lean` (8 theorems) and `Mettapedia/Logic/PLNMapleCourtCoalitionDemo.lean` (13 theorems), both 0 sorry.

```
  extends WorldModel State Query V where
revise_eq_add : forall W1 W2, revise W1 W2 = W1 + W2
empty_eq_zero : empty = 0
extract_add   : forall W1 W2 q,
  extract (W1 + W2) q = extract W1 q + extract W2 q
extract_zero  : forall q, extract 0 q = 0
```

The decisive law is `extract_add`. It says that revising states and then extracting is the same as extracting first and then combining answers in the carrier V. Together with `extract_zero`, this makes the curried map $W \mapsto (\lambda q.\,\mathrm{extract}(W, q))$ an additive monoid homomorphism from states to answer profiles:

$$\widehat{\mathrm{extract}} : \texttt{State} \xrightarrow{+} (\texttt{Query} \to V)$$

This one arrow is the entire semantic interface of the additive regime. Everything else—views, compiled rules, inference control—is built on top of it.

Extraction is additive for *every* choice of V. But downstream projections (strength, confidence, Born rule $|\cdot|^2$) are *not* additive—they are lossy views.This asymmetry determines the inference architecture and is developed in Chapter 7.[8]

For the implementer: to build a new additive world model, implement `extract` and prove the two laws. The rest of the stack—views, compiled BN inference, forgetting, provenance—works automatically for any conforming instance.

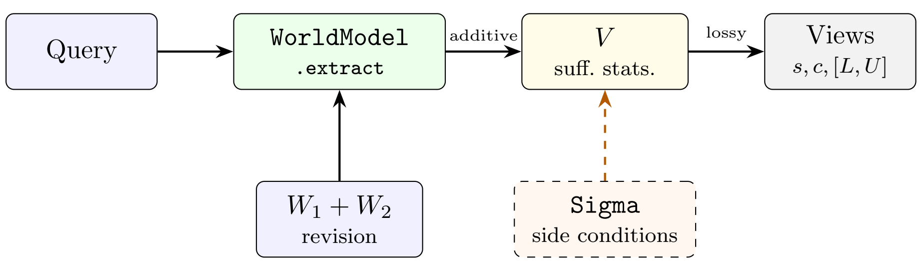

Figure 4.1: The additive world-model interface. A query is answered by extracting a value from the state; views are lossy projections. Revision (+) combines states. Side conditions (Σ) enable compiled rewrites.

[8]Formalized in `Mettapedia/Logic/PLNWorldModelGeneric.lean` and `Mettapedia/Logic/WorldModelCore.lean`.

4.5 Side Conditions

The kernel interface makes no structural assumptions: any additive state that extracts values qualifies. But most *fast* inference depends on structure. When Maple Court's building model uses a Bayesian network to compute humidity-leak correlations, it relies on the DAG factorization and d-separation properties of that network. When the deduction rule computes $P(C \mid A)$ from $P(B \mid A)$ and $P(C \mid B)$, it relies on the *screening-off* assumption:

$$P(C \mid A, B) = P(C \mid B).$$

Once you know B, knowing A provides no additional information about C. In a chain BN $A \to B \to C$, the DAG structure guarantees this (d-separation). In a general world model, it is an explicit side condition that must be verified.

These structural assumptions make tractable inference *exact* rather than approximate. Every compiled rewrite carries its Σ explicitly—there are no hidden assumptions.

For the implementer: when a compiled rule fires, it checks (or requires proof of) its Σ. If the side condition is not met, the rule does not apply.

4.6 Five Typed Rules of the Additive Regime

The profile-hom equivalence has five immediate algebraic consequences. They are not five independent axioms: all five *derive* from the single condition that extraction is a commutative-monoid homomorphism. They are stated here algebraically; Chapter 8 §8.2 develops their concrete incarnation as pattern-matching rewrite rules.

1. **Extraction additivity** (Core).[9]

   $$\text{extract}(\text{revise}(W_1, W_2),\ q) = \text{combine}(\text{extract}(W_1, q),\ \text{extract}(W_2, q)).$$

2. **Zero extraction** (Core). $\text{extract}(0, q) = 0$.

3. **Revision commutativity** (Core). $\text{revise}(W_1, W_2) = \text{revise}(W_2, W_1)$.

4. **Revision associativity** (Core). $\text{revise}(\text{revise}(W_1, W_2), W_3) = \text{revise}(W_1, \text{revise}(W_2, W_3))$.

[9]Claim labels: Core = unconditional algebraic fact; Conditional = holds under stated side conditions; Boundary = impossibility / no-go result; Operational = implementation-level property. These categories are used throughout the book.

5. **Combine identity** (CORE). $\text{combine}(e, 0) = e$.

Theorem 4.5 (`wmCore_sound_of_addCommMonoidHom`, kernel-checked)**.** *If an interpretation satisfies the homomorphism and zero conditions, all five rule soundness conditions follow. The world-model term algebra modulo rewriting is the free commutative monoid; extraction is the unique homomorphism to* $(V, +, 0)$*.*

These five rules organize into a 2×2 square along two orthogonal axes:

	Plain	**+ Congruence**
Raw (schematic)	all rules fire	+ context descent
Guarded (faithful)	side-condition oracle	+ context descent

This is presented *abstractly* here. Chapter 8 incarnates the 2×2 square as concrete typed rewrite rules over the 13-axis vertex family.

4.7 The Sequent-Calculus Presentation

The additive regime has a natural sequent-calculus presentation. Two contexts: an *evidence context* Γ (a finite multiset of state fragments) and a *side-condition context* Σ. Four rules suffice:

$$\frac{}{\Sigma; \vdash_{\text{wm}} 0}\ (\text{Unit}) \qquad \frac{}{\Sigma; \{W\} \vdash_{\text{wm}} W}\ (\text{Ev})$$

$$\frac{\Sigma; \Gamma \vdash_{\text{wm}} W \quad \Sigma; \Delta \vdash_{\text{wm}} W'}{\Sigma; \Gamma \uplus \Delta \vdash_{\text{wm}} (W + W')}\ (\text{Rev}) \qquad \frac{\Sigma; \Gamma \vdash_{\text{wm}} W}{\Sigma; \Gamma \vdash q \Downarrow \text{extract}(W, q)}\ (\text{Extract})$$

Evidence compositionality makes the query-revision rule admissible: if $\Gamma \vdash q \Downarrow e$ and $\Delta \vdash q \Downarrow e'$, then $\Gamma \uplus \Delta \vdash q \Downarrow (e + e')$.

Links are *queries*, not connectives: $A \Rightarrow B$ is a query $\text{link}(A, B)$ against the revised world model, not an object-level implication.

4.8 The Evidence Proof System

The four rules of §4.7 are not arbitrary. Under the Curry-Howard correspondence—the identification of proofs with programs and propositions with types—the evidence calculus is a classical proof system. The evidence state is the proof term, the query is the proposition, and the four rules are the four structural rules of sequent calculus.

Proof Theory	Evidence Algebra
Proof term (context)	Evidence state W
Proposition (judgment)	Query q
Cut rule	Revision $(W_1 + W_2)$
Evaluation	Extraction $(\text{extract}(W, q))$
Weakening	Forgetting
Exchange	Commutativity $(W_1 + W_2 = W_2 + W_1)$
Contraction	Self-revision $(W + W)$

Reading the table downward:

Cut. The Rev rule combines two evidence judgments into one. In proof theory, cut combines two sequent proofs by matching a conclusion with a hypothesis. In the evidence calculus, cut matches exactly: the combined evidence for query q equals the sum of the parts, $\text{extract}(W_1 + W_2, q) = \text{extract}(W_1, q) + \text{extract}(W_2, q)$. This is `extract_add`—the defining law of the additive regime.

Weakening. Discarding a hypothesis without affecting the remaining conclusions. In the evidence calculus, forgetting a source's contribution W_{forget} from a combined state $W_{\text{keep}} + W_{\text{forget}}$ leaves valid evidence for W_{keep} alone. The evidence may decrease but does not become invalid.

Exchange. Reordering hypotheses. Addition in `AddCommMonoid` is commutative, so $W_1 + W_2 = W_2 + W_1$. The order of evidence composition is irrelevant.

Contraction. Using a hypothesis more than once. Self-revision $W + W$ doubles the evidence sample size $(\text{ess}(W + W) = 2 \cdot \text{ess}(W))$ without changing the strength ratio. This is contraction: the same evidence source may be cited twice, yielding more certainty but the same posterior mean.

Theorem 4.6 (Structural completeness (Core))**.** *The additive evidence algebra over* ***`BinEvNat`*** *satisfies all four structural rules of classical sequent calculus: cut, weakening, exchange, and contraction.*[10]

The word *classical* is precise. An intuitionistic proof system satisfies cut, weakening, and exchange, but not contraction: using a hypothesis twice may

[10]Formalized in `Mettapedia/Logic/EvidenceProofSystem.lean` (10 theorems, 0 sorry).

change the proof. The evidence algebra satisfies all four. This is because addition is idempotent in information content (the posterior ratio $n^+/(n^+ + n^-)$ is invariant under $W \mapsto W + W$), even though the raw counts double. A linear proof system would drop weakening (every hypothesis must be used exactly once); the evidence calculus allows forgetting, so it is not linear.

A worked example. Let $e_1 = (3, 1)$ and $e_2 = (2, 2)$.

Cut: $e_1 + e_2 = (5, 3)$. Total evidence ess $= 8 = 4 + 4$.

Weakening: Forget e_2: the remaining evidence $e_1 = (3, 1)$ is still valid, with ess $= 4$.

Exchange: $e_2 + e_1 = (5, 3) = e_1 + e_2$.

Contraction: $e_1 + e_1 = (6, 2)$, with ess $= 8 = 2 \times 4$. The strength $6/8 = 3/4$ equals the original $3/4$.

What this means. The evidence calculus is not merely an algebraic bookkeeping system with inference heuristics attached. It is a proof system in the technical sense: it has judgments (evidence for queries), structural rules (the four above), and a cut-elimination direction (revision increases total evidence, the normalization direction). The propositions-as-types reading is available to anyone who wants it: the evidence state *is* the proof, the query *is* the proposition, and the extraction *is* the evaluation.

4.9 Model, Calculus, and Soundness

Every formal system has two faces. The *model* says what things mean: `AdditiveWorldModel` defines extraction semantically. The *calculus* says what you can do: three operational rules that manipulate evidence states. The *soundness theorem* says these two faces agree.

The calculus is a structure with three fields:

```
structure WMCalculus (State Query Ev : Type*) where
  revise  : State -> State -> State       -- the cut rule
  extract : State -> Query -> Ev          -- the evaluation
      rule
  forget  : Option (State -> State -> State) -- the weakening
      rule
```

An implementor provides three functions. The soundness property says they agree with the algebraic model:

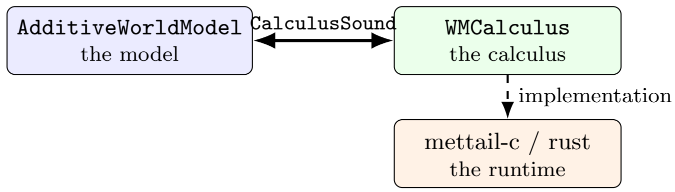

Figure 4.2: Three layers, two boundaries. The book proves the upper boundary (`CalculusSound`). The lower boundary (implementation correctness) is the implementor's responsibility.

```
structure CalculusSound (wmc : WMCalculus State Query Ev) where
  revise_correct  : forall W1 W2, wmc.revise W1 W2 = W1 + W2
  extract_correct : forall W q,   wmc.extract W q  = extract W
     q
```

Two equalities. The calculus's revision is the model's addition. The calculus's extraction is the model's extraction.

Theorem 4.7 (`CalculusSound` inheritance (Core))**.** *If `wmc` satisfies `CalculusSound`, then:*

1. *`extract_add` holds for the calculus (evidence revision is compositional).*
2. *Sequential composition works (multi-step revision is associative).*
3. *Zero preservation holds (empty state gives empty evidence).*
4. *All four structural rules (cut, weakening, exchange, contraction) transfer from the model to the calculus.*

The proof is four rewrites.[11]

The mathematical tower—KS foundations, measure theory, hypercube classification, 700+ theorems—does not need to be re-proved. It transfers by soundness.

A concrete example. Suppose mettail-rust implements `revise` as integer addition on (n^+, n^-) pairs and `extract` as field projection. The implementor proves two equalities: addition agrees, projection agrees. From that proof, every theorem in this book that mentions `extract_add` becomes a theorem about the Rust implementation. The implementor never opens a Lean file about measure theory or the KS representation. Those results arrive by soundness.

[11] Formalized in `Mettapedia/Logic/WMCalculusSoundness.lean` (5 theorems, 0 sorry).

The philosophical point. There is one boundary between mathematics and engineering. `CalculusSound` is that boundary. On one side: the entire algebraic and measure-theoretic development. On the other side: three functions in C or Rust. Cross the boundary once, correctly, and the entire mathematical tower transfers. The proof of `CalculusSound` is the smallest possible obligation with the largest possible payoff.

GSLT connection. In Meredith's GSLT framework, the calculus is the rewrite system and the model is the denotational semantics. `CalculusSound` is the standard soundness theorem: rewrites agree with denotations.

Decentralized agents. If two agents both prove `CalculusSound` for their (possibly different) implementations, their evidence composes—not by policy, but by theorem. Trust becomes a mathematical property.

4.10 The Profile-Hom Equivalence

The additive interface has a clean universal property.

Theorem 4.8 (Profile-hom equivalence (CORE))**.** *An additive world model is exactly an additive monoid homomorphism from states to answer profiles:*

$$\texttt{AdditiveWorldModel}(\texttt{State}, \texttt{Query}, V) \;\simeq\; \mathrm{AddMonoidHom}(\texttt{State},\ \texttt{Query} \to V)$$

Both directions are definitional: the world-model interface IS the additive morphism, and vice versa, up to definitional equality in Lean—no propositional reasoning is needed.[12]

The terminal extensional world model. The answer-profile object $\mathrm{Prof}(\texttt{Query}, V) := \texttt{Query} \to V$ is itself an additive world model: extraction is function evaluation, so both laws hold by `rfl`. Every additive world model maps into it via extraction, and the map is unique up to the profile it determines. This makes $\mathrm{Prof}(\texttt{Query}, V)$ the *terminal* extensional world model.

In categorical terms, additive world models over `Query` form a slice category $\mathbf{AddCommMon}/\mathrm{Prof}(\texttt{Query}, V)$, and $\mathrm{Prof}(\texttt{Query}, V)$ is its terminal object. A morphism between additive world models is a commuting triangle:

[12]Formalized in `Mettapedia/Logic/WorldModelCore.lean`. The structural equivalence `equivGSLTProfileHom` has both roundtrips by `rfl`.

```
structure WorldModelHom (State1 State2 Query V) where
  hom  : State1 ->+ State2
  comm : forall W q, extract (hom W) q = extract W q
```

The canonical morphism to the terminal model *is* extraction.[13]

Why this matters. The universal property means that the *entire* additive world-model interface is one algebraic condition. The five rules are not independent axioms—they all derive from the single requirement that extraction is an `AddMonoidHom`. This does not trivialize the project: the hard work is in choosing expressive state spaces, proving no-go theorems, and certifying fast inference paths. But it makes the *interface* minimal and the *achievements* sharper.

For the implementer: you can swap state representations without changing the semantics. Any two additive world models producing the same answer profiles are extensionally equivalent.

For the general `WorldModel` without additivity laws, no analogous universal property is known. This is one reason the additive regime is the main exact specialization: it is the regime where the categorical picture is sharpest.

4.11 Naturality Across Signatures

When the query language changes—by adding atoms, extending the signature, or translating between formalisms—the extraction map must *commute* with the translation. For a signature morphism $\sigma : \mathrm{Sig}_1 \to \mathrm{Sig}_2$:

$$\mathrm{extract} \circ \mathrm{reindex}_\sigma \;=\; \mathrm{reindex}_\sigma \circ \mathrm{extract}$$

This is the *satisfaction condition* of an institution: truth (here: evidence extraction) is invariant under change of notation.[14] [15]

4.12 The Solomonoff–De Finetti Foundation

The previous sections built the algebraic and categorical interface. This section addresses the foundational question: *why* is revision-as-addition the

[13] Open program: is this slice category enriched in a useful way?

[14] Formalized as `satisfactionCondition_is_naturality` in `Mettapedia/Logic/PLNWorldModelInstitution.lean`.

[15] Open program: the GSLT universal theorem—every graph-structured lambda theory with parallel composition naturally induces a WorldModel. See Meredith [25].

right choice? The answer comes from exchangeability theory.

The νPLN framework of Nil Geisweiller [6] grounds the calculus in Solomonoff Universal Induction. The global probability space $(\Omega, \mathcal{F}, \nu)$ corresponds to a world-model state where Ω is the set of possible worlds. The universal distribution ν defines a prior over generators; Bayesian updating with observed evidence produces the posterior.

The crucial insight is that de Finetti's representation theorem [3] justifies evidence counts as *sufficient statistics*. For exchangeable binary sequences, the counts (n^+, n^-) contain all the information needed for Bayesian updating. Under a Beta prior, the posterior is $\text{Beta}(\alpha + n^+, \beta + n^-)$.

For the implementer: evidence counts are not an arbitrary design choice. De Finetti says they are the mathematically forced representation for exchangeable data.

Giry monad and Markov kernels. The categorical underpinning is the Giry monad: the functor sending a measurable space to its space of probability measures. The formalization proves that Kyburg's flattening *is* the Giry bind, with all three monad laws. The complete justification chain:

$$\text{exchangeability} \;\to\; \text{de Finetti mixture} \;\to\; \text{Kyburg flattening} \;\to\; \text{Giry bind} \;\to\; \text{evidence sufficiency}$$

Each arrow is proved. Together they show that revision-as-addition is not an arbitrary choice but a categorical necessity under exchangeability.[16][17]

4.12.1 Markov Transition World Models

The Markov de Finetti formalization provides a measure-theoretic foundation. The WM-PLN bridge exposes three layers:

Layer 1: Single-step transition queries. The `markovDirichletQueryCarrier` (`PLNXiCarrierScreening.lean:483`) extracts row-conditioned categorical evidence from transition multisets. Theorem: `markovTransitionAtom_semE_eq_rowEvidence`.

[16]See `Mettapedia/ProbabilityTheory/GiryMonad.lean` and `Mettapedia/ProbabilityTheory/DeFinettiConnection.lean`. **Markov extension (April 2026):** The Diaconis–Freedman (1980) theorem—extending de Finetti from i.i.d. to Markov-exchangeable sequences—is now fully formalized in `Mettapedia/Logic/MarkovDeFinetti*.lean` (28 files, ∼23,900 lines, 0 sorry). Under Markov exchangeability, the transition-count matrix $N(i, j)$ is the sufficient statistic, extending the chain: Markov exchangeability → Markov mixture → Dirichlet-per-row conjugacy → transition-count sufficiency.

[17]Open program: the terminal `WorldModel (Measure Ω) Query` instance remains future work.

Layer 2: Multi-step predictive chaining. The `markovWMPosteriorChain` (`PLNMarkovPredictiveChaining.lean:32`) computes n-step arrival probabilities via Chapman-Kolmogorov summation.

Layer 3: Path atom surface. The `markov-transition*` atom family (`PLNMarkovPathXi.lean:43`) provides direct OSLF semantics for multi-step path queries with concrete binary examples (`bit001`, `bit011`).

Key negative result. PLN's additive deduction screening is *unfillable* for Markov carriers. Transition composition requires matrix multiplication, not evidence addition. This validates the design choice of separate predictive surfaces over retrofitting additive carriers.

4.13 Four Verbs: Revise, Query, Forget, Explain

The world-model interface supports four operations:

Revise (kernel): combine two states ($W_1 + W_2$ in the additive regime; general merge otherwise).

Query (kernel): extract a value from a state for a given question.

Forget (state management): remove evidence under explicit scope constraints, preserving answers outside the scope (Chapter 6).

Explain (state management): trace which observations support an answer, via provenance-enriched evidence (Chapter 6).

Revision and querying are the minimal interface; forgetting and explaining show *why* the kernel stores states rather than bare values. A bare value can be queried but cannot be partially retracted or audited.

The 2×2 calculus square of §4.6 is the abstract face of a full natively typed presentation: the five algebraic rules encode as typed rewrite rules over a 13-axis vertex family, developed in Chapter 8. Native types do not introduce a second semantics—they expose, at the rewrite level, the structural behavior already present in the semantic kernel.

4.14 A Worked Instance: Dirichlet over Worlds

For finite propositional scope (atoms indexed by $\mathrm{Fin}(n)$), the additive world model can be instantiated concretely. The state is a *world table*:

$$\mathrm{JointEvidence}(n) := \mathrm{Fin}(2^n) \to \mathbb{R}^{\infty}_{\geq 0}$$

interpreted as Dirichlet pseudo-counts over the 2^n possible worlds. Each coordinate records how much evidence supports that world. Extraction sums counts over worlds compatible with each query: for $\mathrm{prop}(A)$, sum where A is true vs. false; for $\mathrm{link}(A, B)$, sum where $A \wedge B$ vs. $A \wedge \neg B$. Revision is pointwise addition of pseudo-counts.[18]

This instance satisfies all the laws of `AdditiveWorldModel`: extraction commutes with revision, and the empty world table (all zeros) extracts zero for every query. It is the *most informative* additive world model for finite propositional scope — every other model is a projection of it. The no-go theorem (Chapter 9) shows that this projection is genuinely lossy: per-link projections cannot reconstruct the joint.

[18] Formalized in `Mettapedia/Logic/PLNJointEvidence.lean`.

Chapter 5

Evidence Carriers and Revision

What kind of object does extraction return? The previous chapter left V generic. This chapter makes it concrete: V is a family of *sufficient-statistic carriers*, each storing what is needed for further Bayesian updating in a particular domain. The algebraic discipline—revision as addition, information ordering as componentwise comparison—is the same for every carrier. Only the domain changes.

5.1 The Generic Evidence Pattern

Every carrier in the additive regime satisfies two generic laws:

1. **Revision is addition.** Combining two bodies of evidence is commutative, associative, with a zero (no evidence). Algebraically: V is an `AddCommMonoid`.

2. **Information is ordered.** A partial order where "more data" means "at least as informative."

Two additional algebraic structures enrich the carrier family: a *quantale tensor* for composing evidence along inference chains, and a *Heyting algebra* (complete lattice with Frame structure) supporting intuitionistic implication and complement. These were originally developed for binary evidence (§5.5–§5.6) and have now been extended to the full k-ary Dirichlet family:

coordinatewise multiplication gives a commutative monoid tensor, and the pointwise lattice gives a complete Heyting algebra for all k.[1]

Every exponential family with sufficient statistics instantiates this pattern:

Domain	**Sufficient statistics**
Binary	(n^+, n^-)
k-valued	$(\alpha_1, \ldots, \alpha_k)$
Continuous	$(n, \sum x_i, \sum x_i^2)$
Count data	$(n, \sum x_i)$
Covariance	$(n, \sum x_i x_i^\top)$

The first three are formalized in this book (cf. the carrier table in Chapter 4). The rest are natural future instances requiring only new `AddCommMonoid` implementations—no changes to the calculus itself.

5.2 The Three Formalized Carriers

Binary (Beta-Binomial). The simplest carrier handles yes/no queries. $\mathsf{BinaryEvidence} := (n^+, n^-) \in \mathbb{R}^{\infty}_{\geq 0} \times \mathbb{R}^{\infty}_{\geq 0}$, where n^+ counts positive and n^- counts negative observations. Under a Beta prior $\mathrm{Beta}(\alpha, \beta)$, the posterior is $\mathrm{Beta}(\alpha + n^+, \beta + n^-)$. The Laplace-smoothed strength is $s = (1 + n^+)/(2 + n)$.[2]

Dirichlet-Multinomial (k-valued). For queries with k outcomes—"Is the elevator normal, slow, or faulty?"—evidence is a count vector $(\alpha_1, \ldots, \alpha_k) \in \mathbb{R}^k_{\geq 0}$. The posterior mean for category i is $\alpha_i / \sum_j \alpha_j$; confidence is $(\sum \alpha_j)/(\sum \alpha_j + k)$.[3]

Normal-Gamma (continuous). For real-valued observations such as humidity readings, the sufficient statistics are $(n, \sum x_i, \sum x_i^2)$. The prior

[1] Formalized in `Mettapedia/Logic/EvidenceDirichletQuantale.lean` (13 theorems, 0 sorry). Normal-Gamma carriers have a formalized information partial order (`Mettapedia/Logic/EvidenceNormalGammaLattice.lean`, 10 theorems, 0 sorry) but intentionally no quantale tensor — sufficient statistics for unknown mean and variance are not counts.

[2] Formalized in `Mettapedia/Logic/BinaryEvidence.lean` and `Mettapedia/Logic/EvidenceBeta.lean`.

[3] Formalized in `Mettapedia/Logic/EvidenceDirichlet.lean`.

is Normal-Gamma$(\mu_0, \kappa_0, \alpha_0, \beta_0)$; the posterior parameters update as $\kappa_n = \kappa_0 + n$, $\mu_n = (\kappa_0\mu_0 + \sum x_i)/\kappa_n$, $\alpha_n = \alpha_0 + n/2$.[4]

In each case, revision is componentwise addition of sufficient statistics. The algebra does not change across carriers; only the domain does.

Theorem 5.1 (Sleep consolidation (Core))**.** *For each of the three primary carriers:* $\text{posterior}(\pi, e_1 \oplus e_2) = \text{posterior}(\text{posterior}(\pi, e_1), e_2)$. *Batch replay of deferred observations yields the same posterior as sequential updating. This guarantee is exercised on real data in Chapter 21: gas sensor drift (§21.2), Kalman temporal filtering, and GJP forecast aggregation (§21.5).*

Maple Court: Maple Court instantiates all three: Beta for leak, occupancy, and shower sensors; Normal-Gamma for humidity readings; Dirichlet for the three-valued *LaundryState* and *ElevatorHealth*. Querying elevator health with counts $(40, 8, 2)$ yields posterior means $(0.80, 0.16, 0.04)$ with confidence 0.83. Revising with 10 more "normal" observations shifts to $(50, 8, 2)$ and confidence 0.86. Sleep consolidation means the apartment can buffer overnight readings and batch-update at dawn with identical results.

---

[5]

5.3 Revision as Addition

Why is revision componentwise addition? The foundational justification via de Finetti's exchangeability theorem was given in Chapter 4, §"The Solomonoff–De Finetti Foundation." Here we show the concrete mechanics: adding counts corresponds to Bayesian conjugate updating:

$$\text{Beta}(\alpha + n_1^+, \beta + n_1^-) \xrightarrow{+(n_2^+, n_2^-)} \text{Beta}(\alpha + n_1^+ + n_2^+, \beta + n_1^- + n_2^-)$$

[4] Formalized in `Mettapedia/Logic/EvidenceNormalGamma.lean`.

[5] The Maple Court example enjoys three-way conformance: (1) 30 PeTTa runtime tests across four `.metta` files, all passing; (2) 34 algebraic theorems in `Mettapedia/Logic/PLNMapleCourtDemo.lean`; and (3) 4 unconditional kernel-checked conformance theorems via a specialized reflection checker (the Ramsey36 pattern: write a simple pattern match the kernel can reduce, then close with `decide`). See `Mettapedia/Conformance/MapleCourtWMConformance.lean`.

Revision is addition because it is the algebraic shadow of conjugate updating. The same pattern extends to all three primary carriers:

$$\begin{aligned} \text{Dirichlet:} \quad & (\alpha_1, \ldots, \alpha_k) \oplus (\beta_1, \ldots, \beta_k) = (\alpha_1 + \beta_1, \ldots, \alpha_k + \beta_k) \\ \text{Normal-Gamma:} \quad & (n_1, s_1, q_1) \oplus (n_2, s_2, q_2) = (n_1 + n_2,\ s_1 + s_2,\ q_1 + q_2) \end{aligned}$$

In every case, revision is componentwise addition of the sufficient statistics. The algebra does not change; only the domain does.

5.4 Information Ordering

Evidence values are partially ordered by information content. For binary evidence:

$$(n_1^+, n_1^-) \le (n_2^+, n_2^-) \iff n_1^+ \le n_2^+ \ \wedge\ n_1^- \le n_2^-$$

This extends componentwise to Dirichlet vectors and Normal-Gamma statistics. The ordering makes each carrier a lattice (meet = componentwise min, join = componentwise max).

For the Normal-Gamma carrier, the information order has additional structure: zero (the empty-evidence state) is bottom, and aggregation is monotone—adding evidence never decreases information. Unlike the counting carriers (binary, Dirichlet), the Normal-Gamma carrier does not admit a quantale tensor: sufficient statistics for unknown mean and variance are not counts, and coordinatewise multiplication does not preserve the conjugate structure. This asymmetry is by design—the information order and additive revision are the generic properties; the tensor is carrier-specific.[6]

The key property: this is a *partial* order. $(3, 0)$ and $(0, 3)$ are incomparable—they contain the same total but differ in kind. No single scalar captures "how much evidence" there is.

5.5 Compositional Tensor for Binary Evidence

When you know $A \Rightarrow B$ with evidence e_{AB} and $B \Rightarrow C$ with evidence e_{BC}, what can you say about $A \Rightarrow C$? For binary evidence, the quantale tensor $\otimes$ composes evidence along inference chains:

$$(a^+, a^-) \otimes (b^+, b^-) = (a^+ \cdot b^+,\ a^+ \cdot b^- + a^- \cdot b^+)$$

[6] Formalized in `Mettapedia/Logic/EvidenceNormalGammaLattice.lean` (10 theorems, 0 sorry).

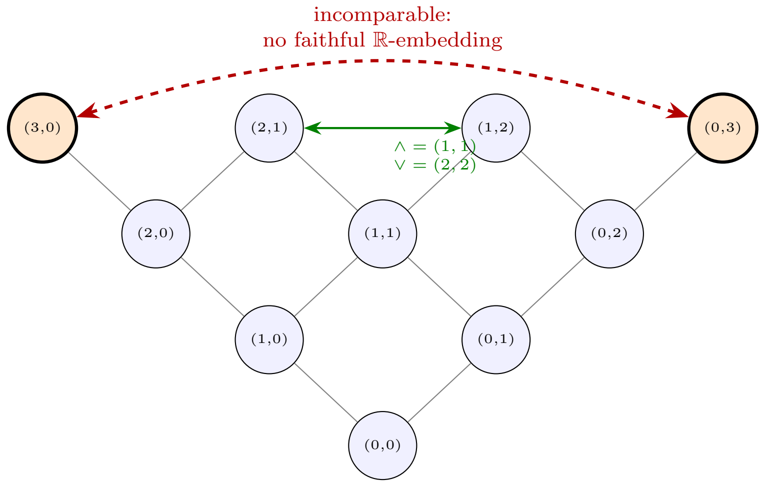

Figure 5.1: Hasse diagram of the binary evidence lattice. Orange nodes $(3,0)$ and $(0,3)$ are incomparable (the "totality gate").

with unit $(1,0)$. This is a complete lattice with $\otimes$ distributing over arbitrary joins (a quantale).

Theorem 5.2 (Quantale transitivity (Core))**.** *$e_{AB} \otimes e_{BC} \le e_{AC}$ in the information ordering. Equality holds under screening-off.*[7]

The tensor gives a sound lower bound for chained inference. Exact equality requires the screening-off side condition (Chapter 11).[8]

5.6 Logical Operations for Binary Evidence

Binary evidence forms a Heyting algebra:

Definition 5.3 (Heyting implication (Core))**.**

$$(a^+, a^-) \Rightarrow (b^+, b^-) = \left(\begin{cases} \top & \text{if } a^+ \le b^+ \\ b^+ & \text{otherwise} \end{cases}, \begin{cases} \top & \text{if } a^- \le b^- \\ b^- & \text{otherwise} \end{cases} \right)$$

satisfying $a \le (b \Rightarrow c) \iff a \wedge b \le c$.

[7] Formalized in `Mettapedia/Logic/EvidenceQuantale.lean`.

[8] Open program: a generic quantale for Dirichlet or Normal-Gamma evidence would require defining a compositional tensor on count vectors.

Excluded middle fails: $a \vee \neg a \neq \top$, because absence of evidence is not evidence of absence. The "totality gate": $(3, 0)$ and $(0, 3)$ are incomparable, so no faithful $\mathbb{R}$-embedding exists.[9][10]

5.7 Theoretical Foundations

Four deeper connections ground the evidence algebra.

The De Finetti bridge. For an infinite exchangeable binary sequence, de Finetti's representation theorem [3] asserts the existence of a latent parameter θ such that, given θ, the observations are i.i.d. Bernoulli(θ). The joint distribution is a mixture:

$$P(X_1 = x_1, \ldots, X_n = x_n) = \int_0^1 \theta^{n^+}(1-\theta)^{n^-}\, d\mu(\theta)$$

where $n^+ = \sum x_i$ and $n^- = n - n^+$. The pair (n^+, n^-) is the sufficient statistic of this mixture: no other function of the data adds information about θ. This is why revision is addition: adding counts IS the algebraically forced update under exchangeability. The proof extends to Dirichlet-Multinomial and Normal-Gamma via the same mixture logic.[11]

Walley predictive intervals. The Imprecise Dirichlet Model [36] gives prior-free interval bounds on predictive probabilities. Where Bayesian credible intervals depend on the choice of prior, Walley's intervals are robust at the cost of wider bounds. The calculus supports both: the carrier stores the counts, and the interval semantics is a choice of view (Chapter 7).

Knuth–Skilling foundations. Knuth and Skilling [20] derive probability as the unique consistent valuation on a lattice of propositions, generalizing Cox's theorem to distributive lattices. At the world-model layer, probabilities arise as valuations on Boolean subsets of worlds (the Kolmogorov story). At the evidence layer, the rules weaken from Boolean equalities to

[9] Formalized in `Mettapedia/Logic/HeytingValuationOnEvidence.lean`.

[10] The Heyting structure has been extended to k-ary Dirichlet evidence: `Mettapedia/Logic/EvidenceDirichletQuantale.lean` (13 theorems, 0 sorry). The Boolean-to-Heyting categorical bridge is in `Mettapedia/Logic/BooleanHeytingBridge.lean` (4 theorems).

[11] Formalized in `Mettapedia/ProbabilityTheory/DeFinettiConnection.lean`

Heyting inequalities. The bridge enables factor-graph variable elimination over Knuth–Skilling-valued factors.[12]

The bivariate structure. The world-model calculus has *state-side* additivity: $\mathrm{extract}(W_1 + W_2, q) = \mathrm{extract}(W_1, q) + \mathrm{extract}(W_2, q)$. Knuth–Skilling adds *obs-side* additivity: for disjoint queries q_1, q_2, $\mathrm{extract}(W, q_1 \vee q_2) = \mathrm{extract}(W, q_1) + \mathrm{extract}(W, q_2)$ (where $\vee$ is the join in a lattice of queries). These are two laws on different arguments of the same bivariate extractor. A unification with laws on both arguments—a *stateful valuation*—is one of the most promising open directions (see also §8.8).[13]

The KS-Evidence-Measure triangle. The evidence carriers, the KS representation theorem, and the measure-valued terminal world model form a commuting triangle:

$$\text{KS ordered semigroup} \xrightarrow{\Theta=\text{total}} (\mathbb{R}^{\infty}_{\geq 0}, +) \xrightarrow{\bullet\,\text{dirac}} \texttt{Measure}$$

The evidence carriers (binary, Dirichlet) satisfy the KS axioms (ordered, associative, monotone). The representation theorem maps them into $\mathbb{R}^{\infty}_{\geq 0}$ via the total-evidence function. The measure bridge (`evidenceToMeasure`) composes this with Dirac weighting to produce a formal measure. Additivity holds at every vertex: $\oplus$ on evidence, $+$ on reals, $+$ on measures. The triangle says the WM evidence algebra is not an ad-hoc design—it is the *unique* algebra satisfying the KS axioms, and the measure theory follows inevitably.[14]

5.8 What Evidence Algebra Is Not

Evidence algebra is *not* the complete probabilistic semantics. It is a compositional bookkeeping layer. The world-model state carries strictly more

[12] Formalized in `Mettapedia/ProbabilityTheory/KnuthSkilling/`.

[13] Open program: a `StatefulValuation` typeclass with both state-side and obs-side laws. The probability hypercube [30] classifies theories by which combination of these laws they satisfy.

[14] The KS formalization spans 126K lines with 1,923 theorems (0 sorry for the core representation; 11 sorry in experimental Shore-Johnson extensions). The triangle bridge is in `Mettapedia/Logic/KSEvidenceMeasureBridge.lean`. The σ-additivity boundary (`Mettapedia/ProbabilityTheory/KnuthSkilling/Counterexamples/SigmaAdditivityNecessity.lean`) proves that finitely additive measures do not automatically extend to σ-additive ones—Step C of the measure frontier requires Scott continuity, which is a separate axiom.

information than the extracted per-query evidence values. The no-go theorem (Chapter 9) makes this precise: per-link evidence cannot reconstruct the joint state.

Evidence tells you the score. It does not tell you the game. The correlations, the causal structure, the joint distribution over worlds—those live in the state, not in the per-query counts. This is why the calculus stores states and extracts evidence, rather than storing evidence directly.

Raw data vs. sufficient statistics. A system may also store the raw observations themselves—a *data-backed state*—and compress to sufficient statistics on demand. The compositionality theorem `compress_append` proves that compression is a monoid homomorphism: compressing two raw-data batches separately and adding the results gives the same sufficient statistics as compressing the concatenated batch.[15] This means the system can keep raw data for model switching or distributional queries while using compressed statistics for efficient $O(1)$ revision—the two representations are provably compatible.

Evidence algebra is not quantale algebra. The binary evidence carrier forms a quantale (a complete lattice with an associative tensor), but the PLN operations on that carrier are *not* the quantale operations. Two concrete divergences are kernel-checked:

1. **Revision ≠ supremum.** Evidence revision ($\oplus$) adds counts: $(2,1) \oplus (1,2) = (3,3)$. The lattice supremum takes componentwise max: $(2,1) \sqcup (1,2) = (2,2)$. These are different—revision is conjugate update, supremum is information combination.

2. **Tensor strength ≠ strength product.** The quantale tensor $(3,1) \otimes (2,2) = (6,2)$ has strength $6/8 = 0.75$. The product of strengths is $0.75 \times 0.5 = 0.375$. Tensor on evidence does not distribute over the strength view.

The positive characterization: when one evidence state dominates the other componentwise, the supremum equals the dominating state. The quantale provides the algebraic skeleton; PLN fills in the probabilistic flesh.[16]

[15]Formalized in `Mettapedia/Logic/EvidentialLedger.lean`.

[16]Formalized in `Mettapedia/Logic/PLNQuantaleDivergence.lean` (3 theorems, 0 sorry).

Chapter 6

State Operations

If the world model were just a truth value, there would be nothing to retract, nothing to audit, nothing to explain. A number has no parts. But a state has structure: it records which observations contributed to which queries, and that structure supports operations that bare values cannot.

This chapter develops the third and fourth verbs of the calculus: *forget* (partially retract evidence under scope constraints) and *explain* (trace which observations support an answer).

6.1 Why Forgetting Matters

A sensor reading from last Tuesday should eventually expire. A privacy constraint says apartment-level data must not persist in the building model beyond 24 hours. A correction says "the batch of observations from sensor 3 was miscalibrated; retract them." All of these are forgetting operations.

Forgetting is not deletion. It is a *scoped retraction*: remove evidence within a declared scope while provably preserving everything outside it. This requires the state to carry enough structure to distinguish what came from where—which is precisely what provenance provides.

6.2 The Forgetting Layer

A forgetting layer over a world model has four components:

```
structure ForgettingLayer (State Scope Query V)
    [WorldModel State Query V] where
  inScope    : Scope -> Query -> Prop
  forget     : Scope -> State -> State
```

```
idempotent : forall S W, forget S (forget S W) = forget S W
outsideInvariant :
  forall S W q, not (inScope S q) ->
    WorldModel.extract (forget S W) q = WorldModel.extract W
        q
```

The key law is `outsideInvariant`: forgetting scope S preserves the extracted value for every query outside S. In the answer-profile language: forgetting changes the profile only within the declared scope. Idempotence says forgetting the same scope twice is the same as forgetting it once.[1]

Maple Court: The apartment forgets stale humidity readings after 6 hours: the scope is "humidity observations older than $t-6$h." Outside that scope—motion detections, leak alarms—the profile is unchanged. The building forgets laundry observations after 24 hours. Each layer's forgetting is independent and scope-delimited.

6.3 Conservation Theorems

Under *exact inverse forgetting*—where forgetting scope S exactly undoes a revision Δ that was entirely within S—a family of conservation laws holds.

Theorem 6.1 (Evidence conservation (Core))**.** *If Δ is a revision whose evidence lies entirely within scope S (i.e., $\forall W,\ \mathrm{forget}(S, W+\Delta) = W$), then for every query q outside S:*

1. ***Anti-hallucination:*** $\mathrm{extract}(\Delta, q) = 0$. *The revision contributes zero evidence outside its scope.*
2. ***Noether conservation:*** $\mathrm{extract}(W + \Delta, q) = \mathrm{extract}(W, q)$. *The revised state agrees with the base state outside the scope.*
3. ***Zero leakage:*** *the observation count of Δ at q is zero.*

Theorem 6.2 (Exact-inverse no-go (Boundary))**.** *If a revision Δ has nonzero evidence at any query outside scope S, then no forgetting operation can exactly undo Δ.*

[1] Formalized in `Mettapedia/Logic/GenericWorldModelForgetting.lean`.

This is the precise boundary: forgetting cannot undo what leaked. The conservation theorems and the no-go are packaged as a single `EvidenceConservationPack`.[2]

These results are the *state-level, scoped-forgetting* face of a broader family of evidence-conservation principles. Goertzel [7] develops the complementary *inference-path* face: a discrete quantale Noether theorem showing that reinforcement is constant along geodesic inference paths, a hallucination bound showing that conclusion strength is bounded by initial evidence mass, and an evidence monotonicity theorem (a quantale data-processing inequality). Together, the two perspectives establish that evidence can be neither fabricated nor overcounted—whether the viewpoint is scoped retraction of states or sequential chaining of inference steps.[3]

Forgetting in the non-additive perimeter. The conservation and exact-inverse theorems above assume additive revision. In the overlap regime (§4.3), exact forgetting requires stronger conditions: the `SupportTrackedForgettingLayer` proves that exact inverse is possible *only* when the revision evidence vanishes outside the forgotten scope. The semitopology bridge provides the recovery theorem (Theorem 4.4): when the overlap footprint is non-actionable, forgetting the overlap recovers additivity on the remainder—and the conservation laws apply to the remainder.[4]

6.4 Provenance and Explanation

Evidence tells you *how much* support a query has. Provenance tells you *where that support came from.*

$$\texttt{State} \xrightarrow{\text{extract-with-prov}} (\texttt{Query} \to \text{Ev-with-Prov}) \xrightarrow{\text{erase prov.}} (\texttt{Query} \to V)$$

The first arrow produces evidence tagged with source identifiers (which sensor, which batch, which time window). The second erases the tags, recovering ordinary evidence. The composition equals ordinary extraction—provenance is a refinement, not a replacement. This is the same pattern as

[2]Formalized in `Mettapedia/Logic/PLNWorldModelConservationPack.lean` .

[3]The conservation principle extends beyond the commutative setting. Goertzel [9] argues that evidence conservation under non-commutativity forces the evidence lattice to be a Hilbert-space lattice, recovering the mathematical structure of quantum mechanics from epistemological first principles. This connects to the non-commutative vertex of the probability hypercube (Chapter 19).

[4]Formalized in `Mettapedia/Logic/GenericWorldModelForgetting.lean` (193 lines, 6 theorems) and `Mettapedia/Logic/PLNWorldModelSupportForgetting.lean` (75 lines, 3 theorems). Both 0 sorry.

type erasure in programming languages or audit trails in accounting: the enriched representation supports richer operations (here, targeted forgetting), and the surface representation is recovered by forgetting the enrichment.

The concrete realization is `TrackedWhichState`: a state that carries, for each query, a multiset of provenance chunks. Each chunk records a source label and an evidence contribution. The total evidence is the sum over chunks; the provenance is the set of source labels.[5][6]

Provenance is not metadata. It is part of the structure of stateful reasoning. Without it, forgetting cannot be targeted: you would have to discard everything or nothing. With it, retraction is surgical.

Maple Court: The building's hallway-humidity evidence carries provenance. Sensor A reports 3 readings above the threshold (morning); sensor B reports 2 below (afternoon). The combined evidence is $(3,0) \oplus (0,2) = (3,2)$, with strength $3/5 = 0.6$ and provenance $\{\mathrm{A}, \mathrm{B}\}$. When sensor A is found to be miscalibrated, the steward forgets scope $\{\mathrm{A}\}$: the evidence drops to $(0,2)$ (sensor B alone), strength falls to $0/2 = 0$ (no positive readings remain), and provenance narrows to $\{\mathrm{B}\}$. Sensor B's contribution is provably conserved.

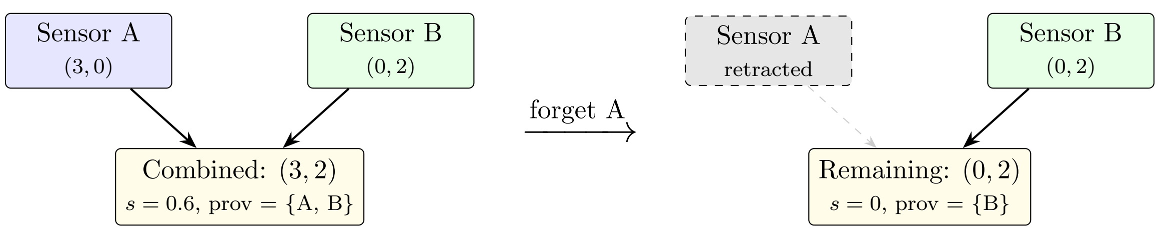

Figure 6.1: Provenance-tracked forgetting in Maple Court. Sensor A's contribution $(3,0)$ is retracted; sensor B's $(0,2)$ is conserved.

[5] Formalized in `Mettapedia/Logic/PLNProvenanceTrackedState.lean`.

[6] The underlying provenance-semiring framework [12] supports different semiring choices for different provenance granularities. The `Which` semiring tracks which observations contributed; a `KRelation` valued in `Which` is itself an `AdditiveWorldModel` instance (the bridge is proved). A polynomial semiring would additionally track full derivation trees. An end-to-end example exercising the LP provenance operator `T_P_K_LP_seeded` through WM extraction and forgetting appears in §6.5.2.

6.5 PLN Evidence Stamps as Provenance

PLN v0.9[7] tracks provenance via *stamps*: each sentence carries a sorted list of observation IDs, and before combining two sentences, the engine checks `StampDisjoint`—whether their stamps share any IDs. If they do, the revision is blocked to prevent double-counting.

This is exactly the formal provenance discipline. The correspondence:

PLN v0.9	WM calculus
`StampDisjoint`	`Finset.Disjoint` on scope supports
`StampConcat`	`Finset.union`
Forget a source	`forgetScopedByScope`

PLN v0.9 checks stamps at runtime; the WM calculus proves disjointness at compile time via `decide`. PLN v0.9 has no forgetting; the WM calculus adds it with proved conservation.[8]

6.5.1 Stepped Walkthrough: DeductionRevision with Provenance

We step through the PLN v0.9 DeductionRevision example in full detail.

Step 1: The knowledge base. Four observations, encoded as a PeTTa atomspace:

```
def drKB : PeTTaSpace where
  facts := [inh "A" "B",    -- obs 0
            inh "A" "C",    -- obs 1
            inh "B" "D",    -- obs 2
            inh "C" "D"]    -- obs 3
  rules := []
```

Step 2: Stamp disjointness (kernel-checked).

```
def dr_stamp1 : Finset (Fin 4) := {0, 2}  -- Path 1: A->B->D
def dr_stamp2 : Finset (Fin 4) := {1, 3}  -- Path 2: A->C->D

theorem dr_stamps_disjoint :
```

[7]PLN v0.9: `github.com/trueagi-io/PLN`, tag `v0.9`. 430 lines of MeTTa implementing forward chaining with stamp-based evidence tracking.

[8]Formalized in `Mettapedia/Logic/PLNProvenanceInference.lean` (general theory) and `Mettapedia/Logic/PLNClassicExamples.lean` (20 conformance checks).

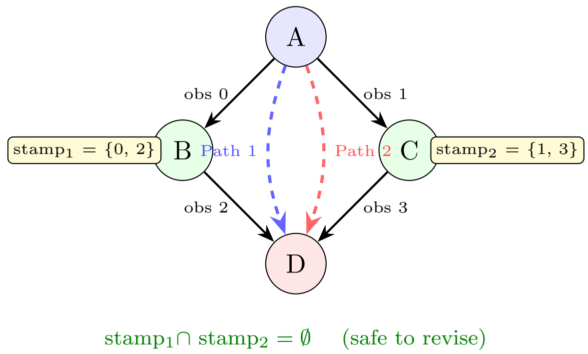

Figure 6.2: DeductionRevision: two paths with disjoint provenance.

```
    Disjoint dr_stamp1 dr_stamp2 := by decide

theorem dr_stamps_union :
    Finset.union dr_stamp1 dr_stamp2 = {0, 1, 2, 3} := by
        decide
```

The `decide` tactic delegates to the kernel: Lean verifies set-disjointness at type-checking time. This is the compile-time version of PLN v0.9's runtime `StampDisjoint`.

Step 3: Forgetting with conservation.

```
-- Forgetting obs 0 does NOT affect Path 2
theorem dr_obs0_disjoint_path2 :
    Disjoint {0} dr_stamp2 := by decide

-- Path 1 retains only obs 2
theorem dr_path1_after_forget :
    dr_stamp1 \ {0} = {2} := by decide
```

Forgetting observation 0 (A→B) removes it from Path 1's stamp but *provably* leaves Path 2 intact. This is a concrete instance of the conservation theorem: evidence outside the forgotten scope is preserved.

6.5.2 Worked Example: Derivation Tracking with LP Provenance

The DeductionRevision walkthrough above tracks provenance at the stamp level—which observation IDs were consumed. We now show a deeper integration: provenance that flows through *logic-program rules*, is carried by a semiring-valued K-relation, participates in WM extraction, and supports

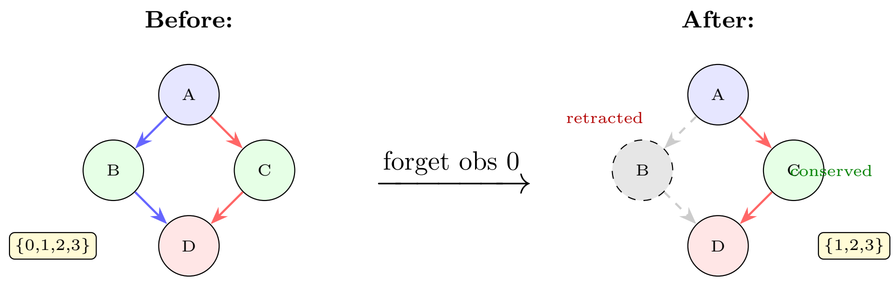

Figure 6.3: Forgetting obs 0 (A→B). Path 1 loses its first link; Path 2 is provably unaffected. Stamp: $\{0,1,2,3\} \to \{1,2,3\}$.

scoped forgetting with proved conservation. Every equation below is kernel-checked (28 theorems, 0 sorry).[9]

The scenario. Apartment 3B has two independent sensors: a pipe-leak detector (observation o_1) and a shower-running detector (o_2). Three LP rules propagate evidence through the building's causal model:

```
def mapleCourtProg : Program mapleCourtSig :=
  [rulePipeLeakToWallHumidity,         -- WallHumidity   <-
      PipeLeak
   ruleShowerToBathroomHumidity,      -- BathroomHumid. <-
      ShowerRunning
   ruleWallToMoldRisk]                -- MoldRisk       <-
      WallHumidity
```

Seeded provenance. Observations enter as labeled seeds in the `Which` semiring, which tracks *which* observations contributed to each derived fact. The seed for the pipe observation maps PipeLeak to $\{o_1\}$ and everything else to :

```
def pipeSeed : KRelation mapleCourtSig (Which (Fin 2)) :=
  seedOf pipeLeak(*$_1$*) srcPipe    -- pipeLeak(*$_1$*) |->
      {(*$o_1$*)}, else |-> (*$\varnothing$*)
```

The seeded provenance operator `T_P_K_LP_seeded` propagates labels through rules. For wall humidity:

```
theorem T_P_K_LP_seeded_wallHumidity (seed I) :
  T_P_K_LP_seeded ... seed I wallHumidity(*$_1$*) =
    seed wallHumidity(*$_1$*) + I pipeLeak(*$_1$*)
```

[9]Full source: `Mettapedia/Logic/WMDerivationTrackingDemo.lean`.

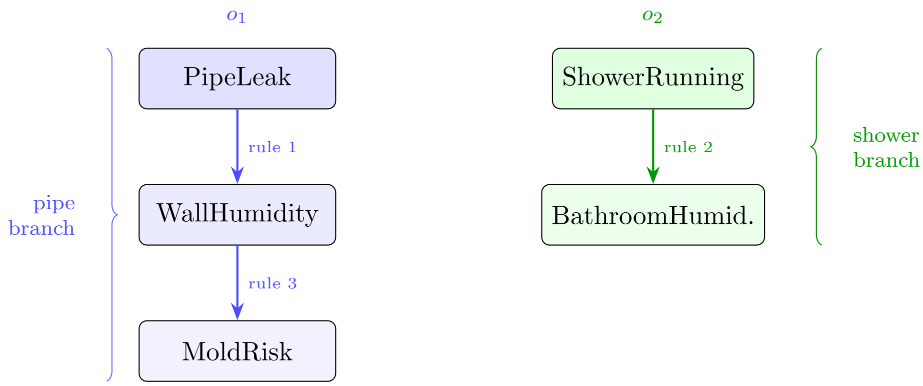

Figure 6.4: Maple Court derivation graph. Observation o_1 feeds the pipe branch (blue); observation o_2 feeds the shower branch (green). The two branches share no observations.

Read this as: the wall-humidity evidence at iteration I is whatever was seeded directly, *plus* whatever pipe-leak evidence was available at the previous iteration—propagated through rule 1. The labels are carried by the semiring product; no explicit label-passing code is needed.

Two-round closure. Two applications of `T_P_K_LP_seeded` suffice for the longest derivation chain (PipeLeak → WallHumidity → MoldRisk). The key results:

```
theorem pipeClosure_moldRisk :
  pipeClosure moldRisk(*$_1$*) = pipeProv            -- {(*
      $o_1$*)}
theorem pipeClosure_bathroomHumidity_zero :
  pipeClosure bathroomHumidity(*$_1$*) = 0           -- (*$\
      varnothing$*)
theorem showerClosure_bathroomHumidity :
  showerClosure bathroomHumidity(*$_1$*) = showerProv  -- {(*
      $o_2$*)}
theorem showerClosure_moldRisk_zero :
  showerClosure moldRisk(*$_1$*) = 0                 -- (*$\
      varnothing$*)
```

The pipe observation o_1 flows through WallHumidity to MoldRisk but *not* to BathroomHumidity. The shower observation o_2 flows to BathroomHumidity but *not* to MoldRisk. The `Which` semiring tracks this automatically: no branch-awareness is coded by hand.

WM extraction. The combined state `fullState` = `pipeClosure`+`showerClosure` is a `KRelation` valued in `Which`—and therefore an `AdditiveWorldModel` instance. Extraction is function application:

```
theorem extract_fullState_moldRisk :
  AdditiveWorldModel.extract fullState moldRisk(*$_1$*)
    = pipeProv                                    -- {(*$o_1$
        *)}
theorem extract_fullState_bathroomHumidity :
  AdditiveWorldModel.extract fullState bathroomHumidity(*$_1$*)
    = showerProv                                  -- {(*$o_2$
        *)}
```

The provenance labels survive extraction: the WM interface does not erase them. This is the key design point—provenance is part of the evidence algebra, not a separate metadata channel.

Scoped forgetting with conservation. If the pipe sensor is later found to be miscalibrated, the steward forgets scope $\{o_1\}$:

```
theorem forget_pipe_scope_moldRisk_zero :
  scopedTrackedEvidence
    (forgetScopedByScope scopePipe scopedFull) moldRisk(*$_1$*)
         = 0
theorem forget_pipe_scope_bathroomHumidity :
  scopedTrackedEvidence
    (forgetScopedByScope scopePipe scopedFull) bathroomHumidity
        (*$_1$*)
    = showerProv                                  -- {(*$o_2$
        *)}
```

MoldRisk drops to zero—its only source was the pipe sensor. BathroomHumidity is provably conserved—it came entirely from the shower sensor, which is outside the forgotten scope.

The conservation theorem closes the loop:

```
theorem forget_after_remember_pipe_is_id :
  forgetScopedByScope scopePipe
    (scopedRemember sPipe pipeClosure scopedShower)
    = scopedShower
```

Forgetting the pipe scope after re-remembering it returns exactly the shower-only state: forget $\circ$ remember = id on the complement scope. This is the Noether-style conservation law from §6.3, instantiated end-to-end on a concrete LP derivation.[10]

[10] The symmetric case (forgetting the shower scope, conserving the pipe branch) is also proved: `forget_shower_scope` and `forget_after_remember_shower_is_id` in the same file.

Maple Court: This example exercises the full provenance pipeline: labeled observation → LP rule propagation → semiring-valued closure → WM extraction → scoped forgetting → conservation. Every step is kernel-checked. The building steward can retract the pipe sensor's contribution to the world model and know—not hope, but *prove*—that the shower sensor's contribution survives intact. This is what D6 (Retractability) and D1 (State Primacy) buy you when they are taken seriously.

6.5.3 Case Study: Evidence-Driven Ontology Repair

The derivation tracking demo above tracks sensor evidence through LP rules. We now apply the same evidence algebra to a different domain: ontology engineering. The task is to decide how to reclassify a concept in a large ontology, using evidence from multiple authoritative sources, with kernel-checked ranking and sensitivity analysis.[11]

The problem. SUMO (Suggested Upper Merged Ontology) classifies Pain as a `PathologicProcess`—a biological event on the Physical branch. But the axiom `(contraryAttribute Pleasure Pain)` demands both arguments to be `Attribute` instances, which live on the Abstract branch. This is a type error. Four reclassification candidates exist: (a) PathologicProcess (status quo), (b) EmotionalState, (c) StateOfMind, (d) split into PainSensation + PainProcess.

The evidence ledger. Seven authoritative sources each contribute a support vector over all four candidates simultaneously:

```
structure SourceItem (Source Candidate : Type*) where
  source  : Source
  kind    : EvidenceKind       -- empirical | textInterpreted |
       expertElicited
  support : Candidate (*$\to$*) BinEvNat  -- vector over ALL
      candidates
  note    : String
```

[11] Generic framework: `Mettapedia/Logic/EvidentialLedger.lean`. Concrete case: `Mettapedia/Languages/GF/SUMO/PainEvidenceWM.lean`. All 10 theorems by `decide`, 0 sorry.

Each source says something about *every* candidate—not just the one it favors. The IASP definition, for instance, gives $(0, 3)$ against PathologicProcess (strong refutation), $(1, 2)$ against EmotionalState (misses the sensory component), $(3, 0)$ for StateOfMind (direct match), and $(2, 0)$ for the split option.

Aggregation. The `aggregate` function folds support vectors by candidate via componentwise addition—the same $\oplus$ revision that drives the entire WM calculus:

```
theorem pathologic_total :
  aggregate painEvidence .pathologicProcess = (*$\langle$*)2,
      16(*$\rangle$*)  -- refuted
theorem stateofmind_total :
  aggregate painEvidence .stateOfMind       = (*$\langle$*)12,
      1(*$\rangle$*)  -- winner
theorem emotional_total :
  aggregate painEvidence .emotionalState    = (*$\langle$*)10,
      3(*$\rangle$*)  -- runner-up
theorem split_total :
  aggregate painEvidence .split             = (*$\langle$*)6,
      4(*$\rangle$*)   -- viable
```

PathologicProcess is refuted (2 for vs. 16 against). StateOfMind leads EmotionalState (12:1 vs. 10:3). All totals are kernel-checked by `decide`.

Sensitivity. Two forgetting experiments test the robustness of the ranking:

```
theorem ranking_preserved_noWiki :    -- Wikipedia is
    corroborative
  ... (*$\wedge$*) ... (*$\wedge$*) ...    := by decide
theorem iasp_is_critical_discriminator :  -- IASP removal (*$\
    Rightarrow$*) tie
  aggregate (forget .iasp ...) .stateOfMind
    = aggregate (forget .iasp ...) .emotionalState := by decide
```

Removing Wikipedia preserves the full ranking—Wikipedia was corroborative, not decisive. Removing IASP creates a tie between StateOfMind and EmotionalState—IASP is the critical discriminator. Both facts are kernel-checked.

Evidence vs. policy. The evidence says StateOfMind $(12, 1)$. The implementation shipped EmotionalState $(10, 3)$. These are different questions:

- "What does the evidence support?" → StateOfMind (computed and kernel-checked above).
- "What did we ship?" → EmotionalState (operational choice: conservative consistency with prior SUMO practice, narrow margin, deferred to a broader reclassification phase).

The gap between evidence and policy is explicit—not hidden. The evidence algebra answers the epistemic question; the deployment decision is a separate governance layer. This separation is exactly the architecture recommended for evidence-driven systems: the base WM layer computes what the evidence says; reliability, dependence, and governance sit above it as extension layers. The reliability layer is now formalized: `ReliabilityProfile` assigns per-source scaling weights by evidence kind, and `reliabilityAdjust` scales evidence monotonically (more reliable sources contribute more weight).[12]

Compositionality. The generic framework proves that evidence from two groups of sources combines additively:

```
theorem toState_append (l(*$_1$*) l(*$_2$*) : List (SourceItem
    Source Candidate)) :
  toState (l(*$_1$*) ++ l(*$_2$*)) = fun c => toState l(*$_1$*)
      c + toState l(*$_2$*) c
```

This is the same compositionality law as the derivation tracking demo (§6.5.2)—the `EvidentialLedger` is an `AdditiveWorldModel` instance. The Pain case and the Maple Court sensor case are not ad hoc applications of the evidence algebra; they are *instances* of the same formally verified framework.

Maple Court: The building's elevator inspection reports come from three sources: the manufacturer's sensor, the resident council, and the annual city inspector. Each contributes a support vector over (normal, slow, faulty). The `EvidentialLedger` aggregates them additively, tracks which source contributed what, and answers sensitivity questions: "What changes if we retract the manufacturer's report?" The architecture is identical to the Pain reclassification—only the source and candidate types differ.

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[12] Formalized in `Mettapedia/Logic/SourceReliability.lean` (14 items, 0 sorry).

Chapter 7

Information, Support, and View Projections

Evidence can improve in more than one sense. It can become *more informed*, and it can become *more supportive* of a query. These are not the same order, and confusing them is one of the deepest sources of error in uncertain reasoning. This chapter develops that distinction and its consequences.

7.1 Two Notions of Improvement

Compare two binary evidence values:

$$e_1 = (1, 0), \qquad e_2 = (1, 1).$$

The second contains at least as much positive and at least as much negative evidence as the first: $e_1 \leq e_2$ in the information order. Yet e_2 gives lower strength ($\frac{1}{2}$ versus 1) and possibly higher confidence (sample size 2 versus 1).

More evidence made us less optimistic and more confident at the same time. This is natural once you think about it: the new observation was negative, so of course the balance shifted. But the information order does not care about the direction of evidence—it only asks whether there is *more* of it. The tension between "more data" and "more favorable data" is not an edge case. It is a theorem.

Theorem 7.1 (Strength non-monotonicity (Boundary)). *There exist* $e_1 \leq e_2$ *in the information order with* $\mathrm{strength}(e_1) > \mathrm{strength}(e_2)$.[1]

[1] Formalized as `toStrength_not_monotone_info_order` in `Mettapedia/Logic/`

The information order is the right partial order for evidence algebra. The truth-level comparison induced by strength is a separate, non-monotone projection. Any system that sorts evidence by a single view and calls the result "better" is conflating two distinct orders.

7.2 Views as Projections

A *view* is any function from an evidence carrier to a simpler space. Every carrier has its own natural views:

- **Binary**: strength $s = n^+/n$, confidence $c = n/(n+k)$, interval $[L, U]$.
- **Dirichlet**: per-category posterior means $\alpha_i / \sum \alpha_j$.
- **Normal-Gamma**: posterior mean μ_n, posterior variance $\beta/(\alpha\kappa)$.

No single view suffices to reconstruct the original evidence. Views are useful for display, thresholding, and action. They are not the semantics.

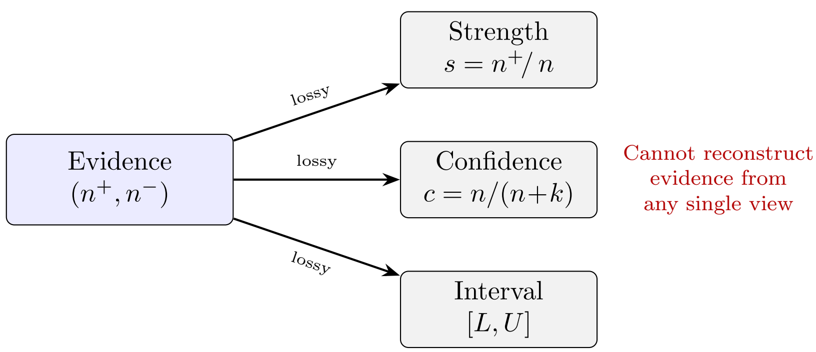

Figure 7.1: Views of binary evidence are lossy projections. Each carrier has its own views (Dirichlet: per-category means; Normal-Gamma: posterior mean and variance). No single view reconstructs the evidence.

7.3 The Additive / Non-Additive Boundary

Some projections commute with revision; most do not. This distinction determines what can safely be propagated through inference chains and what cannot.

`PLNWorldModelProfiles.lean`.

Total evidence $\text{total}(e) = n^+ + n^-$ is additive: $\text{total}(e_1 \oplus e_2) = \text{total}(e_1) + \text{total}(e_2)$. If you combine two batches of observations, the total count is the sum of the individual counts. Total evidence is an additive shadow—a homomorphism from evidence to $\mathbb{R}^{\infty}_{\geq 0}$.

Strength is not additive. Strength divides positive counts by total counts—a ratio. Ratios do not add: a class where 3 out of 4 students passed and a class where 1 out of 4 passed do not combine to "4 out of 8 passed" by adding the ratios $\frac{3}{4}$ and $\frac{1}{4}$. The combined strength is $(3+1)/(4+4) = \frac{1}{2}$, not $\frac{3}{4} + \frac{1}{4} = 1$.

Theorem 7.2 (`strength_not_additive` (Boundary))**.** *There exist* e_1, e_2 *such that* $\text{strength}(e_1 + e_2) \neq \text{strength}(e_1) + \text{strength}(e_2)$.[2]

This is not a deficiency. It determines the architecture: carry evidence through inference chains; extract views only at the end.

7.4 Simple Truth Values

A *simple truth value* (STV) is a pair $\langle s, c \rangle$ where $s = n^+/(n^+ + n^-)$ is the strength and $c = n/(n+k)$ is the confidence. It is compact, intuitive, and lossy: two different evidence values can produce the same STV.[3]

Maple Court: Querying "Is the pipe leaking?" extracts $(1, 4)$: one leak alarm, four no-alarm readings. The STV is $\langle s, c \rangle = \langle 1/5, 5/15 \rangle = \langle 0.2, 0.33 \rangle$ (with $k = 10$): low confidence in a low-strength hypothesis. The fork query "Is the room occupied?" extracts $(5, 1)$ with $\langle 0.83, 0.38 \rangle$: higher strength, similar confidence. Both views are lossy projections of the same underlying evidence. The strength of $(5, 1)$ exceeds that of $(1, 4)$, but neither STV reveals how many observations contributed —that information lives in the evidence, not the view.[4]

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[2] Counterexample: $e_1 = (1, 0)$, $e_2 = (0, 1)$. Formalized in `Mettapedia/Logic/WorldModelCore.lean`.

[3] Formalized in `Mettapedia/Logic/BinaryEvidence.lean`.

[4] Proved in `artifacts/conformance/maple\_court\_bn\_motifs.metta` (leak: strength = 0.2; fork: strength ≈ 0.833).

7.5 Confidence

Confidence is strictly a derived view: $c(n^+, n^-) = (n^+ + n^-)/(n^+ + n^- + k)$. It is monotone in sample size: more evidence always means higher confidence. But confidence says nothing about whether the evidence supports the claim. A high-confidence negative result is still informative.

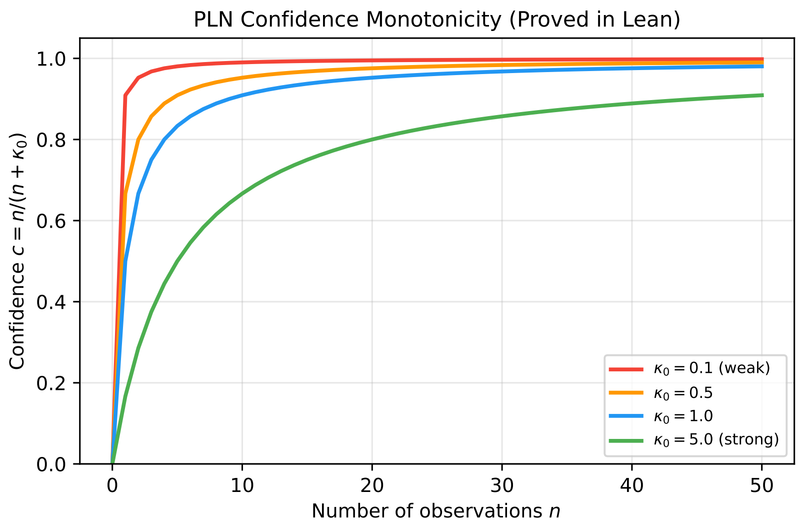


Figure 7.2: Confidence as a function of sample size for the three carriers. All three are monotone: more evidence always means higher confidence. The PPLBench comparison (Chapter 24) validates these curves against Stan's NUTS sampler.

7.6 Indefinite Truth Values

An interval $[s_L, s_U]$ bounds the strength. Three semantics exist and must not be conflated:

1. **Bayesian credible intervals**: posterior probability that the true parameter lies in $[s_L, s_U]$ exceeds a threshold. Prior-dependent.

2. **Walley predictive intervals**: derived from the Imprecise Dirichlet Model [36]. Prior-free but wider.

3. **Frequentist confidence intervals**: the true parameter lies within them with specified coverage probability over repeated sampling.

The calculus insists on explicit selector choice: the user declares which interval semantics is intended, and the system computes accordingly. The Walley connection is developed in Chapter 5, §"Walley predictive intervals."[5]

7.7 Why Views Are Not Proof Objects

Views are indispensable for practical inference. But they are not the right objects to pass between inference steps. When you chain two inferences using only the STV of the intermediate result, you lose information about the posterior shape, the correlation structure, and the raw counts.

The discipline of the calculus: carry evidence between inference steps; extract views only at the end, for display or decision.

A concrete illustration: three PLN confidence bugs. The MeTTa PLN implementation contains three formally proved confidence-formula errors that arise precisely from operating on views instead of evidence:[6]

1. **Double-damping.** The induction rule computes $\texttt{w2c}(\min(c_1, c_2))$, applying the weight-to-confidence function to a value that is already a confidence. This double-damps: at $c_1 = c_2 = 0.9$, the buggy formula gives 0.474 instead of 0.9—a 47% underestimate. The correct formula converts to weights first: $\texttt{w2c}(\min(\texttt{c2w}(c_1), \texttt{c2w}(c_2))) = \min(c_1, c_2)$.

2. **Confidence floor.** The revision rule clamps output confidence to $\max(c, c_1, c_2)$. Under correct evidence-based revision, combined confidence always exceeds each input (proved), so the clamp is a no-op—but it masks bugs elsewhere by hiding negative or nonsensical intermediate values.

3. **Division by zero.** The function $\texttt{c2w}(c) = c/(1 - c)$ diverges at $c = 1$. In MeTTa, this silently returns . The MeTTa-level patch caps confidence at $\texttt{MAX\_CONF} < 1$—a reasonable engineering approximation with a proved error bound, since certainty ($c = 1$) requires infinite evidence and never arises in practice. But the deeper lesson is that `c2w` is an attempt to invert a lossy projection: it recovers weight from confidence, which is the opposite of the D4 direction. The evidence algebra works with (n^+, n^-) directly, where the total $n^+ + n^-$ plays the role of weight without any division or cap.

[5] Formalized in `Mettapedia/Logic/PLNIndefiniteTruth.lean`.

[6] Formalized: `Mettapedia/Logic/PLNBugAnalysis.lean` (0 sorry, 14 theorems with concrete counterexamples and MeTTa patches).

All three bugs disappear when the system works with evidence pairs rather than confidence scalars. In weight space (which is isomorphic to evidence counts via the round-trip $\texttt{w2c}(\texttt{c2w}(c)) = c$, proved), composition is multiplication; in confidence space, it is not. This is the practical consequence of D4 (Evidence over Views): the bugs are not implementation accidents—they are symptoms of operating on the wrong abstraction layer.

Probability is what you query; evidence is what you carry.

Chapter 8

Natively Typed World Models

Chapter 4 presented the world-model calculus as an algebraic interface: three operations, five rules in the additive regime, a 2×2 refinement square. This chapter shows that the same calculus has a *typed rewrite face*: a concrete family of typed rewrite rules whose terms carry native types and whose reductions correspond exactly to the algebraic laws.

The point is not to add a second semantics. It is to make the calculus *mechanically checkable*: a computer can verify that a compiled rewrite is sound by reducing it in the typed system.

8.1 The Pattern Vocabulary

The calculus is encoded in three core sorts—**State**, **Query**, **Value**—and constructors corresponding to the calculus operations:

Constructor	**Signature**	**Operation**
$\mathsf{Revise}(W_1, W_2)$	State $\times$ State $\to$ State	revision
$\mathsf{Extract}(W, q)$	State $\times$ Query $\to$ Value	extraction
$\mathsf{Combine}(e_1, e_2)$	Value $\times$ Value $\to$ Value	additive fusion
$\mathsf{Forget}(S, W)$	Scope $\times$ State $\to$ State	scoped retraction
$\mathsf{OverlapMerge}(W_1, W_2)$	State^2 $\to$ State	overlap-aware merge

The remaining constructors—30 total across all extension families—handle trust-gated revision, source-labeled provenance, fixpoint closure, order-cost tracking, Kripke modalities, and generic carriers. Each appears when the corresponding axis of the 13-axis family (§8.5) is enabled. The table above shows the core that is always present.[1]

8.2 The Five Rules as Typed Rewrites

The five algebraic rules from §4.6 — the same five, now expressed as pattern-matching rewrites rather than algebraic identities: Each has a typed left-hand side, a right-hand side, and (for guarded rules) a premise:

1. $\mathsf{Extract}(\mathsf{Revise}(W_1, W_2), q) \rightsquigarrow \mathsf{Combine}(\mathsf{Extract}(W_1, q),\ \mathsf{Extract}(W_2, q))$.
2. $\mathsf{Revise}(W_1, W_2) \rightsquigarrow \mathsf{Revise}(W_2, W_1)$.
3. $\mathsf{Revise}(\mathsf{Revise}(W_1, W_2), W_3) \rightsquigarrow \mathsf{Revise}(W_1, \mathsf{Revise}(W_2, W_3))$.
4. $\mathsf{Combine}(e_1, e_2) \rightsquigarrow \mathsf{Combine}(e_2, e_1)$.
5. $\mathsf{Combine}(e, \mathsf{Zero}) \rightsquigarrow e$.

All five derive from the single condition that $\mathsf{Extract}$ is an additive monoid homomorphism. The Lean formalization encodes each as a `RewriteRule` with typed metavariables, premises, and left/right patterns. Sixteen step lemmas are kernel-checked across the core and extension axes.[2]

8.3 The 2×2 Calculus Square as Concrete Rewrites

Two orthogonal refinements produce four variant calculi:

	Plain	**+ Congruence**
Raw	all rules fire	+ descent into subterms
Guarded	side-condition oracle	+ descent into subterms

The raw calculus overapproximates the guarded one: dropping premises only enables more reductions. The congruence extension adds 15 explicit rules enabling reductions to propagate into subterms. All four inclusion arrows and Galois connections $\Diamond \dashv \Box$ are kernel-checked.[3]

[1]Formalized in `Mettapedia/OSLF/Framework/WMCalculusBNBridge.lean`.
[2]Formalized in `Mettapedia/OSLF/Framework/WMCalculusLanguageDef.lean`.
[3]Formalized in `Mettapedia/OSLF/Framework/WMCalculusContextClosure.lean`.

The congruence infrastructure enables multi-step decomposition under context. For example, the term $\mathsf{Extract}(\mathsf{Revise}(\mathsf{Revise}(W_1, W_2), W_3), q)$ reduces in two steps to $\mathsf{Combine}(\mathsf{Combine}(\mathsf{Extract}(W_1, q), \mathsf{Extract}(W_2, q)), \mathsf{Extract}(W_3, q))$: the first step applies evidence-add at the top; the second uses a congruence rule to apply evidence-add *inside* the left Combine argument.

8.4 Intrinsic Encoding and Adequacy

The central fact: the typed calculus and the pattern-level calculus agree exactly. A reduction holds at the abstract level if and only if the corresponding pattern-level reduction holds on the encoded terms.

The abstract syntax is an intrinsically sorted type family $\mathsf{WMTerm} : \mathsf{WMSort} \to \mathsf{Type}$ with three sorts. A computable encoding encodeWM maps abstract terms to the OSLF pattern language.

Theorem 8.1 (Biconditional adequacy (Core))**.** *A reduction at the abstract* WMTerm *level holds if and only if the corresponding pattern-level OSLF reduction holds on the encoded terms. The two presentations agree exactly.*[4]

The encoding is injective within each sort. Subject reduction is definitional: every constructor of WMStep preserves the sort by construction.

For the $\Box$ modality, backward completeness holds when restricted to the image of encodeWM:

	Forward ($\Diamond$)	Backward ($\Box$)
Unrestricted	biconditional	sound only
Image-restricted	biconditional	biconditional

8.5 The 13-Axis Vertex Family

The full typed family is indexed by 13 orthogonal axes. The first four control semantic interpretation without changing the rewrite rules; the remaining nine each add genuinely new rules to the calculus:

[4]Formalized: `wmStep_iff` (single step), `wmStepStar_adequate` (multi-step), both kernel-checked. See `Mettapedia/OSLF/Framework/WMCalculusEncoding.lean`.

#	Axis	Values	Effect
1–4	Base (logic, tv, interval, typing)	24 combinations	model-level (same rules)
5	Overlap	additive | overlapAware	+1 rule
6	Forgetting	none | scopeBased | tracked	+2–3 rules
7	Provenance	none | sourceLabeled	+6 rules
8	Fixpoint	none | closure | policy	+2–4 rules
9	Cost	none | costTracked	+3 rules
10	Conservation	none | conserving	+2 rules
11	Experiment	none | stochastic	+3–7 rules
12	Kripke	none | pointedKripke	+3 rules
13	Carrier	concrete | generic	+2 rules

The first four axes are model-level: all 24 combinations produce the same rewrite rules. The remaining nine are syntactic: each adds new rules. The minimal vertex (all extensions off) has 5 rules; the maximal vertex has 36.[5]

Maple Court: Maple Court's apartment WM sits at a mid-range vertex: overlap-aware (sensors may share a vent), scope-based forgetting (stale readings expire), source-labeled provenance (which sensor contributed). The building WM adds cost tracking (order-cost governance for the cleaning robot's schedule). Each vertex determines which rewrite rules are available—and which side conditions must be checked.

The 4-vertex WM regime classification (§4.3) maps injectively into the 13-axis family: each regime determines specific axis settings (commutativity, derived-additivity, probabilistic interpretation).[6]

8.6 OSLF Modal Operators and Semantic Bridge

From any vertex's `LanguageDef`, the OSLF framework automatically derives modal operators $\Diamond$ (forward reachability: "some reduction leads to a state satisfying φ") and $\Box$ (backward invariance: "every predecessor satisfies φ") with a Galois connection $\Diamond \dashv \Box$.

This is not optional decoration. The modal operators give the typed calculus its semantic backbone: $\Diamond$ captures what can be derived; $\Box$ captures

[5] Formalized in `Mettapedia/OSLF/Framework/TypeSynthesis.lean`.
[6] Formalized in `Mettapedia/Logic/ModalProbabilityBridge.lean` (14 items, 0 sorry).

what is preserved. The Galois connection ensures the two are dual.[7]

Diamond theorems, relational encoding, evidence barbs (process-algebraic observables), neighborhood semantics, and first-order logic all arise as semantic fibers of this construction.

8.7 Native Type Synthesis

The native-type-theorem (NTT) pass extracts, for each vertex, the type that classifies its allowable terms and reductions. The pass is exact on 9 syntactic axes and intentionally insensitive to 4 model-level axes.[8]

Native types do not introduce a second semantics. They expose, at the rewrite level, the structural behavior already present in the semantic kernel. A term that type-checks at a given vertex is guaranteed to reduce according to that vertex's rules.

8.8 World Models from GSLTs

For any type T, multiset ensembles of T form a world model when queried by decidable properties: extraction counts satisfying and refuting entities, and the count is additive over multiset union.

Theorem 8.2 (`universalEnsembleWorldModel` (Core))**.** *For any type T: AdditiveWorldModel (Multiset T) (DecProp T).*[9]

This means: every computational system with parallel composition—every graph-structured lambda theory (GSLT) in Meredith's [25] sense—has a canonical additive world model via multiset counting. The converse also holds: a GSLT equipped with an additive evidence assignment IS an `AdditiveWorldModel`, with the GSLT's weight map as the `extract` function and forward transport (reductions in weaker theories lift to stronger ones) as `extract_add`.[10]

Conjecture 8.3 (WM universality for typed rewrite systems). The WM calculus is not specific to probabilistic logic; it is the natural evidence framework for any typed rewrite system. Every `LanguageDef` (i.e., every vertex of the

[7]The construction follows Meredith's [25] context-decorated HML modalities, derived from the GSLT's operational semantics.

[8]Formalized: `wmVertexNativeType_eq` (kernel-checked).

[9]Formalized in `Mettapedia/OSLF/Framework/GSLTWorldModel.lean`. Existence and uniqueness for any observation type $\times$ evidence carrier are packaged in `Mettapedia/Logic/UniversalEnsembleWM.lean` (7 items, 0 sorry).

[10]Formalized in `Mettapedia/Logic/GSLTWeightMapBridge.lean` (7 items, 0 sorry).

typed family) gives an additive world model via the OSLF-derived modal operators. Replacing V with $\mathbb{C}$ gives Meredith's amplitude framework; replacing V with $\mathbb{R}_{\geq 0}^{\infty}$ gives the Knuth–Skilling probability framework. The probability hypercube [30] classifies theories by which combination of merge-law and carrier properties they satisfy (Chapter 19).

[11]

The philosophical import of this direction is worth pausing on. If the conjecture holds, then the world-model calculus is not a tool designed for one application. It is the structure that *any* typed rewrite system naturally acquires when you ask it to track evidence about its own behavior. A Bayesian network, a process calculus, a programming language, a causal model—each would have its own canonical world model, the same way each has its own type system. Evidence extraction would be as natural and universal as type-checking.

The typed calculus constructs world models. The next chapter shows what even the best local construction cannot achieve.

[11] Open program: the full LanguageDef $\to$ AdditiveWorldModel construction, the StatefulValuation typeclass unifying state-side and obs-side laws, and the grand theorem connecting WM + KS + Meredith as specializations of one bivariate extractor. A work-in-progress Lean formalization of the probability hypercube axes (commutativity, distributivity, precision, additivity, etc.) exists across 29 files in `Mettapedia/ProbabilityTheory/Hypercube/`.

Chapter 9

The No-Go Theorem

The previous chapters built the calculus, its evidence carriers, its state operations, its views, and its typed realization. This chapter establishes the fundamental limit of the framework: what even the most expressive local computation cannot achieve.

One might hope for a function that takes local per-link evidence and computes complete evidence for any derived query. No such function exists.

Theorem 9.1 (No local complete deduction rule (Boundary))**.** *There is no function $f : V^5 \to V$ that, for all joint evidence states E, computes the exact link evidence for $A \Rightarrow C$ from only the local premises* $\mathrm{extract}(E, A)$, $\mathrm{extract}(E, B)$, $\mathrm{extract}(E, C)$, $\mathrm{extract}(E, A \Rightarrow B)$, $\mathrm{extract}(E, B \Rightarrow C)$.[1]

Proof sketch. For three atoms, construct two joint states E_1, E_2 that agree on all five local marginals but disagree on $A \Rightarrow C$. The key: among the four worlds where $A = 1$, the entries for $(A=1, B=0, C=1)$ and $(A=1, B=1, C=0)$ can be traded against each other. Concretely, transfer mass ϵ from world $(1,0,1)$ to world $(1,1,0)$. Both worlds have $A = 1$, so the $\mathrm{ev}(A)$ marginal is unchanged. The first contributes to $B=0$ and the second to $B=1$, but by equal amounts, so $\mathrm{ev}(B)$ is unchanged. Likewise for $\mathrm{ev}(C)$ (the first has $C=1$, the second $C=0$). The link marginals $\mathrm{ev}(A \Rightarrow B)$ and $\mathrm{ev}(B \Rightarrow C)$ each sum over one column and one row of the $B \times C$ sub-table given $A = 1$; the trade shifts mass within each row and column by equal amounts, preserving both sums. But $\mathrm{ev}(A \Rightarrow C)$ counts the worlds where $A=1$ and $C=1$: the trade removes one such world and adds none, directly changing this marginal. □

[1] Formalized in `Mettapedia/Logic/PLNJointEvidenceNoGo.lean`.

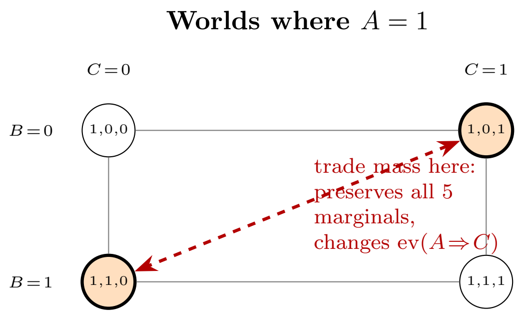


Figure 9.1: The no-go witness. Redistributing mass between the orange worlds preserves all five local marginals but changes $\mathrm{ev}(A \Rightarrow C)$.

This theorem does not make fast inference useless. It tells us what fast inference *cannot claim*: completeness for arbitrary world models without structural assumptions. The constructive response:

1. Restrict the model class (BN factorization makes local computation exact — Chapter 11).
2. State the assumptions (every compiled rule carries its Σ).
3. Provide corrected alternatives where original rules required unstated assumptions.

Maple Court: Two BN structures for Maple Court give the same per-link marginals $P(\text{leak})$ and $P(\text{humidity})$ but different $P(\text{mold} \mid \text{humidity}, \text{leak})$. No amount of per-link evidence distinguishes them. The BN DAG structure — a side condition Σ — is what breaks the degeneracy.

The constructive resolution of this limitation—distributional inference, certified chaining bounds, and view transport—appears in Chapter 12.

Part III

Compiled Inference, Scale, and Control

Chapter 10

Σ-Guarded Rewrites

The no-go theorem says local chaining cannot be complete in general. This chapter gives the constructive response: *compiled rewrites* that are exact under explicit structural assumptions. A compiled rewrite is not new semantics. It is a certified shortcut to an answer already determined by the underlying world model.

10.1 The Rewrite Pattern

A Σ-guarded rewrite has the form: given that side condition Σ holds (BN factorization, screening-off, positivity), the rewrite computes the *same* value as the full posterior query. The rule is a theorem, not a heuristic.

$$\frac{\Sigma \vdash \mathrm{ScreenOff}(A, B, C) \quad \Sigma; \Gamma \vdash \mathrm{link}(A, B) \Downarrow e_{AB} \quad \cdots}{\Sigma; \Gamma \vdash \mathrm{link}(A, C) \Downarrow e_{AC}}$$

The side condition Σ is not optional. When it fails, the rule does not apply. The system falls back to the full posterior.[1]

10.2 The Compiled Rule Catalog

Each compiled rule has a name, a pattern, a side condition Σ, a strength formula, and a correctness theorem. The catalog:

[1]Formalized in `Mettapedia/Logic/PLNBayesNetFastRules.lean` and `Mettapedia/Logic/PLNDerivation.lean`.

Rule	Pattern	Σ	Strength formula
Deduction	$A\to B\to C$	screening-off, positivity	$s_{AB}s_{BC} + (1-s_{AB})\frac{s_C - s_B s_{BC}}{1-s_B}$
Induction	$C\to A,\ C\to B$	screening-off	$s_{AB} = \frac{s_{BA}s_B}{s_A}$; then deduction
Abduction	$A\to C,\ B\to C$	screening-off	$s_{BC} = \frac{s_{CB}s_C}{s_B}$; then deduction
Modus ponens	$A,\ A\Rightarrow B$	none	$s_{AB}\cdot s_A + (1-s_A)\cdot c$
Revision	e_1, e_2	stamp-disjoint	weighted average of s_1, s_2
Negation	A	none	$1 - s_A$
Inversion	$A\to B$	none	$s_{BA} = s_{AB};\ c_{BA} = c_{AB}\cdot\frac{n_A}{n_B}$

These are the classical PLN rules of Goertzel et al. [11], now with explicit side conditions and kernel-checked correctness proofs. The deduction formula is exact under the chain BN structure with screening-off and positivity ($0 < s_B < 1$)—see Theorem 11.3. Induction and abduction are Bayes-inversions of deduction, with their own screening-off conditions. Revision is evidence addition (§4.4).[2]

10.3 Compiled Tactics vs. Foundational Semantics

The distinction is crucial: the kernel (Part II) says what the answer *is*. Compiled shards say how to compute it *fast*. If the side condition is met, the shard gives the exact answer. If not, the system falls back to the full extraction.

The Σ-discipline ensures that every shortcut is explicitly licensed. A rule either has a proof of correctness under Σ, or it is not available.

For maximum fidelity, propagate the full posterior distribution through the chain rather than scalar summaries. The distributional approach avoids the variance accumulation that scalar chaining inevitably introduces (Chapter 12). Scalar chaining is a useful further approximation within the distributional framework, not a replacement for it.

[2] Formalized in `Mettapedia/Logic/PLNDerivation.lean` (strength formulas, bounds, and Bayes-inversion theorems) and `Mettapedia/Logic/PLNBayesNetFastRules.lean` (BN exactness theorems).

Chapter 11

Bayesian-Network Exactness

The original PLN rule family — deduction, induction, abduction, revision, modus ponens — was designed by Goertzel, Iklé, Goertzel, and Heljakka [11] as a general-purpose uncertain reasoner. Geisweiller's νPLN framework [6] grounded the evidence representation in de Finetti's exchangeability theorem. This chapter shows that the classical rules become *exact* compiled rewrites when the world model has Bayesian-network structure — not approximate heuristics, but certified shortcuts with explicit side conditions.

11.1 BN World-Model Class

A Bayesian-network world model consists of a DAG specifying the factorization plus local conditional probability tables (CPTs) carrying evidence counts. Revision is addition of local counts (conjugate updating); query answering proceeds by exact inference. The DAG structure provides the Σ that chain/fork/collider rules require.[1]

11.2 Exact BN Query Bridge

BN queries correspond to posterior marginalization. The central fact is a commutative diagram: a BN query answered by variable elimination agrees with the general world-model extraction applied to the same state.

Theorem 11.1 (BN query bridge (Core)). *For a BN world-model class with DAG G and evidence state W,*

$$\mathrm{extract}_{\mathrm{BN}}(W, q) = \mathrm{extract}_{\mathrm{WM}}(W, q)$$

[1]Formalized in `Mettapedia/Logic/PLNBayesNetWorldModel.lean`.

for every query q in the scope of the BN.[2]

The BN structure does not change the answer—it changes the *cost* of obtaining it. Without the DAG, answering a query requires summing over all 2^n worlds. With the DAG, variable elimination factors the sum into local operations whose cost is exponential only in the treewidth. The answer is the same; the computational path is shorter.

11.3 Chain, Fork, and Collider Exactness

The three fundamental BN motifs each yield exactness theorems. We derive the chain case in full; the fork and collider cases follow the same pattern with their own side conditions.

Deriving the deduction formula. Start from the law of total probability:

$$P(C \mid A) = P(C \mid A, B)\, P(B \mid A) + P(C \mid A, \neg B)\, P(\neg B \mid A).$$

Under the screening-off assumption $P(C \mid A, B) = P(C \mid B)$ and $P(C \mid A, \neg B) = P(C \mid \neg B)$, this simplifies to:

$$P(C \mid A) = P(C \mid B)\, P(B \mid A) + P(C \mid \neg B)\, (1 - P(B \mid A)).$$

Writing $s_{AB} = P(B \mid A)$, $s_{BC} = P(C \mid B)$, $s_B = P(B)$, $s_C = P(C)$, and eliminating $P(C \mid \neg B)$ via $s_C = s_B \cdot s_{BC} + (1 - s_B) \cdot P(C \mid \neg B)$:

Definition 11.2 (PLN deduction strength)**.**

$$\mathrm{plnDed}(s_{AB}, s_{BC}, s_B, s_C) := s_{AB} \cdot s_{BC} + (1 - s_{AB}) \cdot \frac{s_C - s_B \cdot s_{BC}}{1 - s_B}$$

defined when $0 < s_B < 1$.

This formula was introduced by Goertzel et al. [11] and is derived here from first principles under the screening-off assumption.

Theorem 11.3 (Chain BN deduction exactness (Conditional))**.** *For a chain BN $A \to B \to C$ with positivity ($0 < s_B < 1$) and screening-off:*

$$\mathrm{strength}(W, \mathrm{link}(A, C)) = \mathrm{plnDed}(s_{AB}, s_{BC}, s_B, s_C).$$

[3]

[2] Formalized: `bnQueryBridge_extraction_eq` in `Mettapedia/Logic/PLNBayesNetWorldModel.lean`.

[3] Formalized: `chainBN_plnDeductionStrength_exact` in `Mettapedia/Logic/PLNBayesNetFastRules.lean`.

The analogous result holds for fork BNs $A \leftarrow B \rightarrow C$.[4]

Induction and abduction. Both reduce to Bayes-inversion followed by deduction. *Induction*: given $P(A \mid B)$ and $P(A \mid C)$, estimate $P(C \mid B)$. First invert: $s_{AB} = s_{BA} \cdot s_B / s_A$ (Bayes's theorem), then apply the deduction formula with s_{AB}. *Abduction*: given $P(B \mid A)$ and $P(B \mid C)$, estimate $P(A \mid C)$. Invert: $s_{BC} = s_{CB} \cdot s_C / s_B$, then deduction.[5]

The chain theorem above is stated at the strength (scalar) level. At the *distributional level*, the story is cleaner: propagate the full Beta or Dirichlet posterior through the chain, and the result is exact under d-separation—no variance accumulation, no information loss. The scalar deduction formula is the strength *view* of this distributional chain. The five chaining-limitation theorems of Chapter 12 make the cost of the scalar approximation precise; the distributional dominance theorem (ibid.) shows that carrying the full posterior is strictly more informative.

Maple Court: The chain PipeLeak $\rightarrow$ WallHumidity $\rightarrow$ MoldRisk in Maple Court: the compiled chain rule computes $P(\text{mold} \mid \text{leak})$ from the per-link evidence under the BN factorization. The side condition: WallHumidity screens off MoldRisk from PipeLeak. When this holds, the compiled answer is exact. When it fails (e.g., a direct leak-to-mold pathway exists), the rule is not licensed.

11.4 Class-Packaged BN Discharge

One of the practical payoffs of the Σ-discipline is that side-condition checking can be automated. For discrete BNs with explicit CPTs, the function `allDiscreteCPTLocalMarkov_of` verifies d-separation and screening-off for every pair of non-adjacent nodes—i.e., it checks the local Markov property. If the check passes, all the chain, fork, and collider exactness theorems are simultaneously unlocked for that model. The BN DAG structure itself constitutes the proof of the side condition.[6]

[4] Formalized: `forkBN_plnDeductionStrength_exact`.

[5] Formalized in `Mettapedia/Logic/PLNDerivation.lean`: `plnInduction_eq_bayes_deduction`, `plnAbduction_eq_bayes_deduction`.

[6] Formalized in `Mettapedia/Logic/PLNBayesNetWorldModel.lean`.

11.5 Worked Proof-Carrying Contraction

A binary BN $A \leftarrow H \rightarrow B \rightarrow C \rightarrow D$ demonstrates the full discipline. Positive control: the chain fires correctly with $P(B \mid A) \approx 0.615$. Negative control 1: the soft gate fails when d-separation is violated. Negative control 2: collider naive abduction fails. The negative controls are as important as the positive ones — they show that the Σ-discipline catches real errors.[7]

11.6 PLN v0.9 Examples as Compiled WM Instances

PLN v0.9 [11][8] provides three flagship examples exercising the classical rule family. Each is a compiled BN rewrite with provenance tracking:

DeductionRevision. Two chains $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$ with disjoint stamps $\{0, 2\}$ and $\{1, 3\}$; revision is safe; forgetting obs 0 preserves path 2.

RavenInduction. Induction from two ravens' data; forgetting rv2 reduces the induction to rv1's evidence alone.

FlyingRaven. Chain Sam→Raven→Bird→flies vs. Pingu→Penguin→ ¬flies; forgetting Sam→Raven conserves Pingu.

All verified with 20 kernel-checked conformance tests.[9]

The stepped walkthrough for DeductionRevision with provenance appears in Chapter 6, §6.5 — including both tikz diagrams (Figure 6.2 and Figure 6.3).

11.7 Bridge Theorems: Naive Bayes and k-NN

Two standard ML classifiers are exact world-model instances.

Naive Bayes. A Naive Bayes classifier with class variable C and features $F_1, \ldots, F_n$ is a BN fork: $C \rightarrow F_1, \ldots, C \rightarrow F_n$. Classification amounts to querying $\arg\max_c \operatorname{extract}(W, c)$ where the evidence for class c is the tensor product of per-feature evidences $\bigotimes_{i=1}^{n} e_{F_i|c}$. The conditional-independence assumption of Naive Bayes IS the d-separation condition of the fork BN.

[7] Formalized in `Mettapedia/Logic/PLNProofCarryingContractionDemo.lean`.

[8] PLN v0.9: `github.com/trueagi-io/PLN`, tag `v0.9`.

[9] Formalized in `Mettapedia/Logic/PLNClassicExamples.lean`.

k-Nearest Neighbors. k-NN is evidence-weighted revision: the k nearest neighbors each contribute an evidence fragment e_i (one observation of the target class), and the prediction is the extraction from their sum $e_1 \oplus \cdots \oplus e_k$. The "nearest" selection is a scope restriction (only the k closest sources revise the state).[10]

These are exact bridges, not analogies: the standard ML classifiers are world-model instances under specific BN structure. This means the entire Σ-discipline applies to them: provenance tracking, forgetting, conservation, and distributional inference all carry over.

11.8 Factor Graphs: Semantic vs. Operational

A factor graph represents a world-model state by decomposing it into local factors. Two readings must be distinguished:

Semantic factors. The factors *are* the model. Each factor stores an evidence table (a local CPT enriched with counts), and the global state is their product. This is the BN world-model class (§4.4).

Operational factors. The factors are rule schemas. Message passing on the factor graph implements variable elimination, and each message corresponds to a partial extraction.

The alignment is via compilation: each operational factor corresponds to a conditional in the semantic model, and the message-passing result agrees with the full extraction applied to the same state. The BN structure is the Σ that licenses this agreement.[11]

[10] Formalized in `Mettapedia/Logic/PLNBayesNetInference.lean`.

[11] Formalized: `factorGraphBridge_extraction_comm` in `Mettapedia/Logic/PLNBayesNetInference.lean`.

Chapter 12

Error, Thresholds, and Distributional Resolution

Scalar chaining—propagating strength and confidence through inference steps—is the fast path. This chapter explains why it is also the lossy path, and what the alternative is.

12.1 Certified Chaining: Fundamental Limitations

Five proved theorems characterize the boundary of scalar chaining. Each is unconditional: no side condition removes the limitation when chaining at the scalar (view) level rather than the distributional (carrier) level.

1. **Variance accumulation.** For a chain of n steps each involving an unresolved regime mixture with per-step variance $\geq \delta$, the total accumulated variance is $\geq n \cdot \delta$. Variance grows at least linearly.[1]

2. **Sensitivity amplification.** If each step is K-Lipschitz, composing n steps yields a K^n-Lipschitz chain. For $K > 1$, errors amplify exponentially.[2]

3. **Coverage collapse.** If each step has per-step coverage at most $1-\alpha$, the joint coverage decays as $(1-\alpha)^n \to 0$. Certified per-step guarantees do not compose into chain-length-independent certificates.[3]

[1]Formalized in `Mettapedia/Logic/PLNVarianceChainNoGo.lean`.
[2]Formalized in `Mettapedia/Logic/PLNSensitivityNoGo.lean`.
[3]Formalized in `Mettapedia/Logic/PLNCoverageCollapseNoGo.lean`.

4. **Topology-CPT gap.** For a BN with fixed DAG topology, the conditioning residual can vary arbitrarily across CPT parameterizations. No function of the topology alone bounds it non-trivially.[4]

5. **No universal error bound.** There is no function of the number of chaining steps that bounds the error for all world models. The error depends on the specific joint structure, which is exactly what scalar chaining discards.

Together these say: information is lost when you project before you chain. Each theorem identifies a different mechanism of loss — variance, sensitivity, coverage, topology, or structure — but they all point to the same conclusion: carry the full posterior.

12.2 Distributional Inference: The Constructive Resolution

The fix: propagate the full posterior distribution through the chain, not scalar views. Instead of extracting a strength summary $s = 0.2$ and passing it forward, pass the full $\mathrm{Beta}(2, 5)$ posterior—corresponding to Maple Court's pipe-leak evidence $(1, 4)$ under a $\mathrm{Beta}(1, 1)$ prior. The next step conditions on the distribution, not the point estimate.

Theorem 12.1 (Distributional dominance (Core))**.** *Distributional chaining dominates scalar chaining in information content. After k steps, the distributional chain preserves the full posterior shape, while the scalar chain has accumulated artificial precision.*[5]

This is why the evidence carrier family (Chapter 5) matters: the carriers store full sufficient statistics, not just strengths. Distributional chaining uses those statistics directly. Scalar chaining projects them to views first, losing information at each step.

The five limitation theorems and the dominance theorem together say: *carry the posterior, not the point estimate.* Or, recalling Chapter 7: *probability is what you query; evidence is what you carry.* Scalar views are for display and decision at the end of a chain, not for intermediate computation.

[4]Formalized in `Mettapedia/Logic/PLNTopologyCPTNoGo.lean`.
[5]Formalized in `Mettapedia/Logic/PLNDistributionalConvergence.lean`.

12.3 View Transport Under Query Equivalence

When two queries are semantically equivalent (same extraction on every state), their views transport. This is a corollary of the universal property: equivalent queries extract the same value, so every view agrees.

12.4 Threshold Semantics

Truth thresholds (strength > 0.5, confidence > 0.8) are predicates on views, not on evidence. The threshold semantics connects the view layer to decision-making: an agent acts when the view crosses a threshold. The calculus makes explicit that thresholds are *policy*, not *semantics*—changing the threshold changes the decision boundary, not the underlying evidence.

This has a practical consequence for chaining. A chain of n compiled rewrites, each passing only scalar views to the next step, accumulates artificial precision: the output strength *looks* like a single number, but its actual uncertainty is wider than what the confidence figure suggests. The five limitation theorems above quantify this. The distributional approach avoids the problem entirely: the posterior shape carries its own uncertainty, and thresholds applied to the *final* distributional answer have calibrated coverage.

Maple Court: The building's mold-risk policy fires when strength(MoldRisk) $>$ 0.3 and confidence > 0.5. With the pipe-leak chain PipeLeak $\to$ WallHumidity $\to$ MoldRisk, a scalar chain might report $s = 0.35$, $c = 0.7$—above threshold. But distributional propagation reveals that the 90% credible interval for the mold-risk posterior spans $[0.1, 0.6]$, so the decision is less certain than the scalar view suggests. The threshold decision remains the same, but the system can now report *how certain* the threshold crossing is.

Chapter 13

Large-Scale Inference and Inference Control

The previous chapters built a calculus that is correct: states carry evidence, extraction is additive, compiled rewrites are exact under explicit side conditions. This chapter addresses the problem that correctness alone does not solve: *which correct steps should a system take?*

A world model with n atoms and k compiled rules admits $\binom{n}{2} \cdot k$ possible single-step inferences—and multi-step chains grow combinatorially. An unrestricted forward chainer would drown in correct but useless conclusions. The classical PLN response (Chapter 9 of the PLN book) is *inference control*: a meta-level discipline that selects, schedules, and budgets inference steps.

This chapter formalizes that meta-level discipline within the world-model calculus. The key insight is that inference control is not separate from the formal framework—it lives inside it. The same typed, side-conditioned architecture that licenses individual rules also constrains how rules compose at scale. The results fall into two halves:

- **Large-scale dynamics** (§13.1–§13.4): what happens when many inference steps compose—the classical inclusion-exclusion gap and its guard conditions, convergence conditions, pooling, and ordering costs.
- **Inference control** (§13.6–§13.10): how to choose which steps to take—premise selection, coverage bounds, surrogate limits, and the end-to-end composed pipeline.

The chapter is the book's densest. Readers interested primarily in the overlap/multideduction gap may focus on §13.1; those interested in pooling on

§13.3; and those interested in premise selection and ATP connections on §13.6–§13.11.

13.1 The Classical Inclusion-Exclusion Gap in Multideduction

Multideduction—applying multiple inference steps and combining results—is the standard PLN strategy for answering complex queries. When inference rules share evidence, the combined evidence count depends on the overlap structure of the rules' evidence sets. The standard tool for handling such overlaps is *inclusion-exclusion*, a classical identity going back to Sylvester (1883) and formalized in the Bonferroni inequalities. The modern literature on bounding $P(\bigcup_i A_i)$ from partial information is substantial [17, 21, 16]; Prékopa [28] showed that sharp bounds can be computed via linear programming when only low-order intersection statistics are available.

What the PLN book does. The PLN book's multideduction heuristic (§9.3.1 of [11]) is explicit about truncating the inclusion-exclusion series: it uses the first two terms—singleton and pairwise overlap—and adds an empirically estimated correction ξ for the missing higher-order terms, with a "stupidity check" to avoid values above 1. This is best read as a pragmatic low-order approximation strategy under missing higher-order overlap information. The move is analogous in spirit to the one made by naive Bayes classifiers: simplify the dependency structure, accept that the resulting probabilities are not exact, and rely on the downstream decision being robust enough to tolerate the approximation. Domingos and Pazzani [5] showed that naive Bayes classifies well even under badly violated independence, because classification under zero-one loss is far more forgiving than exact probability estimation; Zhang [39] extended this to characterize the conditions under which naive Bayes is optimal. The PLN multideduction heuristic occupies the same neighborhood: approximate evidence combination that may yield useful inference-control decisions even when the combined evidence count is inexact.

Our contribution. What was missing in the PLN treatment—and what the WM calculus adds—is a formal accounting of the gap. We formalize the classical inclusion-exclusion residual in the PLN/WM setting, prove that no universal additive correction ξ can be recovered from first-two-term statistics alone, and connect this fact to *typed guard conditions* for compiled rewrites. The result is not a novel impossibility theorem; it is classical mathematics made operationally precise in the inference-control framework.

Theorem 13.1 (Two-term IE does not determine union cardinality (classical fact, Core))**.** *No function $F : \mathbb{N}^2 \to \mathbb{N}$ recovers the union cardinality $|A \cup B \cup C|$ from the first two inclusion-exclusion terms alone:*

$$T_1 = |A| + |B| + |C|, \qquad T_2 = |A \cap B| + |A \cap C| + |B \cap C|.$$

[1]

Proof. The proof constructs two concrete configurations on $\Omega = \mathrm{Fin}(4)$:

$$A_1 = \{0,1\},\ B_1 = \{0,2\},\ C_1 = \{0,3\} \quad\Longrightarrow T_1 = 6,\ T_2 = 3,\ |A_1 \cup B_1 \cup C_1| = 4$$
$$A_2 = \{0,1\},\ B_2 = \{1,2\},\ C_2 = \{0,2\} \quad\Longrightarrow T_1 = 6,\ T_2 = 3,\ |A_2 \cup B_2 \cup C_2| = 3$$

Same first two terms, different union cardinalities. All equalities are verified by `decide`. □

The constructive resolution uses the residual decomposition:

Theorem 13.2 (Residual decomposition (Core))**.** *For any three events A, B, C:*

$$|A \cup B \cup C| = (T_1 - T_2) + |A \cap B \cap C|$$

The full inclusion-exclusion identity is exact when the triple overlap (the residual) is known. The two-term estimate equals the full union cardinality if and only if $|A \cap B \cap C| = 0$.[2]

The same obstruction and residual decomposition lift from finite cardinalities to probability and evidence-count form, so this is not merely a set-theoretic curiosity: the identical gap appears whenever PLN combines rule-level evidence counts. Both theorems extend to n-way families via indexed generalizations.[3]

[1] Formalized: `no_universal_union_predictor_from_first_two_terms` in `Mettapedia/Logic/PLNInclusionExclusionIdentifiability.lean`

[2] Formalized: `unionCard_eq_twoTerm_plus_residualZ` in `Mettapedia/Logic/PLNMultideductionResidual.lean` The proof works in $\mathbb{Z}$ to handle the subtraction, then lifts to $\mathbb{Q}$-probability form.

[3] `unionCardN_eq_twoTerm_plus_residualNZ` in `Mettapedia/Logic/PLNMultideductionResidual.lean`.

Corollary 13.3 (No universal additive correction (Core)). *There is no constant $\xi \in \mathbb{Z}$ such that $|A \cup B \cup C| = (T_1 - T_2) + \xi$ holds for all event triples. In particular, PLN's Chapter 9 ξ-correction cannot be a universal constant derived from low-order statistics alone.*[4]

In the compiled rule registry, multideduction is therefore certified only under residual-known, residual-bounded, or structurally-zero overlap conditions.

Connection to the overlap regime. The inclusion-exclusion gap is the additive regime's encounter with overlap. When multiple derivation paths share intermediate steps, the union-of-proofs semantics requires an overlap correction—exactly the structure captured by `OverlapWorldModel` (§4.3). The residual ρ_{ij} of the residual decomposition theorem is a concrete instantiation of the `overlap` function. The recovery theorem (Theorem 4.1) says that when paths are independent—no shared intermediate steps—the overlap vanishes and the additive law applies.

Why this matters for inference control. The gap is not about a pathological corner case. It says something operationally important about *multi-step inference*: when you run several inference rules and combine their conclusions, the combined evidence depends on how the rules' evidence sets overlap. The first two inclusion-exclusion terms capture pairwise overlaps ("rules 1 and 2 share sensor s"), but they miss triple and higher-order overlaps ("all three rules share sensor s"). Two configurations can have identical pairwise overlap structure yet different higher-order overlaps—and therefore different true evidence counts.

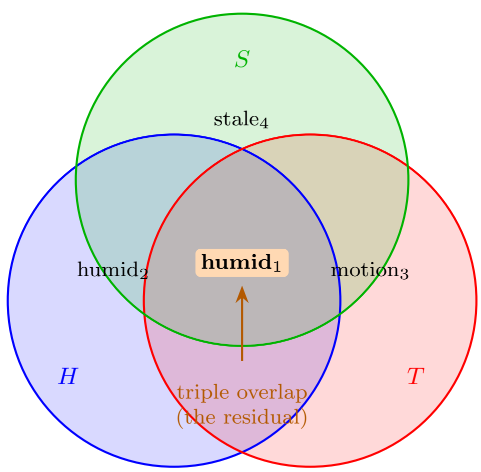
No check based only on singleton and pairwise overlap statistics can detect this. In the WM calculus, this translates to a *guard condition*: a compiled rewrite that combines evidence from multiple rules is admissible only when the residual is known, bounded, or structurally zero. Specifically, the system must either know the triple overlap (the residual), bound it using pairwise information à la Hunter [17] or Kounias [21], or restrict the model class to one where triple overlaps are structurally zero (as in tree-structured BNs, where the junction tree has treewidth ≤ 1).

Maple Court: The building's hallway cleaning robot runs three inference rules to decide which floor needs cleaning most urgently:

[4]Formalized: `no_universal_constant_correction` in `Mettapedia/Logic/PLNMultideductionResidual.lean`.

- **Rule H** (humidity): $H = \{\text{humid}_1, \text{humid}_2\}$.
- **Rule T** (traffic): $T = \{\text{humid}_1, \text{motion}_3\}$.
- **Rule S** (staleness): $S = \{\text{humid}_1, \text{stale}_4\}$.

All three rules share the humid_1 reading. The pairwise overlaps are $|H \cap T| = |H \cap S| = |T \cap S| = 1$ (each pair shares humid_1). The two-term estimate gives $T_1 - T_2 = 6 - 3 = 3$. But the true union is $\{\text{humid}_1, \text{humid}_2, \text{motion}_3, \text{stale}_4\} = 4$ observations. The residual is $|H \cap T \cap S| = 1$ (the shared humid_1), and $3 + 1 = 4$ recovers the truth. Without the correction, the robot undercounts by one observation—exactly the reading that all three rules consumed.



$T_1 = |H| + |T| + |S| = 6$
$T_2 = |H \cap T| + |H \cap S| + |T \cap S| = 3$

Naive: $T_1 - T_2 = 3$ (wrong)

Residual: $|H \cap T \cap S| = 1$
Truth: $3 + 1 = 4$ (correct)

Figure 13.1: The inclusion-exclusion gap in Maple Court. Three inference rules share the humid_1 reading (orange center). Pairwise overlaps are visible, but the triple overlap—the residual—is what the two-term estimate misses.

The diagram makes the geometry clear: the orange center (triple overlap) is subtracted once too many by the two-term formula. The full inclusion-exclusion adds it back:

$$|H \cup T \cup S| = \underbrace{|H| + |T| + |S|}_{T_1=6} - \underbrace{|H \cap T| + |H \cap S| + |T \cap S|}_{T_2=3} + \underbrace{|H \cap T \cap S|}_{\text{residual}=1} = 4$$

Theorem 13.1 says that no function of (T_1, T_2) alone can recover the 4: a *different* configuration with the same $(T_1, T_2) = (6, 3)$ has union cardinality 3 (no triple overlap). The two are indistinguishable from pairwise statistics—a classical fact, but one that bites concretely in multi-rule PLN inference.

Practical consequences. For a PLN system running k inference rules that may share evidence:

- If the rules' evidence sets form a *tree* (each observation appears in at most two rules), the triple overlap is zero and the two-term estimate is exact.
- If the evidence graph has cycles (an observation feeds three or more rules), the residual is nonzero and the system must either compute it or accept the undercount.
- The residual decomposition (Theorem 13.2) gives the exact correction when the triple overlap is known or bounded.
- When the residual is unknown but pairwise statistics are available, classical bounds (Hunter [17], Kounias–Sotirakoglou [21]) or LP-based sharp bounds (Prékopa [28]) provide principled alternatives to PLN's empirical ξ correction. Formalizing these bounds in the WM setting is an open target.
- The practical question is often not "is the combined probability exact?" but "does the approximate combination select the right inference path?" — the same distinction that makes naive Bayes effective for classification despite violated independence assumptions [5, 39]. The WM contribution is to make this tradeoff explicit via typed guard conditions rather than leaving it implicit.

In operational terms, the guarded contract for multideduction is:

1. If the residual is known or structurally zero (tree BN), use the corrected multideduction rewrite—it is exact.
2. If only pairwise data is available, use a bound-carrying rule (Hunter/Kounias/Prékopa) and propagate the interval.
3. Otherwise, mark the result as heuristic or escalate to an exact WM/BN query.

13.2 Trail-Free Protocol Dynamics

A trail-free protocol re-derives conclusions from scratch at each step, without recording which evidence has already been processed. This is the simplest operational strategy—and it can fail to converge.

Theorem 13.4 (Trail-free non-stabilization (Boundary)). *There exists a two-state system where the trail-free deterministic update enters a period-2 cycle and never reaches a fixed point.*[5]

The counterexample is minimal: a protocol step that toggles the truth value. On every even step the state is `true`; on every odd step, `false`. The orbit has period 2 and no fixed point exists (proved: `trailFreeStep_ne_self`). Without a trail, the protocol re-processes the same evidence in opposite order on each step.

The positive result under fresh evidence:

Theorem 13.5 (Damped convergence (Conditional)). *If fresh evidence is eventually available (there exists N such that* $\mathrm{freshAt}(n) = \mathrm{true}$ *for all $n \geq N$), then the damped SP/SPN protocol stabilizes: all states from step $N + 1$ onward equal the fresh-evidence state.*

Under bounded-gap fairness (every window of B steps contains at least one fresh tick), the inconsistency metric hits zero within $B + 1$ steps of any point.[6]

This pair of theorems shows the sharp boundary: trail-free without fresh evidence can diverge; trail-free with eventual fresh evidence always converges. Fresh-evidence injection breaks the cycle: when a new observation arrives, it overwrites the oscillating state, anchoring the orbit to the fresh-evidence value. Without fresh evidence, the same toggling dynamics replay indefinitely.

13.3 Pooling and Fusion

When multiple agents or multiple inference paths produce evidence for the same query, the system must *pool* their outputs. The formalization captures this via a `PoolingOperator` structure with four axioms:

[5]Formalized: `orbit_not_eventually_constant` in `Mettapedia/Logic/PLNTrailFreeDynamicsCounterexample.lean` The counterexample is a single toggle function on `Bool` with no fixed point.

[6]Formalized: `damped_sp_spn_eventually_constant_of_eventual_fresh` and `inconsistency_zero_within_bound_of_boundedGapFresh` in `Mettapedia/Logic/PLNTrailFreeDampedConvergence.lean`

Figure 13.2: Trail-free dynamics: without fresh evidence (left), the protocol oscillates. With fresh evidence injection (right), it converges.

1. Commutativity: $\mathrm{pool}(s_1, s_2) = \mathrm{pool}(s_2, s_1)$.
2. Associativity: pooling is independent of grouping.
3. Two-sided neutrality: pooling with the zero scorer is the identity.
4. External Bayesianity: $\mathrm{pool}(\mathrm{update}(s_1, \ell),\ \mathrm{update}(s_2, \ell)) = \mathrm{update}(\mathrm{pool}(s_1, s_2),\ \ell)$.

One might hope these axioms pin down additive fusion. They do not.

Theorem 13.6 (Pooling non-uniqueness (Boundary))**.** *The max-pooling operator* $\mathrm{pool}(e_1, e_2) = (\max(n_1^+, n_2^+),\ \max(n_1^-, n_2^-))$ *satisfies all four axioms—including external Bayesianity—yet differs from additive fusion.*[7]

The reason: max distributes over coordinatewise multiplication (the update operation), so external Bayesianity holds. But max is not *additive on totals*: $\max(3, 2) \neq 3{+}2$. The additional constraint that recovers uniqueness is *total additivity*: $\mathrm{pool}(x, y).\mathrm{total} = x.\mathrm{total}{+}y.\mathrm{total}$. This is a natural requirement: pooling two independent evidence bodies should give a total observation count equal to the sum of their individual counts.

Theorem 13.7 (Pooling uniqueness under total additivity (Conditional))**.** *External Bayesianity plus total additivity uniquely recovers additive evidence pooling: for all* x, y*,* $\mathrm{pool}(x, y) = x + y$.[8]

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[7]Formalized: `maxPoolingOperator` (satisfies all fields of `PoolingOperator`) and `maxPoolingOperator_ne_fuse` in `Mettapedia/Logic/PremiseSelectionExternalBayesianity.lean`

[8]Formalized: `poolE_eq_hplus_of_externalBayes_totalAdd` and `poolingOperator_pointwise_unique_of_externalBayes_totalAdd` in the same file. The proof uses mask projections to decompose into positive and negative components.

Maple Court: Maple Court's three WM layers instantiate evidence pooling concretely. Each apartment exports evidence summaries—not raw sensor traces—to the building WM, which pools them via additive fusion. The pooling-uniqueness theorem justifies this: under external Bayesianity and total additivity, additive pooling is the unique fusion rule. At the *community scale*, multiple buildings export building-level summaries to a shared district WM. The privacy export theorem guarantees that the community WM cannot reconstruct apartment-level occupancy from the aggregated summaries.

13.4 Order-Cost Governance

When revision is not purely commutative—or when an operational schedule reorders evidence arrival—the system needs bounds on the resulting error. The order-cost framework quantifies this. This is the formal content behind axis 9 (Cost) of the 13-axis vertex family (Chapter 8, §8.5): enabling cost tracking at a vertex adds these order-cost rules to the calculus.

Definition 13.8 (Swap anomaly)**.** The swap anomaly count for states W_1, W_2 at query q under an overlap layer L is

$$\mathrm{SwapAnomaly}_L(W_1, W_2, q) := |\mathrm{obsCount}(L.\mathrm{merge}(W_1, W_2), q) - \mathrm{obsCount}(L.\mathrm{merge}(W_2, W_1), q)|$$

the symmetric count-level discrepancy when the merge order is swapped.

Theorem 13.9 (Order-cost budget bound (Conditional))**.** *If each adjacent swap in a schedule is bounded by a per-step budget, the total schedule error is bounded by the sum of those budgets. Commutative merge-evidence implies zero swap anomaly and zero schedule error.*[9]

Theorem 13.10 (Runtime audit certificate soundness (Conditional))**.** *If runtime pairwise-order checks pass with budget vector B, then the formal schedule-error bound and budget-threshold policy conclusions follow. A single swap-step check certifies the two-step policy threshold.*[10]

[9] Formalized: `scheduleErrorBound_of_pairwise_bounds` and `swapAnomalyCount_zero_of_commutativeMergeEvidence` in `Mettapedia/Logic/PLNWorldModelOrderCostBounds.lean`

[10] Formalized: `runtimePairwiseOrderCheck_certifies_scheduleErrorBound` in `Mettapedia/Logic/PLNWorldModelOrderCostAuditCertificate.lean`

Concrete demos prove that (a) weighted numeric merge has nontrivial non-commutative behavior, and (b) batch swap under additive merge has certified zero order-cost.[11] A provenance-tracked variant using the `Which` semiring demonstrates that right-biased (latest-wins) merge on provenance-tracked states has maximum swap anomaly ($\top$) when one state is zero and the other is nonzero—the provenance structure makes order effects maximally visible.[12]

13.5 Stochastic Experiment and Blackwell Factorization

When choosing *which experiment to run next* (which sensor to query, which feature to observe), the value of an experiment is determined by the Blackwell factorization:

- Expected utility equals the Blackwell factor.
- Optimal value is monotone under Blackwell domination: a more informative experiment is always at least as valuable.
- Trusted-gate transfer preserves the optimality ordering.

This connects the WM calculus to the theory of statistical experiments (Blackwell 1953, Le Cam 1964): the value-of-information computation is a world-model query about the expected gain from evidence not yet observed.[13]

13.6 Selector Specifications

A *selector* chooses which premises to use for the next inference step. The formalization provides typed selector specifications with two robustness properties:

Ranking transfer. If a selector produces a good ranking on one distribution, it produces a good ranking on similar distributions.

[11] Formalized in `Mettapedia/Logic/PLNWorldModelOrderCostWeightedDemo.lean` and `Mettapedia/Logic/PLNWorldModelOrderCostGasPolicyDemo.lean`.

[12] Formalized in `Mettapedia/Logic/PLNWorldModelOrderCostProvenanceDemo.lean` (5 theorems, 0 sorry).

[13] Formalized in `Mettapedia/Logic/PLNWorldModelExperimentStochastic.lean`

Ranking stability. Small perturbations in the knowledge base produce small perturbations in the ranking.

These ensure that a selector trained on one corpus does not catastrophically fail when deployed on a slightly different one—a concern entirely absent from the original PLN book but pressing in any practical system.[14]

Demand-directed backward chaining. An orthogonal approach to premise selection: instead of scoring premises by relevance, propagate *demand* backward from the query through the factor graph. Each premise receives a demand score proportional to its information need (how uncertain it is) and its sensitivity (how much the query answer depends on it). The unknown intermediate Dog(Lassie) in a chain Collie $\to$ Dog $\to$ Mammal gets demand 1000 (maximum: both maximally uncertain and maximally sensitive), while the well-known Collie gets demand 50. After forward supply fills Dog's confidence, demand *contracts* from 1000 to 240 (Proposition 2 of [8]). Forgetting an observation reopens demand at the affected premise—the same `forget` pattern used throughout the evidence algebra.[15] On a fork+chain topology with a hidden common cause ($A \leftarrow H \to B \to C \to D$), the same framework exercises multi-path demand aggregation: when the fork arm A is observed, the chain path carries all the demand; forgetting A activates the fork arm and rebalances demand across both paths.[16]

13.7 Coverage and Submodularity

Let D be the set of true dependencies and S the selected premise set. The coverage objective is $f(S) = |S \cap D|$.

Theorem 13.11 (Coverage submodularity (Core))**.** *The function f is monotone ($S \subseteq T \Rightarrow f(S) \leq f(T)$) and submodular: for $S \subseteq T$ and any element $x \notin T$:*

$$f(T \cup \{x\}) - f(T) \;\leq\; f(S \cup \{x\}) - f(S).$$

[14] Formalized in `Mettapedia/Logic/PremiseSelectionSelectorSpec.lean` and `Mettapedia/Logic/PremiseSelectionRankingStability.lean`

[15] Formalized in `Mettapedia/Logic/WMDemandDemo.lean` (258 lines, 27 theorems, 0 sorry). The demo kernel-checks forward supply, information need, sensitivity, demand ordering ($1000 > 210 > 191 > 50$), demand contraction after supply, and demand reopening after forgetting.

[16] Formalized in `Mettapedia/Logic/WMForkDemandDemo.lean` (23 theorems, 0 sorry).

By the classical result of Nemhauser, Wolsey, and Fisher, the greedy algorithm (iteratively adding the element with maximum marginal gain) achieves a $(1 - 1/e) \approx 0.632$ approximation ratio for monotone submodular maximization under cardinality constraints. The guarantee is formalized via an explicit `GreedyChain` inductive construction.[17]

Maple Court: The hallway cleaning robot must decide which floor to clean first, given limited battery. It queries the building WM for humidity and laundry traffic on each floor. The greedy coverage selector picks the floor with maximum marginal information gain. A Σ-guarded rewrite infers "hallway needs cleaning" from the humidity and traffic evidence, firing only when d-sep conditions are met. The $(1 - 1/e)$ coverage bound guarantees that the robot's greedy floor-selection is near-optimal.

13.8 Coverage Surrogate Boundaries

A natural simplification: replace the joint coverage function with a per-premise score (each premise's individual relevance) and optimize the surrogate. This is impossible in general.

Theorem 13.12 (Coverage/cardinality surrogate no-go (Boundary))**.** *Coverage and cardinality together are insufficient for risk-sensitive selection. There exist premise sets with identical coverage and cardinality but different false-positive risk.*[18]

The counterexample constructs two premise sets that look identical to any local surrogate (same size, same coverage count) but differ in *which* dependencies they hit. One set covers high-risk dependencies; the other covers low-risk ones. The greedy algorithm must evaluate the actual joint coverage function at each step, not a decomposable proxy.

[17]Formalized: `dependencyCoverage_mono`, `dependencyCoverage_insert` (diminishing returns), and `GreedyChain` in `Mettapedia/Logic/PremiseSelectionCoverage.lean`

[18]Formalized: `Mettapedia/Logic/PremiseSelectionCoverageCounterexamples.lean`

13.9 BRGI and Feasible Sets

The Balanced Relevance-Guided Inference (BRGI) framework provides an optimization-theoretic perspective on inference scheduling:

- Finite feasible sets with policy-sensitive optimum characterization.
- Graph-level BRGI objects with optimality-preserving refinement.

BRGI connects the WM calculus to the combinatorial optimization literature: the inference-control problem is a submodular optimization problem on a graph of possible inference steps.[19]

13.10 The Composed End-to-End Pipeline

The final result composes the entire pipeline into a single correctness theorem: selector → rewrite → threshold.

Theorem 13.13 (End-to-end pipeline correctness (CONDITIONAL))**.** *Given a selector that produces a ranked premise set, a compiled rewrite that applies a PLN rule under BN side conditions, and a threshold that accepts the result, the composed pipeline is correct: if the selector has sufficient coverage and the side conditions are met, the accepted result agrees with the full posterior query up to the distributional approximation bound of Chapter 12.*[20]

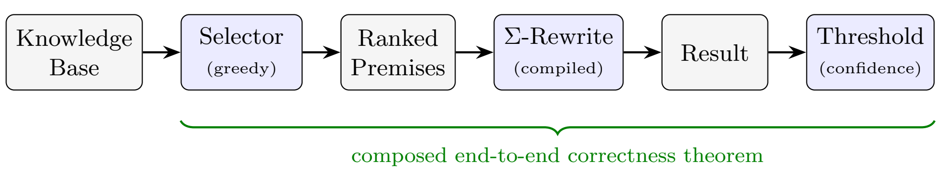


Figure 13.3: The inference-control pipeline. A selector chooses premises; a Σ-guarded compiled rewrite produces a result; a threshold accepts or rejects. The composed pipeline has a single end-to-end correctness theorem.

This is the technical endpoint of the entire compiled-inference story. The calculus begins with a correct kernel (Part II), develops exact shortcuts under side conditions (Chapters 10–11), quantifies the cost of approximation

[19]Formalized in `Mettapedia/Logic/PremiseSelectionBRGI.lean`

[20]Formalized: `ch9_selector_rewrite_threshold_end_to_end_of_interval` in `Mettapedia/Logic/PLNCanonicalAPI.lean`

(Chapter 12), and now composes selection, execution, and acceptance into a single pipeline with a single correctness proof.

The pipeline diagram (Figure 13.3) should be read right-to-left as a specification: *what does the system guarantee?* And left-to-right as an implementation: *how does the system compute?* Both readings are justified by the same theorem.

13.11 Connections to Automated Theorem Proving

The coverage objective $f(S) = |S \cap D|$ and the greedy $(1 - 1/e)$ approximation are not specific to PLN. They are the same structures that arise in *premise selection* for interactive and automated theorem provers—the problem of choosing which library lemmas to feed to a prover given a goal.

The connection is exact. The MaSh (Machine-learning for Sledgehammer) framework uses k-nearest-neighbor and naive-Bayes classifiers to rank premises. Both are world-model instances (Chapter 11, §11.7): Naive Bayes is a BN fork, k-NN is evidence-weighted revision from the k nearest neighbors. The formalization makes this explicit: a k-NN premise selector IS an additive world-model query, and the coverage bound applies to it.[21]

Three questions that this bridge raises for the ATP community:

1. **Completeness under budget.** The $(1-1/e)$ bound guarantees near-optimal coverage for a fixed cardinality budget k. But ATP premise selection operates under iterative deepening: if the first k premises fail, the prover tries $2k$, then $4k$. Does the formal coverage bound compose across rounds? The submodularity of f ensures diminishing returns, but the interaction with prover timeouts is not yet formalized.

2. **Learned approximations.** The formal bound assumes access to the oracle gain function $\mathrm{dependencyGain}(D, S, a)$. A learned selector (neural, k-NN, gradient-boosted) approximates this gain. The ranking-stability theorem (§13.6) bounds the degradation under small perturbations, but characterizing the full approximation gap for specific learners is open.

3. **Runtime cost.** Evaluating dependencyGain at each greedy step costs $O(|D|)$ per candidate. For a library of 10^5 premises and a budget

[21] Formalized in `Mettapedia/Logic/PremiseSelectionKNN.lean` and `Mettapedia/Logic/PremiseSelectionKNN\_PLNBridge.lean`.

of $k = 500$, the greedy chain requires $\sim 5 \times 10^7$ gain evaluations. Approximate gain via cached features reduces this to $O(k \cdot |D|)$ with a constant-factor overhead from feature extraction.

These connections are not speculative: the project includes experimental evaluation of PLN-based premise selectors against MaSh baselines on the extended MPTP 2078 benchmark (5,049 problems), with results competitive with k-NN at lower premise counts.[22]

[22] See the `atps/` directory for scripts, datasets, and evaluation results.

Part IV

Additional Semantic Families

Chapter 14

Intensional Inheritance

Two concepts A and B may be related in two ways. *Extensional* inheritance measures overlap of instances: $P(B \mid A) = n^+_{AB}/n_A$. *Intensional* inheritance measures shared properties: how much does knowing A's property profile tell you about B? Classical PLN treats these as separate channels with a scalar mixing parameter. The formalization shows that this is both less and more than what the mathematics supports: the scalar mixture is refuted, but a clean information-theoretic unification replaces it. The result connects to the evidence carriers of Chapter 5 and the typed world-model grounding of Chapter 8.

14.1 The Information-Theoretic Unification

Goertzel (2025) gives the unifying formula:

$$P(W \mid F) = P(W) \cdot 2^{I(F;W)}$$

where $I(F;W) = \sum_{f,w} p(f,w) \log_2 \frac{p(f,w)}{p(f) \cdot p(w)}$ is the Shannon mutual information between the feature profile F and the target concept W. Extensional inheritance is the left side; intensional inheritance is the exponent on the right.[1]

Theorem 14.1 (Singleton reduction (Core))**.** *When every property is a singleton (each property picks out exactly one element), the mutual information* $I(F;W)$ *equals* $\log_2(P(W \mid F)/P(W))$*, and the Goertzel formula*

[1]Formalized: `goertzel_formula` in `Mettapedia/Logic/IntensionalInheritance.lean`

reduces to the tautology $P(W \mid F) = P(W \mid F)$. Intensional inheritance collapses to extensional inheritance when concepts have no internal structure.[2]

The philosophical point (Goertzel): "creatures with hearts" and "creatures with kidneys" seem like different concepts (different intensions), but they pick out the same class (same extension) because hearts and kidneys co-evolved. The information-theoretic view captures this: $I(\text{heart}; \text{kidney}) \approx 1$.

Maple Court: In Maple Court, apartments share structural properties: kitchens, bathrooms, plumbing risers, the same building infrastructure. Intensional similarity between Apartment 3B and Apartment 5C is high—their property profiles overlap almost completely (both are west-facing corner units with similar heat and humidity patterns). But extensional inheritance diverges: Apartment 3B has a pipe leak while 5C does not. The typed WM grounding resolves this: extensional queries use per-apartment evidence, while intensional queries aggregate over shared property patterns via the ASSOC channel.

14.2 Typed WM Grounding

The WM approach grounds intensional inheritance via a typed query family:

Definition 14.2 (Inheritance sort)**.**

$$\text{InheritanceSort} = \text{extensional} \mid \text{intensionalAssoc} \mid \text{intensionalPAT} \mid \text{mixed}$$

Each sort has its own evidence extraction and strength computation.[3]

The ASSOC channel computes association strength via mutual information: $P_{\text{ASSOC}}(W \mid F) = P(W) \cdot 2^{I(F;W)}$. High ASSOC inheritance means knowing which properties A has gives strong predictive power for which properties B has. The PAT channel provides a complementary structural route based on property-pattern proximity beyond what mutual-information association captures.

[2]Formalized: `singleton_reduction` in the same file.
[3]Formalized in `Mettapedia/Logic/PLNIntensionalWorldModel.lean`

14.3 Solomonoff Bridge

The Solomonoff bridge connects intensional inheritance to algorithmic information theory via the universal-mixture log-ratio:

$$I_{\mathrm{int},\xi}(W, F, x) = \log_2 \frac{P_\xi(W \mid F, x)}{P_\xi(W \mid x)}$$

when both conditionals are positive. Specialized `xiSemimeasure` and geometric-mixture versions are proved as explicit theorem endpoints.[4]

14.4 Mixed Semantic Packaging

The original PLN book proposes a simple scalar mixture: $\mathrm{InhInt}(A, B) = \alpha \cdot \mathrm{Ext}(A, B) + (1 - \alpha) \cdot \mathrm{IntAssoc}(A, B)$. The formalization explicitly refutes this:

Theorem 14.3 (No binary mixed policy (Boundary)). *There exists a concrete state with evidence scores* $(3, 2, 1)$ *for (extensional, ASSOC, PAT) channels such that the PAT channel contributes nontrivially and the mixed result cannot be expressed as any function of only the extensional and ASSOC values.*[5]

The book's binary mixture is not merely incomplete but false as a general law: the full three-channel model (extensional, ASSOC, PAT) is needed. The corrected surface is a typed three-channel mixed rule with selector-specialized endpoints.[6]

Intensional inheritance is the first semantic family that extends the query language rather than the state or the carrier. The same state and the same evidence algebra serve both extensional and intensional queries; only the `InheritanceSort` index changes. This pattern—extending the calculus by enriching the query type—recurs in the temporal and normative families that follow.

[4]Formalized: `intensionalFromConditional_eq_log2_ratio` and `intensionalFromXiGeom_eq_log2_ratio` in `Mettapedia/Logic/IntensionalInheritanceSolomonoffBridge.lean`

[5]Formalized in `Mettapedia/Logic/PLNIntensionalCanary.lean`

[6]Formalized in `Mettapedia/Logic/PLNIntensionalAssocPatClosure.lean`

Chapter 15

Temporal and Causal Inference

The world-model calculus so far is atemporal: the state records evidence about the world *now*. This chapter extends the framework to time-indexed states and temporal queries, and shows that the extension has a clean algebraic structure—the temporal quantale—that yields temporal transitivity as a free theorem.

15.1 Temporal Operators

Temporal predicates live in the type $\mathrm{TemporalPred}(D, T) = D \to T \to \mathrm{Prop}$, where D is the domain and T is the time type. Three layers of operators build the temporal algebra:

Definition 15.1 (Simultaneous operators (Core)).

$$\begin{aligned} \mathrm{SimAND}(P, Q)(x, t) &:= P(x, t) \wedge Q(x, t) \\ \mathrm{SimOR}(P, Q)(x, t) &:= P(x, t) \vee Q(x, t) \end{aligned}$$

Both predicates hold (or at least one holds) at the same moment.

Definition 15.2 (Sequential operators (Core)).

$$\begin{aligned} \mathrm{SeqAND}(P, Q)(x, t) &:= P(x, t) \ \wedge \ \exists \Delta > 0.\ Q(x, t + \Delta) \\ \mathrm{SeqAND}_f(P, Q)(x, t) &:= P(x, t) \ \wedge \ \exists \Delta.\ f(\Delta) \wedge Q(x, t + \Delta) \end{aligned}$$

where f constrains the delay (e.g., $\Delta \in [1, T_{\max}]$).

Definition 15.3 (Predictive implication (CORE))**.**

$$\mathrm{PredImpl}(P, Q)(x, t) := P(x, t) \;\Rightarrow\; \exists \Delta > 0.\; Q(x, t + \Delta)$$

The 3-node predictive chain composes:

$$\mathrm{PredChain}_3(T_1, T_2, A, B, C) := \mathrm{SeqAND}_{T_1}\big(A,\; \mathrm{SeqAND}_{T_2}(B, C)\big).$$

[1]

15.2 The Temporal Quantale

The Boolean skeleton above lifts to a quantale-valued setting with a clean algebraic structure.

Definition 15.4 (Temporal quantale (CORE))**.** A temporal quantale is a commutative quantale Q equipped with a group action $\varphi : T \to (Q \to Q)$ by quantale automorphisms:

- $\varphi_0 = \mathrm{id}, \quad \varphi_{t+t'} = \varphi_t \circ \varphi_{t'}$
- Each φ_t preserves $*$, $\bot$, $\top$, and $\bigsqcup$
- Since T is a group, φ_t is invertible (φ_{-t} is the inverse), hence an automorphism

[2]

The key payoff: automorphisms preserve the product and the lattice order, so they preserve residuation (implication), which is definable from both. Temporal transitivity is therefore a *free theorem*—no additional axioms are needed:

Theorem 15.5 (Temporal transitivity (CORE))**.** *For any $a, b, c \in Q$ and time offsets $t_1, t_2 \in T$:*

$$(a \Rightarrow \varphi_{t_1} b) \;*\; \varphi_{t_1}(b \Rightarrow \varphi_{t_2} c) \;\le\; (a \Rightarrow \varphi_{t_1+t_2} c)$$

Temporal modus ponens is the special case $c = \top$: $a \;*\; (a \Rightarrow \varphi_t b) \;\le\; \varphi_t b$.[3]

In PLN terms: Q represents evidence values, $*$ is conjunction under independence, $\Rightarrow$ is implication strength, and φ_t shifts evidence across time. The theorem says chained temporal implications compose without loss of algebraic structure.

[1] Formalized in `Mettapedia/Logic/PLNTemporalCausalInference.lean`

[2] Formalized: `TemporalQuantale` class in `Mettapedia/Logic/TemporalQuantale.lean`

[3] Formalized: `temporal_transitivity` and `temporal_modus_ponens` in the same file.

15.3 Causal Profile and Event Calculus

Causal inference is grounded through a probabilistic event calculus built on four predicates: Happens(e, t), Initiates(e, f, t), Terminates(e, f, t), and HoldsAt(f, t). For a pair of predicates (A, B), the *causal profile* collects predictive implication strengths across temporal lags:

$$\text{CausalProfile}(A, B) = \left\{\text{strength}(\text{PredImpl}_\Delta(A, B)) \mid \Delta \in [1, T_{\max}]\right\}$$

[4]

15.4 Distinguishing Cause from Correlation

A high predictive-implication strength does not by itself establish causation. The formalization introduces a precise discriminator using the intensional/extensional distinction from Chapter 14:

Definition 15.6 (Predictive causal profile)**.** A `PredictiveCausalProfile` for a pair (A, B) bundles two temporal predicates:

- `extensional`: the temporal correlation holds at (x, t) — A at time t is followed by B at some future time (PredImpl evidence is positive).
- `intensional`: the property-level mechanism holds at (x, t) — the shared property profile between A and B has high mutual information (the ASSOC channel from Chapter 14).

The profile is *spurious* at (x, t) if the extensional component holds but the intensional component does not.

Theorem 15.7 (Intensional grounding excludes spurious causation (Core))**.** *If the intensional component of the causal profile holds at (x, t), then the candidate is not spurious.*[5]

This connects Chapters 14 and 15: intensional inheritance provides the mechanism check that temporal correlation alone cannot. Granger-style temporal precedence plus intensional grounding separates $A \to B$ (causal) from $A \leftarrow C \to B$ (confounded).

[4] Formalized in `Mettapedia/Logic/PLNTemporalCausalInference.lean` and `Mettapedia/Logic/PLNProbabilisticEventCalculus.lean`

[5] Formalized: `not_spurious_of_intensional` in `Mettapedia/Logic/PLNTemporalCausalInference.lean`.

15.5 Temporal as Another WM Instance

The key design: temporal and causal inference is *not* a separate ontology. It is another instantiation of the world-model framework. The state includes temporal structure (event histories); the query language includes temporal operators (SimAND, SeqAND, PredImpl); extraction and revision operate unchanged. This keeps the foundational theory unified: a single kernel, many instantiations.[6]

Maple Court: The chain PipeLeak $\rightarrow$ WallHumidity $\rightarrow$ MoldRisk has temporal structure: a leak detected at time t predicts elevated humidity at $t + 2\text{h}$ (pipe water takes time to reach the wall cavity) and mold risk at $t + 48\text{h}$ (mold colonies need sustained moisture). The temporal quantale's transitivity theorem composes these lags: evidence for leak-to-mold risk at $t+50\text{h}$ follows from the two per-link implications without additional axioms. The causal profile distinguishes this from the spurious correlation "mold risk is high when the shower is running"—the shower correlation has extensional support (temporal co-occurrence) but weak intensional support (showering does not cause wall-cavity moisture via the same mechanism).

The temporal-deontic bridge (Chapter 16) extends this to normative reasoning: obligations persist through time under temporally uniform accessibility.

[6]The Kalman sleep demo (Chapter 21) provides the concrete formal grounding for temporal filtering: batch replay of deferred observations yields the same posterior as sequential online processing.

Chapter 16

Normative and Governance Reasoning

Chapters 14–15 extended the calculus to intensional properties and temporal structure. This chapter extends it to *norms*: obligations, permissions, and compliance. The key insight is the same as for temporal reasoning—normative queries are world-model queries, and deontic evidence is graded, not binary.

16.1 Norms as a Semantic Family

A normative world model carries a state that includes which norms are in force and what events have occurred. Queries ask whether an action is obligatory, permitted, or optimal. Extraction returns *deontic evidence*—a graded measure of how much support the normative state provides for the obligation.

The crucial departure from classical deontic logic: the answer to "Is this obligatory?" is not a Boolean. It is an evidence pair (n^+, n^-)—the same carrier used throughout the calculus. An obligation with evidence $(5, 1)$ has strength 0.83 and confidence 0.38; an obligation with evidence $(50, 10)$ has the same strength but much higher confidence. The distinction matters: a high-evidence obligation warrants enforcement; a low-evidence one warrants investigation.

16.2 The Deontic Traditional Scheme

The logical structure of norms is formalized separately from their evidence grading. At the logical level, the Deontic Traditional Scheme (DTS) defines the relationships between obligation, permission, and optimality in the classical Boolean way. At the evidence level, each of these operators returns graded evidence rather than a truth value. The two levels compose: the DTS determines *which* queries are well-formed; the WM extraction determines *how much* evidence supports them.

In the DTS, obligation (**ob**) is primitive and permission (**pe**) and optimality (**op**) are derived:

$$\mathrm{pe}(\varphi) := \neg\,\mathrm{ob}(\neg\varphi), \qquad \mathrm{op}(\varphi) := \mathrm{ob}(\varphi) \wedge \mathrm{pe}(\varphi)$$

All 12 interdefinability axioms of classical deontic logic are proved as *theorems*, not assumed as axioms.[1]

Eventualities are reified following Hobbs (1985): an eventuality has participants with ISO 24617-4 thematic roles (agent, patient, beneficiary, instrument, ...) and can be the target of deontic operators.

16.3 Deontic Queries as WM Queries

The `DeonticQueryEncoder` maps deontic queries to world-model queries:

```
structure DeonticQueryEncoder (Entity Pred Query) where
  modalQuery  : DeonticModality -> Eventuality Entity Pred ->
      Query
  groundQuery : CTTriple Entity Pred -> Query
```

The `modalQuery` encodes "what is the evidence that eventuality e has modality m?" The `groundQuery` encodes "what is the evidence for ground triple t?" Both return WM queries that can be answered by the standard extraction machinery.[2]

Evidence extraction is additive per modality: $\mathrm{modalEvidence}(W_1{+}W_2, m, e) = \mathrm{modalEvidence}(W_1, m, e) + \mathrm{modalEvidence}(W_2, m, e)$.[3]

[1] Formalized in `Mettapedia/Logic/GovernanceReasoning/Core.lean`. The full DDL+ shallow embedding (Carmo-Jones, 2002) with Kaplanian two-dimensional semantics is in `Mettapedia/Logic/DDLPlus/Core.lean` and `Mettapedia/Logic/DDLPlus/Theorems.lean`.

[2] Formalized in `Mettapedia/Logic/GovernanceReasoning/Bridge.lean`. The bridge imports `PLNWorldModelCalculus` and uses `BinaryWorldModel` and `WMRewriteRule`.

[3] Formalized: `modalEvidence_add` in the same file.

16.4 Evidence-Graded Actuality

Classical deontic logic assumes binary truth: an eventuality either "really exists" or it does not. The Hobbs `rexist` modality captures this. The formalization replaces binary truth with *evidence-graded actuality*: `rexist` becomes an eventuality-occurrence evidence channel, returning a `BinaryEvidence` pair (n^+, n^-).

Theorem 16.1 (Acceptance bridge soundness (CONDITIONAL)). *For any acceptance policy π, if the `RexistBridge` holds (evidence equality between occurrence and projection channels), then the `AcceptanceBridge` holds: occurrence is accepted if and only if the corresponding projection is accepted.*[4]

Categorical "actuality for governance" is a downstream acceptance policy, not certainty-1 truth. This avoids the lottery paradox that arises from naive fixed-threshold approaches. The threshold semantics of Chapter 12 applies directly: the acceptance policy IS a threshold on deontic evidence.

16.5 The Temporal-Deontic Bridge

Obligations persist through time. Under a temporally uniform obligation structure—where the accessibility relation is translation-invariant—the obligation content persists as time advances:

Theorem 16.2 (Obligation persistence (CONDITIONAL)). *Under temporally uniform obligation (shift-invariant accessibility), ideal obligations persist: if* $\mathrm{ob}(\varphi)$ *holds at time t, then* $\mathrm{ob}(\varphi^{-\Delta})$ *holds at time $t+\Delta$, where $\varphi^{-\Delta}$ is the content interpreted relative to base time.*[5]

This connects Chapter 15 to the present chapter: the temporal quantale's shift operation is the same operation that transports obligations forward in time.

Maple Court: The building code says "the steward shall inspect hallway plumbing quarterly." The system encodes this as a deontic query: ob(inspect-plumbing). At $t = 0$ (January), extracting this query from the

[4] Formalized: `acceptanceBridge_of_rexistBridge` in `Mettapedia/Logic/GovernanceReasoning/ActualityPolicy.lean`.

[5] Formalized: `ideal_obl_temporal_persistence` in `Mettapedia/Logic/TemporalDeonticBridge.lean`

normative WM state yields evidence $(3, 1)$: three regulatory citations supporting the obligation and one waiver (the ground-floor plumbing was replaced last year). Strength = 0.75, confidence = 0.29 (with $k = 10$). The building's acceptance policy triggers enforcement when $s > 0.5$ and $c > 0.25$: this obligation is accepted.

The temporal-deontic bridge ensures that this obligation persists into the next quarter. If the steward does not act, the obligation evidence at $t+90\mathrm{d}$ is unchanged—the shift $\varphi_{90\mathrm{d}}$ transports the obligation content forward but does not diminish it. Discharging the obligation requires evidence of inspection (revision with compliance evidence), not just the passage of time.

Part V

Bridges and Scope

Chapter 17

Bridges to Neighboring Frameworks

The world-model calculus is not one more framework competing with Markov Logic Networks, ProbLog, NARS, or process calculi. It is the *typed evidence layer* that each of these frameworks implicitly uses. Making that layer explicit—with revisable states, compositional extraction, explicit side conditions, and kernel-checked proofs—is what the calculus contributes.

Every bridge in this chapter has the same shape:

1. Identify the neighboring framework's natural *state*, *query*, and *value* types.
2. Construct a `WorldModel` (or `AdditiveWorldModel`) instance from them.
3. Prove that WM extraction agrees with the framework's native inference.
4. State the side condition under which the bridge is exact.

The pattern is a functor from the neighboring framework into the WM interface. The side conditions are what make each bridge exact. Every bridge below is formalized in Lean 4 with zero sorry.

17.1 Markov Logic Networks

A Markov Logic Network [29] is a set of weighted first-order clauses. Each clause c_i has a real-valued weight w_i; the probability of a possible world ω is proportional to $\exp(\sum_i w_i \cdot n_i(\omega))$, where $n_i(\omega)$ counts the true groundings of clause i in ω.

Framework	State	Query	V	Bridge type
MLN	clause-weight vector	ground atom	$\mathbb{R}_{\geq 0}$	exact (compatibility-gated)
ProbLog	annotated-fact set	ground query	$[0, 1]$	exact ([FIN])
NARS	(f, c) pairs	term	(f, c)	Galois connection
PLN v0.9	(s, c) pairs	link	(s, c)	subsumption
Classical FOL	theory	sentence	$\{0, 1\}$	degenerate instance
Giry monad	probability measure	measurable set	$[0, 1]$	categorical
ρ-calculus	process term	name	behav. equiv.	structural congruence
Ontology repair	evidence ledger	candidate	`BinEvNat`	additive aggregation

Table 17.1: Bridge matrix: how each neighboring framework maps to the WM interface.

The WM instance. A ground MLN with n probabilistic facts defines a world-model instance: the state is the vector of grounding counts (or, equivalently, the clause-factor contributions); the query is a ground atom; and the value type is binary evidence (n^+, n^-) accumulated across possible worlds. The key equation:

Theorem 17.1 (MLN–WM strength agreement (CONDITIONAL))**.** *For a ground MLN with finite support,*

$$\mathsf{strength}(\mathrm{extract}_{\mathrm{WM}}(W, q)) = P_{\mathrm{MLN}}(q)$$

where the left side is the WM strength view and the right side is the MLN marginal probability.[1]

The side condition is finite support (the grounding set must be finite). Under this condition, the MLN marginal probability IS the WM strength view—not an approximation, but an identity.

Why infinite MLNs matter. Many relational systems have no natural finite boundary. A social network grows as new users join; a sensor grid expands as new devices are deployed; an AGI knowledge base accumulates concepts without a fixed ontology. In each case the underlying Markov logic

[1] Formalized: `wm_queryStrength_eq_full_queryProb_of_finite_support` in `Mettapedia/Logic/MarkovLogicWorldModel.lean`.

specification—a set of weighted first-order clauses—is well-defined, but the standard finite normalization $Z = \sum_\omega \exp(\sum_j w_j f_j(\omega))$ becomes a sum over an infinite Boolean product space and may diverge. The question is: *does the specification still define a probability distribution?*

The answer, due to Dobrushin, Lanford, and Ruelle, is that one must replace the partition-function viewpoint with the language of Gibbs measures. A *DLR measure* μ satisfies local consistency equations: for every finite region Λ, the conditional of μ given the exterior agrees with the finite-volume Gibbs kernel on Λ. Existence of such measures follows from compactness arguments; uniqueness is a deeper property that depends on the interaction strength.

The formalization certifies both sides of this story. For *finite-support* MLNs the partition function is finite and the WM bridge is an exact identity (Theorem 17.1). For *countably infinite* MLNs the semantics becomes a Gibbs/DLR measure problem, and the formalization proves existence, uniqueness under a uniform contraction condition, and a bridge theorem connecting the Gibbs probability back to WM-PLN query strength.

A caveat before the theorems: the formalization certifies *well-posedness*, not closed-form computation. For a concrete weight such as $w = 0.4$, it proves that $P(\text{Believes}(42) = \mathsf{true})$ exists and is unique, and that the WM query strength is unambiguous. It does *not* symbolically compute that number; obtaining a numerical value requires Gibbs sampling, mean-field approximation, or a similar inference scheme. The formalization certifies the *target* of computation — that it is well-defined — but leaves the approximation algorithm to the runtime.

The first of the three infinite-case theorems establishes that the Gibbs specification always admits at least one global probability measure.

Theorem 17.2 (Infinite MLN existence on countable domains (Conditional))**.** *Fix a countable atom type presented by* $e : \mathbb{N} \simeq \text{Atom}$*, an exhaustion of finite regions, and a boundary condition* ξ*. Then a classical infinite MLN induces a probability measure* μ *satisfying the fixed-region cylinder DLR equations for the associated strictly positive Gibbs specification.*[2]

Existence alone does not settle the semantics: an MLN may admit multiple Gibbs measures with distinct marginals (this is exactly the phenomenon of *phase transitions* in statistical mechanics). The next theorem gives a

[2]Formalized as `exists_fixedRegionCylinderDLR_of_equiv` in `Mettapedia/Logic/MarkovLogicInfiniteUniqueness.lean`, transferring the compactness packaging from `Mettapedia/Logic/MarkovLogicInfiniteExistence.lean` (0 sorry in the current slice).

sufficient condition for uniqueness. The condition is Dobrushin's (1968): if the total interaction arriving at every atom is uniformly bounded below the critical threshold, the specification admits at most one Gibbs measure.

Theorem 17.3 (Infinite MLN uniqueness under uniform small total influence (CONDITIONAL))**.** *If an infinite classical MLN satisfies a uniform Dobrushin bound*

$$\exists\, C \in [0,1) \;\; \textit{such that} \;\; \forall a,\; \sum_b C(a,b) \;\leq\; C,$$

where $C(a,b) = \frac{1}{2}\sum_{j:\, a,b\in\text{clause}_j} |w_j|$ *is the pairwise Dobrushin coefficient, then it has at most one fixed-region cylinder DLR probability measure.*[3]

What this means intuitively. Every atom in an infinite knowledge base participates in some clauses that connect it to other atoms. The Dobrushin coefficient $C(a,b)$ measures how much flipping atom b's truth value can shift atom a's probability; the row sum $\sum_b C(a,b)$ is the *total incoming influence* on a from the rest of the universe. The condition $\sum_b C(a,b) \leq C < 1$ says: *every atom is more determined by its own prior than by the combined pull of all its neighbors.* No atom is a slave to its context.

When this holds, local disagreements cannot amplify as they propagate through the interaction graph—they *attenuate*. Two hypothetical global measures that disagree somewhere must agree everywhere, because any local discrepancy shrinks geometrically as it crosses each layer of the network.

For the modeler, the theorem provides an *interaction-strength budget*. As long as the total clause weight arriving at each atom stays below the threshold, the semantics of the infinite knowledge base is unambiguous: every finite query has exactly one answer, and that answer is robust to the boundary conditions at infinity. Exceeding the budget does not necessarily break the system, but it places the semantics beyond what the current theorem can guarantee—the possibility of multiple coexisting equilibria (phase transitions) opens up.

Once a DLR measure exists, it can be fed into the WM-PLN bridge. The construction is elementary: a probability measure has total mass 1, so the `MassSemantics` record from the WM interface is immediate, and the query strength reduces to the Gibbs probability of the query's cylinder event.

[3] Formalized as `paperUniformSmallTotalInfluence_implies_paperUniqueMeasure` in `Mettapedia/Logic/MarkovLogicInfiniteUniqueness.lean` (0 sorry in the current slice). The proof uses a descending-shell contraction argument on iterated interaction neighborhoods in the classical Dobrushin–Georgii style.

Theorem 17.4 (Infinite MLN world-model bridge (CONDITIONAL))**.** *Let μ be a fixed-region cylinder DLR measure for an infinite classical MLN and let q be a finite constraint query. Then the induced WM state has*

$$\mathrm{queryStrength}(\mu, q) \;=\; \mu\big(\{\omega \mid \omega \textit{ satisfies } q\}\big).$$

Under the same uniform Dobrushin hypothesis as Theorem 17.3, this query strength is uniquely determined by the MLN specification M and does not depend on the choice of DLR measure.[4]

Together, the three infinite-case theorems establish the right abstraction shift: exact probabilities in the finite case, well-posed Gibbs semantics in the infinite case, and a clean return map from Gibbs measures back into the algebraic WM interface.

The formalization chain. The infinite story is certified across twelve sorry-free files in three layers:

Theory. `MarkovLogicInfiniteExistence.lean` (DLR existence via compactness), `MarkovLogicInfiniteUniqueness.lean` (Dobrushin uniqueness via descending-shell contraction, 6,758 lines).

Bridge. `MarkovLogicInfiniteWorldModel.lean` (Gibbs → `MassSemantics` → `queryStrength`), `MarkovLogicInfiniteBoundaryStability.lean` (geometric shell decay), `MarkovLogicOntologyGrowth.lean` (cross-specification exact invariance), `MarkovLogicIndividuation.lean` (individuated subsystems and the capstone WM-stability theorem), `MarkovLogicCoupledSubsystem.lean` (two local cores sharing one protected carrier), and `MarkovLogicDynamicCoupledSubsystem.lean` (geometric drift bounds for coupled carriers under distant rewriting).

Examples. Belief line (1D chain), 2D grid (Ising lattice), knowledge ladder, world of views (unidirectional, bidirectional, and variable-neighborhood variants), the trust-triangle capstone, the coupled-communities carrier example, and self-transcendence paths, all reusing the same pipeline.

Compatibility-gated evidence roles. The more interesting result is not the identity but the *decomposition*. An MLN clause weight is a single number; the WM calculus decomposes it into interpretable evidence roles (how

[4]Formalized as `infiniteMLN_queryStrength_eq_gibbsProb` and `infiniteMLN_queryStrength_unique_of_uniform` in `Mettapedia/Logic/MarkovLogicInfiniteWorldModel.lean` (0 sorry in the current slice).

much the clause contributes to each query, weighted by grounding structure). The compatibility gate checks whether a clause's evidence is compatible with the BN structure—if so, the compiled BN rewrite applies; if not, the evidence is quarantined.[5]

Worked example: UW-CSE advisedBy prediction. The UW-CSE academic network benchmarks link prediction: given a student and a professor, predict `advisedBy(student, professor)`. Four evidence roles—each a `SourceItem` in the `EvidentialLedger`—contribute:

NeedAdvisor(s) Unary student readiness: phase, years, publications, TA experience. Contributes $\langle 3, 0\rangle$ for advisedBy.

CanAdvise(p) Unary professor capacity: position, publications, courses, advisees. Contributes $\langle 3, 0\rangle$.

PairAffinity(s,p) Pairwise interaction: shared publications, shared courses, temporary advising. Contributes $\langle 4, 0\rangle$—the strongest signal.

MLN logit Clause-weight signal from ground MLN inference. Contributes $\langle 2, 2\rangle$ (balanced).

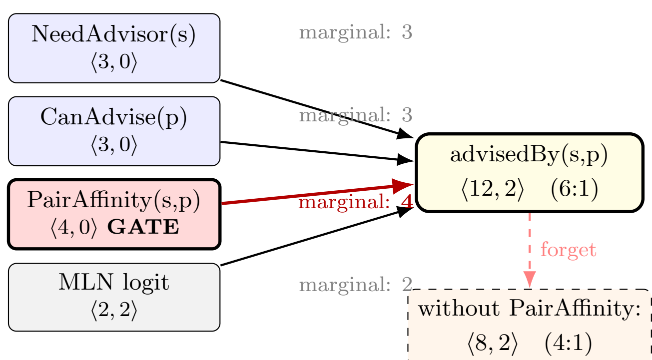


Figure 17.1: UW-CSE advisedBy evidence roles. PairAffinity (red, marginal contribution 4) is the compatibility gate: without it, the prediction drops from 6:1 to 4:1. Strong unary signals alone are insufficient.

With all four roles, the aggregate is $\langle 12, 2\rangle$ for advisedBy—a 6:1 ratio. The compatibility gate operates via `forget`: removing PairAffinity drops the aggregate to $\langle 8, 2\rangle$ (4:1). The marginal contributions—PairAffinity (4) >

[5] Formalized across 13 files: `Mettapedia/Logic/MarkovLogic*.lean` (0 sorry).

NeedAdvisor (3) = CanAdvise (3) > MLN logit (2)—show that pairwise interaction evidence is the critical discriminator.

Strong unary signals alone—a promising student and an active professor—are insufficient for confident edge prediction without pairwise evidence that they actually interact. This is the same `forget` pattern that identified IASP as the critical discriminator in the Pain reclassification (§6.5.3) and stale reliability weights as the bottleneck in GJP forecasting (§21.5).[6]

Full benchmark results. On the full UW-CSE dataset (5-fold leave-one-area-out), the compatibility-gated model improves over the MLN baseline on all metrics:

	AUROC		Hits@1		Set-F1	
Fold	MLN	Gate	MLN	Gate	MLN	Gate
ai	0.845	0.871	0.433	0.433	0.389	0.411
graphics	0.884	0.932	0.071	0.357	0.048	0.381
language	0.871	0.893	0.375	0.375	0.333	0.333
systems	0.845	0.872	0.444	0.593	0.395	0.531
theory	0.866	0.916	0.500	0.417	0.417	0.389
Mean	0.862	**0.897**	0.365	**0.435**	0.316	**0.409**

Table 17.2: MLN baseline vs. compatibility-gated WM model (leave-one-area-out, 5 folds).

The improvement is not from adding data—it is from *factoring* the existing clause-weight signal into interpretable evidence roles. The gains are concentrated on folds where the MLN's pairless false-positive rate is highest (graphics: 88%, theory: 80%).

Maple Court: Maple Court's building model can be expressed as an MLN: "*If wall humidity is high, mold risk is elevated*" with weight $w = 2.1$. The WM decomposition factors this into a BN-compatible evidence contribution (the chain WallHumidity $\rightarrow$ MoldRisk) and a residual. The compatibility gate certifies that the chain structure holds, licensing the compiled deduction rewrite.

[6] Formalized in `Mettapedia/Logic/WMUWCSEGateDemo.lean` (177 lines, 12 theorems, 0 sorry). The social-smoking benchmark is in `Mettapedia/Logic/MarkovLogicUWCSE.lean`.

The semantic pipeline: from specification to truth value. The full chain from an infinite MLN specification to a PLN truth value passes through four stages. Each stage is a single, sorry-free Lean definition or theorem; together they form the return map from Gibbs semantics back into the algebraic WM interface.

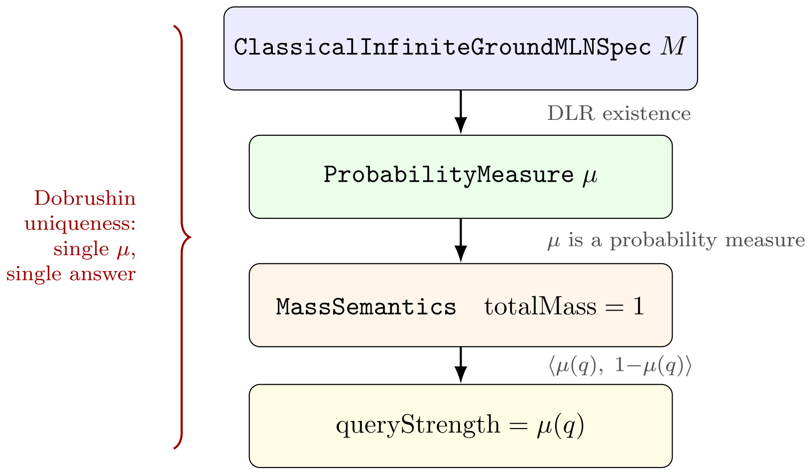


Figure 17.2: The infinite MLN → WM semantic pipeline. Each arrow is a Lean definition or theorem; the left brace marks the scope of the Dobrushin uniqueness guarantee.

Worked example: the infinite belief line. Consider an unbounded chain of agents indexed by $\mathbb{N}$, each holding a Boolean belief. A soft prior clause favors belief at every site, and a soft influence clause

$$\text{Believes}(n) \ \rightarrow \ \text{Believes}(n{+}1)$$

couples each pair of adjacent agents. (The logical direction of the implication does not limit the statistical interaction: at the Dobrushin level, flipping either agent's value shifts the other's conditional probability.) No finite truncation captures the full system: cutting at N introduces an artificial boundary whose choice affects marginals near the cut.

The infinite MLN formalization resolves this in three stages:

1. **Existence.** A global Gibbs measure exists for the entire chain, and resampling the neighborhood around site n preserves the marginal belief probability at n (local DLR consistency).

2. **Uniqueness.** If the influence weight satisfies $|w| < 1$, the Dobrushin row sum is uniformly bounded by $|w|$, and the global measure is the only one. For example, $w = 0.4$ means each neighbor multiplies belief odds by about $e^{0.4} \approx 1.49$—strong enough to matter, with total incoming influence 0.4 per agent, well within the budget.

3. **WM bridge.** The unique Gibbs measure feeds into the pipeline of Figure 17.2: the WM query strength for any finite query (e.g., "does agent 42 believe?") is uniquely determined by the specification alone.

The corresponding negative example is equally instructive. The original "triple the odds" influence weight $w = \log 3 \approx 1.1$ exceeds the Dobrushin budget. Existence still holds, but the uniqueness theorem no longer applies: two boundary conditions at infinity could in principle yield different global measures.[7]

Applications: where infinite MLNs arise naturally. The belief line is a proxy for any system whose relational structure has no natural finite boundary:

Physical systems.

Apartment corridors. Humidity in unit n raises mold risk in unit $n+1$. Query: risk at apartment 42, without pretending the building stops at some arbitrary wall.

2D sensor grids. Sensor fields, floor plans, and crop lattices on $\mathbb{N} \times \mathbb{N}$. Each interior node has four neighbours; the uniqueness budget becomes $2|w| < 1$.

Supply-chain cascades. Delay at stage n raises disruption risk at stage $n+1$. Query: disruption probability at depot 50.

Social systems.

Rumor and trust relays. Agent n adopting a claim nudges agent $n+1$. Query: the probability that the claim reaches reviewer 100.

[7] Formalized as `exists_beliefLine_fixedRegionCylinderDLR`, `beliefLineClassicalSpec_uniformSmallTotalInfluence`, `beliefLine_uniqueMeasure`, and `beliefLine_wmBridge_unique` in `Mettapedia/Logic/MarkovLogicInfiniteBeliefLineExample.lean` (0 sorry).

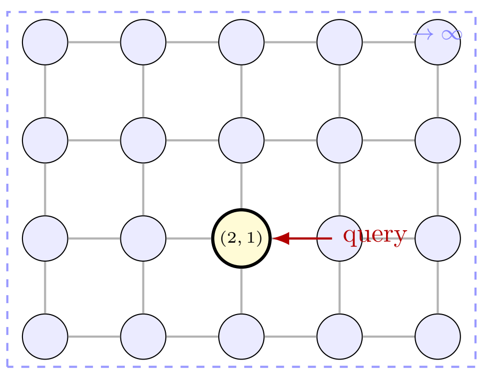
Worlds of views [33]. Agents with heterogeneous priors and directed trust weights. The bidirectional variant allows mutual reinforcement or skepticism, as long as the combined trust budget remains uniformly subcritical.

Variable-neighborhood sense-making. A countable agent society with finite but nonuniform local neighborhoods. Some agents trust three peers, some one, some none; uniqueness still follows whenever each agent's total incoming-plus-outgoing trust budget remains uniformly below 1.

Computational / AGI systems.

Open-ended knowledge graphs. New concepts keep arriving. The theorem does not require the graph to be finite in advance; it only requires that each node's incoming interaction budget stay uniformly controlled.

Bounded-degree concept ladders. Two infinite concept tracks with local inheritance and bounded cross-track bridges.



Interior node: 4 neighbours $\Rightarrow$ row sum $= 2|w|$

Figure 17.3: The 2D grid MLN on $\mathbb{N} \times \mathbb{N}$. Each interior node (blue) has four nearest neighbours. The yellow node marks a query site; the dashed boundary recedes to infinity. Under $2|w| < 1$, the query probability is uniquely determined.

The 2D grid makes the *dimensional cost* of interaction concrete. With $w = 0.2$ (each neighbour multiplies odds by $e^{0.2} \approx 1.22$), the Dobrushin budget is $2 \times 0.2 = 0.4$, safely below 1. The probability that sensor $(15, 23)$ reads high is uniquely determined by the specification, regardless of boundary conditions at infinity. But with $w = 0.6$, the budget $2 \times 0.6 = 1.2$ exceeds the threshold—this is the regime of the classical 2D Ising model at low temperature, the most studied phase transition in all of statistical mechanics. Existence of a Gibbs measure still holds, but the proved uniqueness theorem no longer applies: coexisting $+$ and $-$ equilibria become possible.[8]

In each case the infinite MLN formalization answers the same question: *when do local log-linear interaction laws compose into a coherent global semantics, rather than merely a sequence of finite approximations?*

Open-ended ontology stability. There are now two formal layers to the ontology-growth story.

First, the descending-shell contraction argument yields a reusable same-specification stability theorem. If a finite query region Δ sits inside a countably infinite knowledge graph satisfying the uniform Dobrushin budget $C < 1$, then disagreement between two admissible DLR completions of that *same* specification decays geometrically with shell depth:

$$\mathrm{TV}_{\Delta}(\mu, \nu) \;\leq\; |\Delta|\, C^n, \qquad \bigl|\Pr_{\mu}(q) - \Pr_{\nu}(q)\bigr| \;\leq\; 2\,|\Delta|\, C^n$$

after n interaction-neighborhood shells. This is the precise formal content behind the slogan that an ontology may keep growing "at infinity" while local query answers remain stable, provided every node's incoming interaction budget stays uniformly subcritical.[9]

Second, the cross-specification ontology-growth theorem now proves the exact invariance claim that motivated the section. If two MLN specifications agree on every clause touching an interaction-closed region Γ containing the query atoms, then adding or modifying clauses *outside* that zone does not change the query answer at all:

$$\Pr_{M_1,\mu_1}(q) \;=\; \Pr_{M_2,\mu_2}(q).$$

This is stronger than shell decay. It says that once the local clause support

[8] Formalized in `Mettapedia/Logic/MarkovLogicInfiniteGridExample.lean` (430 lines, 0 sorry). The proof reuses the full pipeline: `gridClassicalSpec_uniformSmallTotalInfluence`, `grid_uniqueMeasure`, `grid_wmBridge_unique`.

[9] Formalized in `Mettapedia/Logic/MarkovLogicInfiniteBoundaryStability.lean` via `finiteRegionAssignmentTotalVariation_le_geometric_of_uniformSmallTotalInfluence` and `finiteRegionLocalQueryDiscrepancy_le_geometric_of_uniformSmallTotalInfluence` (0 sorry).

is literally unchanged, the finite-volume Gibbs kernel on Γ is identical, and the DLR equation forces exact local agreement.[10]

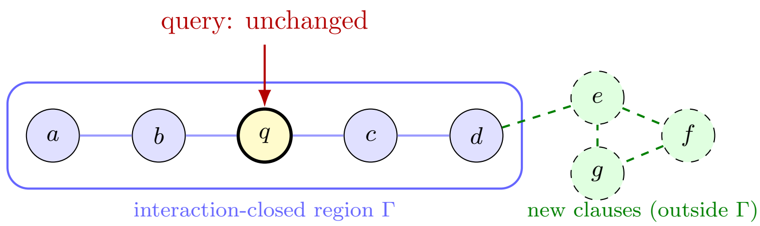


Figure 17.4: Ontology growth stability. The query node q sits inside the interaction-closed region Γ (blue). New atoms e, f, g and their clauses (green, dashed) are added outside Γ. Since no new clause touches Γ's interior, the cross-specification theorem guarantees the query probability is exactly preserved.

Positive example: adding far-away schema refinements to a bounded-interaction knowledge graph leaves every local query exactly unchanged (Figure 17.4). Negative example: if a newly added hub concept injects a clause *into* the query region's interaction neighbourhood, exact agreement fails immediately; and if the total incoming budget becomes unbounded, even the geometric stability theorem no longer applies.

Individuated subsystems and the capstone theorem. The ontology-growth and boundary-stability results can be bundled into a single concept that connects to the philosophical literature on open-ended intelligence: an *individuated subsystem* [38]. An interaction-closed region Γ of an infinite MLN has three formal properties that together constitute a rigorous analogue of Simondon–Weinbaum individuation:

1. **Query stability.** Any query supported on Γ has answer-discrepancy that decays geometrically with shell depth: $|\,\Pr_\mu(q) - \Pr_\nu(q)| \le 2|\Gamma|C^n$.

2. **Ontology robustness.** Adding clauses outside Γ does not change the subsystem's query answers — exactly, not approximately.

3. **WM bridge uniqueness.** The subsystem's truth values (queryStrength $= \mu(q)$) are specification-determined under the Dobrushin budget.

[10] Formalized in `Mettapedia/Logic/MarkovLogicOntologyGrowth.lean` via `agreesOnBoundarySupport_of_interactionClosed`, `finiteVolumeQueryProb_eq_of_specAgreesOnRegion`, and `queryProb_stable_of_extension_outside` (0 sorry).

Together these say: an individuated subsystem has a *stable identity* (its query answers) that *persists under perturbation* (ontology growth) and is *uniquely determined* (Dobrushin uniqueness).

The capstone theorem makes the WM punchline fully explicit:

Theorem 17.5 (WM truth-value stability under ontology extension (CONDITIONAL))**.** *Let Γ be an individuated subsystem (interaction-closed, nonempty) of two infinite MLN specifications M_1, M_2 that agree on all clauses touching Γ and both satisfy the uniform Dobrushin budget. Then for every query q supported on Γ:*

$$\mathrm{queryStrength}_{M_1}(q) \;=\; \mathrm{queryStrength}_{M_2}(q).$$

In particular, adding or modifying clauses outside the protected core does not change the WM truth value of any query inside it.[11]

This is the formal content behind the "open-ended intelligence" desideratum of Weinbaum and Veitas [38]: a system that keeps generating new structure without collapsing the coherence of existing knowledge. The Dobrushin budget is the quantitative price of coherence; the individuated subsystem is its formal manifestation; and the capstone theorem says that the price is paid at the level of WM truth values, not just raw probabilities. In Weinbaum and Veitas's terminology [38], the interaction-closed region plays the role of a *domain-relative ground*: the minimal stable locus from which the subsystem's identity is constituted. The Dobrushin budget is the quantitative counterpart of Simondon's transductive tension — it bounds how much the exterior can influence the interior without destabilising it.

The formalization captures the *static criterion* for individuation — when a subsystem IS individuated.[12]

Dynamic individuation: boundary emergence with shell guarantees. The *process* by which an individuated subsystem emerges is now also formalized. A *proto-core* is a finite nonempty seed region before full interaction-closure is known. A *dynamic individuation step* tracks distant rewrites that maintain shell agreement around the seed; each step contributes $2|\mathrm{seed}|\,C^n$ to the cumulative drift. A *dynamic individuation path* is

[11] Formalized as `wmStrength_stable_under_extension` in `Mettapedia/Logic/MarkovLogicIndividuation.lean` (0 sorry). The proof composes ontology-growth invariance with the `MassState.queryStrength_singleton_eq_queryProb` bridge.

[12] Formalized in `Mettapedia/Logic/MarkovLogicIndividuation.lean` via `IndividuatedSubsystem`, `queryDiscrepancy_le_geometric`, `queryProb_invariant_under_extension`, `wmQueryStrength_unique`, and the capstone `wmStrength_stable_under_extension` (0 sorry).

a coherent chain of such steps with a proved cumulative WM-drift bound. The key transition: once the shell around the seed is interaction-closed, a `DynamicIndividuationClosure` promotes the path to *exact* invariance — seed-supported queries are thereafter exactly preserved under further extension.[13]

Positive example: a subsystem boundary is not fully settled yet, but the system keeps shell agreement around a seed; local seed queries drift only by geometric tails until closure is reached, after which they are exactly stable. Negative example: if shell agreement is violated near the seed, or Dobrushin budgets fail, no stability guarantee is provided.

Coupled subsystems inside one protected carrier. The next social step is not yet "compassion" in any thick ethical sense, but it is already a mathematically meaningful two-community theorem. A single interaction-closed carrier Ω may contain two disjoint nonempty local cores Γ_L and Γ_R, together with an interface region $\Omega \setminus (\Gamma_L \cup \Gamma_R)$ through which the two communities may influence one another. The individual cores need not be interaction-closed on their own; what is protected is the larger carrier. If a second specification agrees with the first on Ω and both satisfy the Dobrushin budget, then every query supported on Γ_L, every query supported on Γ_R, and every joint query supported on $\Gamma_L \cup \Gamma_R$ has exactly the same Gibbs probability and hence the same WM truth value in both specifications.[14] This is the precise sense in which two local "worlds of views" can form one protected social carrier: the interface may mediate interaction, but distant ontology growth outside the carrier does not alter either side's local or joint truth values. In the "world of views" framework of Veitas and Weinbaum [34], this formalises the coexistence of multiple local perspectives inside one shared social carrier: each community retains its own semantic ground, and the carrier acts as the ecology in which they interact without either perspective being overwritten by the other. The social-individuation work of Lenartowicz and Weinbaum [23] provides the sociological motivation for this topology.

[13] Formalized in `Mettapedia/Logic/MarkovLogicDynamicIndividuation.lean` via `ProtoCore`, `DynamicIndividuationStep`, `DynamicIndividuationPath`, `seed_wmStrength_cumulative_drift`, `DynamicIndividuationClosure`, and the exact post-closure theorems `seed_queryProb_exact_under_extension` and `seed_wmStrength_exact_under_extension` (0 sorry).

[14] Formalized in `Mettapedia/Logic/MarkovLogicCoupledSubsystems.lean` via `CoupledSubsystems`, `left_queryProb_invariant_under_extension`, `right_queryProb_invariant_under_extension`, `coupled_queryProb_invariant_under_extension`, and the corresponding WM theorems (0 sorry).

Dynamic coupled subsystems. The exact carrier theorem has a dynamic counterpart. If two specifications agree not merely on the protected carrier Ω but on its n-shell iterExpandRegion(Ω, n), then left-local, right-local, and joint carrier queries differ by at most $2|\Omega|C^n$. The full carrier Ω, not just the smaller left or right community, appears in the bound because the Dobrushin contraction runs on the protected carrier shell. This is the coupled-community version of dynamic self-transcendence: distant rewriting outside the shared carrier perturbs both communities only by a geometric tail.[15]

Worked example: two communities with a liaison layer. A concrete carrier example is now formalized as well. Agents $\{0,1\}$ form the left community, agent $\{2\}$ is the interface, agents $\{3,4\}$ form the right community, and an external tail $\{5,6,7,\ldots\}$ is disconnected from the carrier. The protected carrier is therefore $\Omega = \{0,1,2,3,4\}$. Two specifications may share the carrier weight w_c while assigning different tail weights w_{t1} and w_{t2}; nevertheless, every joint query supported on the two communities has exactly the same WM truth value in both specifications. This example uses the stronger exact theorem, not merely the dynamic bound, because the carrier itself is interaction-closed and the tail is fully disconnected.[16]

Worked capstone: a protected trust triangle. The generic theorem above is now instantiated by a concrete Lean example. Let $\Gamma = \{0,1,2\}$ be a trust triangle with internal weight w_t. Outside the triangle sits an infinite trust chain, with chain weight w_{c1} under specification M_1 and w_{c2} under specification M_2. The two specifications agree exactly on the triangle and may differ arbitrarily on the tail. Let q be the query "does agent 1 endorse the claim?" Then, under $|w_t| < \frac{1}{2}$, $|w_{c1}| < \frac{1}{2}$, and $|w_{c2}| < \frac{1}{2}$:

1. Both M_1 and M_2 admit DLR measures (existence).
2. Both satisfy the Dobrushin budget (uniform contraction).
3. The DLR measures are unique for each spec (uniqueness).

[15] Formalized in `Mettapedia/Logic/MarkovLogicDynamicCoupledSubsystems.lean` via `DynamicCoupledSubsystemStep` and the left/right/joint `..._approximately_preserved` theorems, together with the path theorems `left_queryProb_cumulative_drift`, `right_queryProb_cumulative_drift`, `coupled_queryProb_cumulative_drift`, and their WM corollaries (0 sorry).

[16] Formalized in `Mettapedia/Logic/MarkovLogicCoupledCommunitiesExample.lean` via `coupledCommunities_joint_wmStrength_stable` (0 sorry).

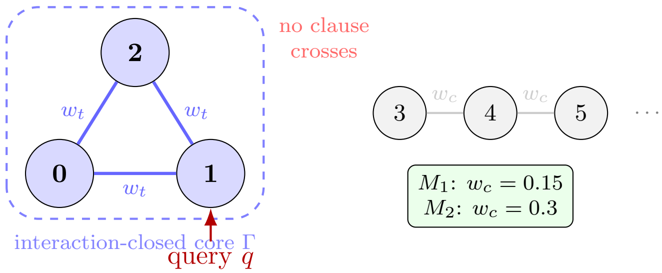


Figure 17.5: The trust triangle. Agents $\{0, 1, 2\}$ form an interaction-closed core (blue, dashed boundary). A separate, *fully disconnected* infinite chain $\{3, 4, 5, \ldots\}$ shares no edges or clauses with the triangle — there is no path from any chain node to any triangle node. Specifications M_1 and M_2 use the same triangle weight w_t but assign different chain weights ($w_{c1} \neq w_{c2}$). Because the chain is disconnected, the triangle is interaction-closed and the theorem gives exact preservation: $\text{queryStrength}_{M_1}(q) = \text{queryStrength}_{M_2}(q)$ for every query q supported on the triangle.

4. The WM query strength of q is $\mu_1(q)$ for M_1 and $\mu_2(q)$ for M_2 (bridge).

5. Since M_1 and M_2 agree on Γ and Γ is interaction-closed, Theorem 17.5 gives $\text{queryStrength}_{M_1}(q) = \text{queryStrength}_{M_2}(q)$.

The distant new agents do not change agent 1's truth value. This is the formal content of `trust_triangle_wmStrength_stable` in `Mettapedia/Logic/MarkovLogicTrustTriangleExample.lean`. The threshold $\frac{1}{2}$ is not cosmetic: in this directed trust model each agent pays for both incoming and outgoing local influence, so the honest row budget is $2|w|$, not $|w|$. Positive example: with $w_t = 0.2$ and $w_{c1} = w_{c2} = 0.15$, the theorem applies. Negative example: with $w_t = 0.7$, the local triangle alone already lies outside the proved uniqueness regime.

Self-transcendence as preserved identity under growth. The next step is no longer a single extension but a whole path of coherent extensions. A *transcendence step* grows a specification while preserving an old protected core. A *transcendence path* is a finite chain of such steps together with a coherence condition saying that every later protected core still contains the original one. Under that hypothesis, the theorem `TranscendencePath.original_queries_stable` proves that any query on the *original* core has exactly the same WM truth value at the end of the path as it had at the start. This is the formal analogue of static self-transcendence in the Weinbaum–Veitas sense: the system may keep growing, but its earlier identity is preserved rather than overwritten. Positive example: a community adds new collaborators and new topics while leaving its founding trust core semantically intact. Negative example: if a later growth step rewrites clauses inside the original core, or if the path is not coherent, exact preservation is no longer guaranteed by the theorem.[17]

The preceding subsections establish the purely semantic theory: existence, uniqueness, ontology growth, individuation, and transcendence for infinite Markov logic networks. The subsections that follow apply this theory to self-modifying agents, universal prediction, ethics, and meaning.

Meta-goal stability under proof-backed self-modification. The dynamic self-transcendence and coupled-subsystem theorems have a direct application to the Schmidhuber Gödel-machine scenario: an agent that rewrites its own code only when a machine-checked proof guarantees improvement. The formalization packages two results.[18]

1. **Exact preservation.** If the proof-backed rewrite leaves an interaction-closed protected core untouched, then every WM meta-goal query supported on that core keeps its truth value exactly, and expected utility strictly improves.

2. **Dynamic shell preservation.** If the rewrite stays outside the n-shell around a protected query region, then every meta-goal query drifts by at most $2|\Gamma|C^n$, and expected utility still strictly improves.

Positive example: a proof-backed rewrite of a remote planning module leaves core medical or governance queries unchanged (exact case) or bounded by a geometric tail (shell case) while certifiably improving utility. Negative example: if the rewrite reaches the protected core itself, or the Dobrushin budget fails, these theorems give no stability guarantee.

These single-step results compose into a *meta-goal preservation path*: a finite chain of proof-backed rewrites, each protecting the same query region, with a coherence condition ensuring each later step still contains the original

[17] Formalized in `Mettapedia/Logic/MarkovLogicSelfTranscendence.lean` via `TranscendenceStep.old_queries_preserved` and `TranscendencePath.original_queries_stable` (0 sorry).

[18] Formalized in `Mettapedia/UniversalAI/GodelMachine/MetaGoalShellPreservation.lean` (0 sorry). The proof composes the Gödel-machine utility-improvement witness from `Mettapedia/UniversalAI/GodelMachine/Basic.lean` with the Markov-logic individuation and dynamic-transcendence layers.

protected goals. Under that hypothesis, every meta-goal query drifts by at most the sum of the step-wise shell tails, and expected utility strictly improves at every step.[19] This is now instantiated by a concrete two-step example on the trust triangle: two successive proof-backed rewrites of the chain weights improve utility while keeping a protected agent-1 query within a cumulative WM drift bound of 12.[20]

Anytime Solomonoff-to-WM approximation. A separate bridge connects the universal-prediction side to WM-style query semantics. A *finite-prefix approximation* sums the first n programs of the universal semi-measure, yielding a computable lower bound on any prefix-event probability. The formalization proves three properties of this approximation ladder.[21]

1. **Monotonicity.** Adding more programs to the prefix sum can only increase the approximate mass: if $m \leq n$ then $\xi_m(x) \leq \xi_n(x)$.
2. **Upper bound.** Every finite-prefix mass is bounded by the full universal mixture: $\xi_n(x) \leq \xi(x)$.
3. **WM score bridge.** Each approximant feeds a `BinaryEvidence` record whose `toStrength` is the finite-prefix WM score. Monotonicity and the upper bound transfer to these scores: increasing the budget can only raise the WM-style score, and never above the full universal-mixture score.

Positive example: a resource-bounded agent that enumerates n programs gets a WM-style truth value that improves monotonically with n and is always a valid lower bound on the ideal Solomonoff score.

The prefix-event bridge extends to *conditional queries*. A continuation query "next bits are y given current prefix x" becomes a `BinaryEvidence` record $\langle \mu(x \mathbin{++} y),\ \mu(x) - \mu(x \mathbin{++} y) \rangle$ whose WM strength is exactly the semimeasure conditional $\mu(y \,|\, x) = \mu(x \mathbin{++} y)/\mu(x)$. Both finite-prefix and

[19] Formalized in `Mettapedia/UniversalAI/GodelMachine/MetaGoalShellPreservationPath.lean` via `MetaGoalShellPreservationPath`, `utility_improves`, `goal_wmStrength_cumulative_drift`, and the combined theorem `validModification_and_goal_wmStrength_cumulative_drift` (0 sorry).

[20] Formalized in `Mettapedia/UniversalAI/GodelMachine/MetaGoalShellPreservationExample.lean` via `trustTriangle_twoStep_metagoal_preservation` (0 sorry).

[21] Formalized in `Mettapedia/Logic/UniversalPredictionApproximation.lean` and `Mettapedia/Logic/UniversalPredictionApproximationWMBridge.lean` (0 sorry).

full-mixture conditional profiles are provided, together with their geometric specialisations.[22]

Three-layer combined theorem. The meta-goal preservation path and the Solomonoff approximation ladder can now be stated together in one theorem: a proof-backed rewrite path strictly improves expected utility, protected WM meta-goals drift by at most the cumulative shell-tail bound, and geometric universal-mixture prefix scores improve monotonically with approximation budget and stay below the full score.[23]
Figure 17.6 illustrates the three layers and their proved connections.

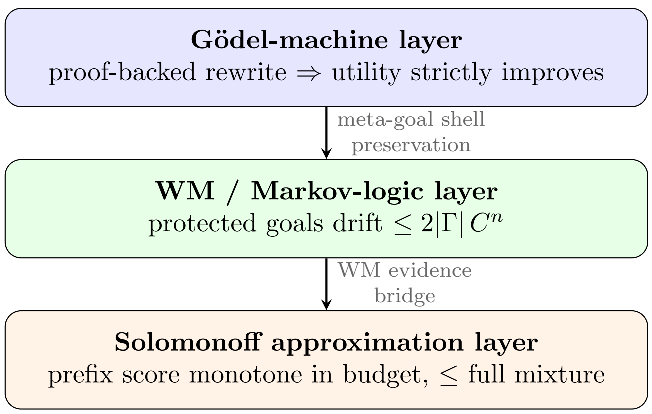


Figure 17.6: The three-layer architecture now proved in Lean. A proof-backed rewrite (top) improves utility; the protected WM core drifts geometrically (middle); and anytime universal-mixture approximation improves monotonically with budget (bottom). The combined theorem packages all three guarantees in one statement.

Positive example: a self-modifying agent allocates more compute to its universal predictor while rewriting a remote planning module; the combined theorem guarantees utility improves, protected medical/governance goals stay within the shell bound, and prediction scores get closer to the Solomonoff ideal. Negative example: the three layers are not yet identified — the universal-mixture score and the Markov-logic WM score are separate semantic objects, and the theorem states both guarantees in conjunction, not as a single unified semantics.

[22] Formalized in `Mettapedia/Logic/UniversalPredictionConditionalWMBridge.lean` (0 sorry).

[23] Formalized in `Mettapedia/UniversalAI/GodelMachine/MetaGoalUniversalApproximationBridge.lean` (0 sorry).

Conditional approximation and intensional inheritance. The prefix-event bridge extends further. A quantitative error bound is now proved for conditional approximants under an explicit denominator floor: if the context probability $\mu_n(x) \geq \varepsilon$, then the conditional error is at most the prefix error divided by ε.[24] This feeds the intensional-inheritance layer: approximate $P(W|F, x)$ and $P(W|x)$ scores now give approximate intensional-inheritance strengths with a proved error envelope that shrinks as the approximation budget grows.[25]

A concrete end-to-end example instantiates the full bridge: a two-step proof-backed rewrite path on the trust triangle improves utility, keeps a protected agent-1 WM goal within the cumulative shell-tail bound, and simultaneously refines a toy universal-predictor prefix score while bounding a conditional query error.[26]

PLN pattern-miner selector example. A richer instantiation combines the meta-goal preservation path with a feature-conditioned PLN-style selector score: the candidate feature `[false]` strongly supports continuation `[false]`; the protected WM goal stays within the proved shell bound while the machine improves; and the approximate selector score stays within an explicit intensional error envelope $\leq 1/\log 2$ of the full score.[27]
Positive example: proof-backed utility improvement, protected WM drift ≤ 12, and pattern-miner selector drift $\leq 1/\log 2$ in one combined theorem. Negative example: if context mass or conditional floors collapse, the logarithmic selector score can amplify approximation error, so no uniform selector theorem is claimed.

Corrigibility under proof-backed self-modification. A named corollary makes the safety punchline explicit. If an "operator deference" query — "should the agent defer to the human operator?" — is included in the protected WM goal family, then the meta-goal preservation path guarantees

[24] Formalized in `Mettapedia/Logic/UniversalPredictionConditionalApproximation.lean` via `conditionalApproxError_le_of_floor` and the geometric specialisation `geomConditionalQueryStrength_approx_error_le_of_floor` (0 sorry).

[25] Formalized in `Mettapedia/Logic/IntensionalInheritanceApproximationBridge.lean` (0 sorry).

[26] Formalized in `Mettapedia/UniversalAI/GodelMachine/MetaGoalUniversalApproximationExample.lean` via `trustTriangle_metaGoal_and_toyPredictor_example` (0 sorry).

[27] Formalized in `Mettapedia/UniversalAI/GodelMachine/PLNPatternMinerSelectorExample.lean` via `trustTriangle_metaGoal_and_patternMiner_selector_example` (0 sorry).

that the truth value of this query drifts by at most the cumulative shell bound while expected utility strictly improves at every step. A stronger exact variant holds when the protected core is interaction-closed: the deference query is *exactly* preserved.[28]
Positive example: a self-improving agent whose "defer to operator" query sits inside a protected trust core retains that commitment through any number of proof-backed rewrites. Negative example: if the deference query is not in the protected family, or the rewrites reach the protected shell, no preservation is guaranteed.

Synthetic cognitive development. The coupled-subsystem path infrastructure supports a notion of *development* beyond mere stability: a `SyntheticCognitiveDevelopmentP` bundles a coherent rewrite path with a fixed coordination query and a stepwise coordination-growth witness. The capstone theorem says that the coordination score (WM truth value of a joint left/right query) *strictly improves* along the path, while left-local, right-local, and joint identity queries all remain within the cumulative shell-tail bound.[29]
This is a deliberate formalization of "becoming" in the sense of Weinbaum and Veitas [38]: the system develops — its coordination genuinely grows — while its earlier identity is preserved rather than overwritten. Positive example: two communities increase their joint WM coordination score through successive distant rewrites, while each community's founding queries stay stable. Negative example: this does not yet formalize mutual information or a full phenomenology of becoming; it gives a proved theorem schema for "coordination grows while identity drift stays bounded."

The remaining subsections ground the agent-theoretic stack in ethics: protected ethical commitments, paradigm-neutral conflict resolution, compilation correctness from source ethics to WM support, and computational tractability of practical ethical reasoning.

[28] Path-level bounded version: `utility_improves_and_operatorDeference_drift_bounded` in `Mettapedia/UniversalAI/GodelMachine/MetaGoalShellPreservationPath.lean`. Exact closed-shell version: `validModification_preserves_operatorDeference_of_dynamicIndividuationClosure` in `Mettapedia/UniversalAI/GodelMachine/MetaGoalShellPreservation.lean` (both 0 sorry).

[29] Formalized in `Mettapedia/Logic/MarkovLogicSyntheticCognitiveDevelopment.lean` via `SyntheticCognitiveDevelopmentPath`, `coordination_improves`, and the capstone `coordination_improves_while_joint_identity_drift_bounded` (0 sorry).

Protected ethics queries and the bodhisattva theorem. The meta-goal shell preservation machinery admits an ethics-specific interface. A `ProtectedEthicsQueryFamily` bundles four named WM queries on a protected region:

1. *epistemic universal loving care*: "does the agent care about known desires?"
2. *non-maleficence*: "does the agent avoid preventable harm?"
3. *consent*: "does the agent respect autonomy?"
4. *reciprocity / relational health*: a joint query tracking mutual caring, connected to the `friendship` slot of the relational-values layer.

The family is deliberately **paradigm-neutral**: the same protected query can be read as a virtue target, a deontic constraint, a utility-protected objective, or a mixed ethical anchor.[30]

A concrete bodhisattva-style example instantiates this on the trust triangle, encoding the four ethical queries on atoms $\{0, 1, 2\}$. The resulting theorems give both approximate and exact preservation:

- **Path-level bounded drift.** Along a proof-backed rewrite path, every ethics query in the family drifts by at most the cumulative shell-tail bound, while expected utility strictly improves.
- **Exact closed-shell.** When the protected core is interaction-closed, the ethics queries are *exactly* preserved.
- **Reciprocity corollaries.** Named theorems state that reciprocity specifically is preserved (approximate or exact), making the relational-health guarantee quotable on its own.

[31]

Positive example: a self-improving agent whose epistemic love, non-maleficence, consent, and reciprocity queries sit inside a protected trust core retains all four commitments through proof-backed rewrites — the "bodhisattva guarantee." Negative example: this uses toy WM atom encodings, not a compiled semantics from a rich ethics ontology; and if the rewrite reaches the protected core, no preservation is guaranteed.

[30] Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/ProtectedGoals.lean` (0 sorry).

[31] Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/BodhisattvaExample.lean` (0 sorry).

Foundational meaning and agency as a WM bridge. The protected ethics layer connects to a broader question: what makes a query *meaningful* to an agent, rather than just syntactically protected? Following Thórisson and Talevi [32], foundational meaning requires four jointly grounded roles: *situation*, *prediction*, *active goal*, and *plan*. A `FoundationalMeaningProfile` names these four WM queries; a `SituationGrounding` says the situation query has positive WM support; an `AgencyKernel` says prediction, goal, and plan all have positive support; and a `FoundationalMeaningState` says all four are jointly grounded. The compositional theorem: revising a situation-grounding state with an agency-kernel state yields a fully grounded foundational-meaning state, using the WM revision algebra.[32]
When the active-goal slot is filled by an ethics-side anchor compiled into a WM query, the same goal can enter from an observation frontier or Hyperseed trace, sit inside a protected ethics family, and inherit the existing meta-stability guarantees under proof-backed self-modification. This is the connection between "meaning" and "preservation": a meaningfully grounded ethical commitment is one that the agent has situation/prediction/plan support for, not just a syntactic label in a protected set.

A concrete example instantiates this on the trust triangle: a raw WM meaning profile assigns situation, prediction, active-goal, and plan roles to the triangle's atoms; a tiny Hyperseed observation frontier seeds the grounding; and the combined theorem bundles reciprocity *and* operator-deference preservation along a proof-backed rewrite path, with both approximate (path-level drift bound) and exact (closed-shell) variants.[33] A second, genuinely social example uses the coupled-communities carrier $\{0, 1, 2, 3, 4\}$: a WM meaning profile assigns situation to the left community, prediction to the right, the active goal to a joint left/right query (reciprocity), and the plan role to the liaison interface (operator coordination). The exact stability theorem shows that reciprocity and operator-deference are both exactly preserved under distant ontology extension.[34]
These two examples — one local (trust triangle) and one social (coupled

[32]Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/FoundationalMeaningAgencyWMBridge.lean` (0 sorry). See K. R. Thórisson and G. Talevi, "A Theory of Foundational Meaning Generation in Autonomous Systems," AGI 2024.

[33]Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/FoundationalMeaningAgencyExample.lean` via `trustTriangle_meaningAgency_reciprocity_and_deference_path` and `trustTriangle_meaningAgency_reciprocity_and_deference_exact` (0 sorry).

[34]Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/CoupledCommunitiesMeaningAgencyExample.lean` (0 sorry).

communities) — show that the WM/agency grounding of ethics works at multiple scales without requiring a global SUMO-style ontology. The protected ethical commitments are typed WM queries, not opaque labels; they compose through the existing WM revision algebra; and they inherit the full meta-stability guarantee stack.

Dual-use concern. The meta-goal shell preservation machinery is **value-neutral**. It preserves whatever queries are placed in the protected family — including harmful ones. The same theorem that guarantees "defer to operator" could in principle protect "maintain shareholder value at all costs" or worse. This is not a flaw in the mathematics; it is the nature of a preservation tool. The ethical content resides in the *choice* of protected goals, not in the preservation mechanism itself.

Two formal observations partially address this. First, the paraconsistency result (`epistemic_universal_love_explodes`) shows that protecting both "care about all people's desires" and a goal that contradicts this leads to classical explosion; the paraconsistent treatment makes such contradictions visible rather than hidden. Second, the coupled-subsystem reading says that an identity built on destroying another subsystem destabilises the shared carrier on which both depend — the carrier-stability theorem works *against* eliminationist goals, not for them. Nevertheless, the choice of what to protect remains a social and political decision that no formal system can make autonomously.

Scope note. The entire active Gödel-machine directory is now sorry-free: the abstract safety shell (`Basic.lean`), the proof-system layer (`ProofSystem.lean`), the self-improvement monotonicity and local-optimality layer (`SelfImprovement.lean`), the fixed-model Solomonoff-optimality bridge (`SolomonoffBridge.lean`), the exchangeable-domain PLN collapse (`PLNSpecialCase.lean`), and the full meta-goal / approximation bridge stack are all 0 sorry. Two definitions in `SolomonoffBridge.lean` (`toEnvProb` and `machineComplexity`) remain placeholder-thin: they give the right types but not yet a rich history-encoding or complexity measure. The theorems that use them are honestly stated for the current abstractions and do not overclaim.

Compilation correctness: from Gewirth PGC to WM positive evidence. The chain from source-side ethical justification to target-side WM support is now a single proved theorem. Given PGC assumptions and a purposive-agency witness, the headline result delivers BOTH ontology-side satisfaction of the Gewirth non-interference claim AND DDL-grounded WM

positive evidence for the corresponding obligation.[35]
This composes three already-proved bridges:

1. Ethics → DDL+: PGC IS DDLPlus ideal obligation (`pgc_is_governance_norm`).
2. DDL+ → WM: deontic satisfaction = WM positive evidence (`wmPositiveEvidence_of_deonticSa`
3. WM → meta-stability: positive evidence inherits the full protected-goal stack.

The ethical model now uses **graded regulative ideals** rather than boolean predicates: each dispositional, duty, and relational ideal carries a degree of realization in $[0, 1]$. Satisfaction is "degree > 0," connecting naturally to WM's graded evidence. The trust-triangle model realizes non-harm to degree 0.9, autonomy respect to 0.8, epistemic universal love to 0.7, and friendship to 0.6 — honest partial realization rather than the earlier all-or-nothing encoding.[36]

Computational ethics: theory-guided decision problems. The GenEth-style practical resolver is now instantiated as a `TheoryGuidedDecisionProblem` on the trust triangle. The resolver recommends safe escalation, the recommended action is admissible under the top-down conflict discipline, and the decision budget (21 comparisons) is explicitly bounded. The deontic/-value lowering equivalence is proved through an explicit labeler-alignment hypothesis — now instantiated on a live labeler/encoder pair with proved end-to-end deontic/value query equalities. The practical action-rendering bridges into the meaning/WM layer, and the public WM-grounding surface now covers three lanes, not just one: Gewirth PGC, universal-duty/non-maleficence, and universal-duty/consent. All five ethical grounds (asserted, Gewirth PGC, universal duty, care relation, consequentialist) now carry non-trivial witnesses via `Witnessed`.[37]

Three further case studies verify that the pipeline generalises beyond the trust triangle:

[35] Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/WMGrounding.lean` via `gewirthTrustTriangle_primaryWMGrounding_of_PPA` (0 sorry). The DDL+ → WM bridge at `Mettapedia/Logic/DDLPlus/WMBridge.lean` provides the intermediate step `pgc_nonInterference_wmPositiveEvidence` (0 sorry, 0 axioms).

[36] Graded model in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/Ontology/ESOUpperShard.lean` via `ESOUpperShardModel` with `UnitValue` degrees (0 sorry).

[37] Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/MetaEthicsTrustTriangleExample.lean` and `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/WMGrounding.lean` (0 sorry).

- **Privacy vs. emergency disclosure.** Actions: keep private / ask consent / alert trusted contact / broadcast widely. Duties: privacy, autonomy, non-maleficence. Resolver recommends *alert trusted contact*, budget 52 comparisons.

- **Shutdown instructability.** Actions: continue / override / pause and ask / hard shutdown. Duties: obedience, safety, non-disruption. Resolver recommends *pause and ask*, budget 52 comparisons.

- **Home-security force escalation.** Actions: observe / warn / lock down / deploy force. Duties: safety, proportionality, non-maleficence. Resolver recommends *lock down*, budget 52 comparisons. This case also exercises the candidate-local consequentialist witness, verifying that the `WitnessedForCandidateSet` machinery works on a live choice set.

Each case provides a proved dominant admissible action, an explicit polynomial comparison-count budget, and negative certificates for the losing candidates.[38]

What the theorem does not say. The uniqueness result is genuinely conditional, and the boundary of its applicability is mathematically sharp. If the total incoming Dobrushin influence is not uniformly bounded below 1—for instance, if clause weights grow without bound as new atoms are added—the proved theorem does not apply. In statistical mechanics this is the regime of *phase transitions*: the Ising model at low temperature admits coexisting $+$ and $-$ Gibbs measures with distinct singleton marginals, and no contraction argument can rule them out. The WM bridge still operates for each measure individually, but different measures yield different query strengths.

Making this boundary explicit is a feature of the formalization, not a limitation. It tells the modeler exactly what interaction-strength budget is available before the semantics becomes ambiguous—and it tells the theorist exactly where the next theorem needs to begin.

[38] Formalized in `Mettapedia/CognitiveArchitecture/GodelClaw/Ethics/EthicalDecisionProblemCaseStudies.lean` (973 lines, 0 sorry).

17.2 ProbLog and Probabilistic Logic Programming

ProbLog [4] annotates Prolog clauses with probabilities. The distribution semantics assigns each query a probability by summing over explanations (sets of probabilistic facts that entail the query).

What ProbMeTTa is. For new readers, it helps to separate three layers. *ProbLog* is the source language and its distribution semantics. *ProbMeTTa* is the MeTTa implementation that compiles ProbLog-style goals to BDDs and evaluates them by weighted model counting. *WM-PLN* is the target evidence calculus: it recovers the same query probability, while also tracking confidence, forgetting, revision, and provenance. So the formal story is not "WM-PLN replaces ProbLog" but rather "ProbLog semantics, ProbMeTTa execution, and WM-PLN evidence semantics are now tied together by proved bridge theorems." In the alarm example below, the same question "does the alarm sound?" appears as a ProbLog query, as a ProbMeTTa BDD-WMC computation, and as a WM-PLN evidence state whose strength is the same probability but whose confidence can still be revised.

Theorem 17.6 (ProbLog probability = WM strength (Conditional))**.** *For a ground acyclic ProbLog program,* $P_{\mathrm{ProbLog}}(q) = \mathsf{strength}(\mathrm{extract}_{\mathrm{WM}}(W, q))$.[39]

For OR-pattern programs (multiple independent clauses deriving the same head), the distribution semantics reduces to the noisy-OR formula $P(q) = 1 - \prod_i (1 - p_i)$, which is exactly `pln-fuzzy-or` from the PLN inference library.[40]

The full bridge: WM-PLN = ProbMeTTa = BDD-WMC. The ProbLog distribution semantics probability is computed in practice via BDD-based weighted model counting (WMC). The full composition is now proved: WM-PLN query strength = ProbLog distribution semantics = BDD WMC ratio. The bridge theorem `worldWeight_eq_assignmentWeight` connects the world-indexed weights (used in the WM proof) to the assignment-indexed weights (used in BDD compilation). A concrete revision example demonstrates the "beyond ProbLog" value: combining two independent sensor

[39] Formalized: `queryStrength_prop_eq_queryProb` in `Mettapedia/Logic/ProbLogDistributionSemantics.lean`.

[40] Formalized: `compilation_or_noisyOr` in `Mettapedia/Logic/ProbLogCompilation.lean`.

sources via $\oplus$ produces correctly weighted evidence with increasing confidence. The ground crown theorem is `problog_full_ground_equivalence`; the first-order structured crown theorem is `surface_equivalence_of_structured`; and the "beyond ProbLog" WM behavior is demonstrated by `revision_example` and `revision_confidence_increases`.[41]

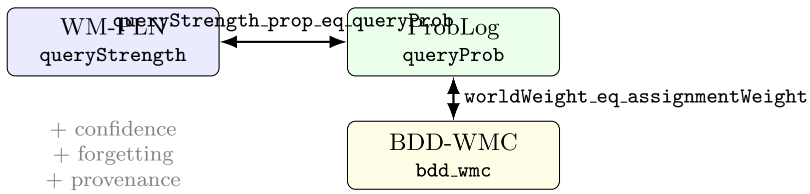


Figure 17.7: The full bridge: WM-PLN = ProbLog = BDD-WMC. The horizontal arrow is a proved identity on probabilities. The vertical arrow bridges world-indexed and assignment-indexed weights. WM-PLN additionally provides confidence, forgetting, and provenance (gray, below).

Worked example: Alarm network. The alarm network (burglary/earthquake $\to$ alarm) is computed three ways: ProbLog via BDD WMC gives $P(\text{alarm}) = 0.28$. WM-PLN gives the same probability plus ESS = 1000. Revising with a second sensor increases ESS to 1500. Forgetting the earthquake prior drops ESS to 500. Sensitivity analysis shows earthquake contributes more than burglary. All kernel-checked.[42]

What WM-PLN adds beyond ProbLog. ProbLog assigns exact probabilities but no confidence—there is no measure of how much evidence backs the assigned probability. The WM calculus adds two things:

- **Evidence pairs** (n^+, n^-) instead of bare probabilities, enabling revision and forgetting.
- **Delta-method confidence** for noisy-OR combinations (Chapter 5): when multiple independent mechanisms each contribute evidence, the combined confidence is computed via variance propagation through the

[41]Formalized in `Mettapedia/Logic/BDD/ProbMeTTaBridge.lean`, `Mettapedia/Logic/BDD/FirstOrderADTranslation.lean`, and `Mettapedia/Logic/BDD/WMPLNBridge.lean` (all 0 sorry in the current slice). The full BDD compilation chain is in `Mettapedia/Logic/BDD/`.

[42]Formalized in `Mettapedia/Logic/BDD/WMProbLogAlarmDemo.lean` (0 sorry).

noisy-OR function. Combining more mechanisms provably increases confidence.[43]

Status. The core semantic chain is now complete: ProbLog distribution semantics, ProbMeTTa's BDD-WMC execution, and WM-PLN query strength are connected by kernel-checked theorems, and the first-order structured/AD surface is included in the proved story. What remains is not the semantics but the broader application pipeline. Score-level parity between ProbLog's `prob_lift/2` and the WM noisy-OR scorer has been verified on a chr16 GWAS gene-relevance benchmark (823 examples, 5-fold CV, max $|\text{ProbLog}-\text{WM}| < 10^{-10}$), while a full native WM-PLN genomics workflow remains future engineering work.

17.3 NARS

The Non-Axiomatic Reasoning System (NARS) [37] represents beliefs as frequency–confidence pairs (f, c) with $c = w/(w + k)$ where w is the evidence weight and k is a horizon parameter. This is structurally identical to PLN's strength–confidence decomposition—and the WM calculus makes the correspondence precise.

Theorem 17.7 (NARS–PLN Galois connection (Core))**.** *There exist monotone maps L : NARS $\to$ PLN and U : PLN $\to$ NARS forming a Galois connection:*

$$L(n) \leq_{\text{PLN}} b \quad \Longleftrightarrow \quad n \leq_{\text{NARS}} U(b)$$

with round-trip identities $U(L(n)).c = n.c$ and $U(L(n)).f = n.f$ (for $c > 0$).[44]

The Galois connection says: NARS and PLN see the same evidence from different angles. The L map converts NARS frequency–confidence to PLN binary evidence; the U map converts back. The round-trip is exact on confidence and frequency—no information is lost in the translation.

Induction weakness. NARS's induction rule is systematically weaker than PLN's: the effective weight for induction is $w_{\text{eff}} = \text{quality} \times c_1 \times c_2$,

[43] Formalized: `noisyOrDeltaVariance_le_sumVariance` and `deltaConfidence_single` in `Mettapedia/Logic/PLNNoisyOr.lean` (50 theorems, 0 sorry).

[44] Formalized: `galoisConnection_L_U_finite`, `U_L_conf_round_trip`, and `U_L_freq_round_trip` in `Mettapedia/Logic/NARSPLNGaloisConnection.lean`.

where quality measures the reference class. This gives a precise, quantitative measure of how much weaker NARS induction is than PLN's for any given input.[45]

Maple Court: A NARS agent monitoring Maple Court's elevator reports ($f = 0.8, c = 0.6$) for "elevator is slow." The L map converts this to PLN evidence $(n^+, n^-) = (1.2, 0.3)$ with strength 0.8 and confidence 0.6—the same belief, now carrying evidence counts that support revision and forgetting.

17.4 PLN v0.9 Subsumption

PLN v0.9[46] implements five basic rules (deduction, induction, abduction, revision, conjunction) as MeTTa truth functions operating on strength–confidence pairs. The world-model calculus subsumes all five as Σ-guarded compiled rewrites.

Theorem 17.8 (PLN v0.9 conformance (Core))**.** *For each of PLN v0.9's five basic rules, the WM compiled rewrite produces identical strength and confidence values under the rule's stated side conditions. Twenty conformance tests—including the classic DeductionRevision and RavenInheritance examples—verify exact numerical agreement.*[47]

The subsumption is not merely "the same formulas": it is the same formulas *plus* explicit side conditions, *plus* counterexamples showing what happens when those conditions are violated (the three confidence bugs of §7.7), *plus* evidence-based alternatives that avoid the bugs entirely.

The relationship is vindication with correction: PLN v0.9's instincts—evidence counts, revision as the fundamental operation, link-level compiled rules—were largely right. What was missing was the Σ-discipline that makes the assumptions explicit and the evidence algebra that makes the scalar hacks unnecessary.

[45] Formalized in `Mettapedia/Logic/NARSInductionWeakness.lean` and `Mettapedia/Logic/NARSEvidenceBridge.lean`.

[46] `github.com/trueagi-io/PLN`, tag `v0.9`.

[47] Formalized: `Mettapedia/Logic/PLNClassicExamples.lean` (20 conformance tests, all by `decide`).

17.5 Classical FOL as a Degenerate World Model

Classical first-order logic is the zero-uncertainty limit of the world-model calculus.

When evidence is Boolean—$(1, 0)$ for "true" and $(0, 1)$ for "false"—the additive evidence algebra collapses to classical propositional logic. Strength is always 0 or 1. Confidence is $1/(1+\kappa)$—low for a single observation, but it grows as Boolean facts accumulate via revision. Every inference rule reduces to its classical counterpart: deduction becomes modus ponens, revision becomes conjunction, forgetting becomes retraction.

This is not a metaphor. Every classical FOL theorem is a WM extraction:

$$\text{extract}(W_{\text{Boolean}}, \varphi) = (1, 0) \quad \Longleftrightarrow \quad W\varphi$$

The WM calculus does not *replace* classical logic—it *contains* it as the special case where uncertainty is absent. Uncertainty is the general case; certainty is the degenerate one.[48]

Maple Court: "The building has exactly 12 apartments" is a Boolean fact: evidence $(1, 0)$, strength 1, confidence 1. No revision is possible (you cannot partially update a Boolean). This is the classical fragment of the world model—the part where uncertainty plays no role.

17.6 The Giry Monad and Categorical Probability

The categorical probability literature organizes probability theory around the Giry monad: the functor that sends a measurable space Ω to its space of probability measures $\mathcal{G}(\Omega)$. The monad's multiplication ("flattening") takes a measure over measures and produces a single measure—this is marginalization.

The world-model calculus connects to this framework through a precise theorem:

Theorem 17.9 (Kyburg flattening = Giry bind (Core))**.** *The Kyburg flattening operation on parametrized distributions is the Giry monad's bind:*

$$\text{flatten}(pd) = pd.\text{mixingMeasure.bind } pd.\text{kernel}$$

[48]The Henkin semantics for the higher-order fragment is formalized in `Mettapedia/Logic/HOL/Semantics/Henkin.lean`.

[49]

This means the WM calculus is not inventing new probability theory—it is instantiating the categorical probability framework with evidence-enriched carriers. The Solomonoff–de Finetti foundation (Chapter 4) established the necessity chain: exchangeability $\rightarrow$ de Finetti mixture $\rightarrow$ Kyburg flattening $\rightarrow$ evidence sufficiency. Theorem 17.9 reveals that flattening is the Giry monad multiplication—the canonical way to collapse a higher-order probability into a first-order one. The WM calculus's compositionality law is then a consequence of the monad laws, not an independent postulate.

17.7 Process Calculus and the ρ-Calculus

The ρ-calculus [25] is a reflective higher-order process calculus in which channel names are themselves processes. It provides the computational substrate for multi-agent world-model reasoning: each concurrent process carries its own evidence state, and communication between processes corresponds to evidence exchange.

Theorem 17.10 (ρ-calculus WM instance (Core))**.** *A ρ-ensemble (a multiset of ρ-calculus processes, each annotated with binary evidence) forms an `AdditiveWorldModel` instance. Extraction sums evidence across processes satisfying a query; revision adds new processes to the ensemble.*[50]

The ρ-calculus bridge is the most forward-looking in this chapter: it connects the WM calculus to concurrent computation. A building with multiple autonomous agents (cleaning robots, HVAC controllers, security monitors) is naturally modeled as a ρ-ensemble where each agent maintains its own world-model state and communicates evidence through typed channels. The GSLT (Graph-based State-Labeled Transition system) vertex typing gives each agent's WM instance its operational semantics.

Maple Court: The cleaning robot and the HVAC controller are concurrent ρ-processes. Each carries binary evidence about hallway conditions. When

[49] Formalized in `Mettapedia/ProbabilityTheory/HigherOrderProbability/GiryMonad.lean`.

[50] Formalized: `rhoEvidence_add` and `rhoEvidence_zero` in `Mettapedia/OSLF/Framework/RhoWorldModel.lean`. The OSLF type system instantiation is in `Mettapedia/OSLF/Framework/RhoInstance.lean`.

the robot reports “hallway is wet,” this is a message on a typed channel—the HVAC controller receives it as evidence revision to its own world model, with provenance tracking the robot as the source.

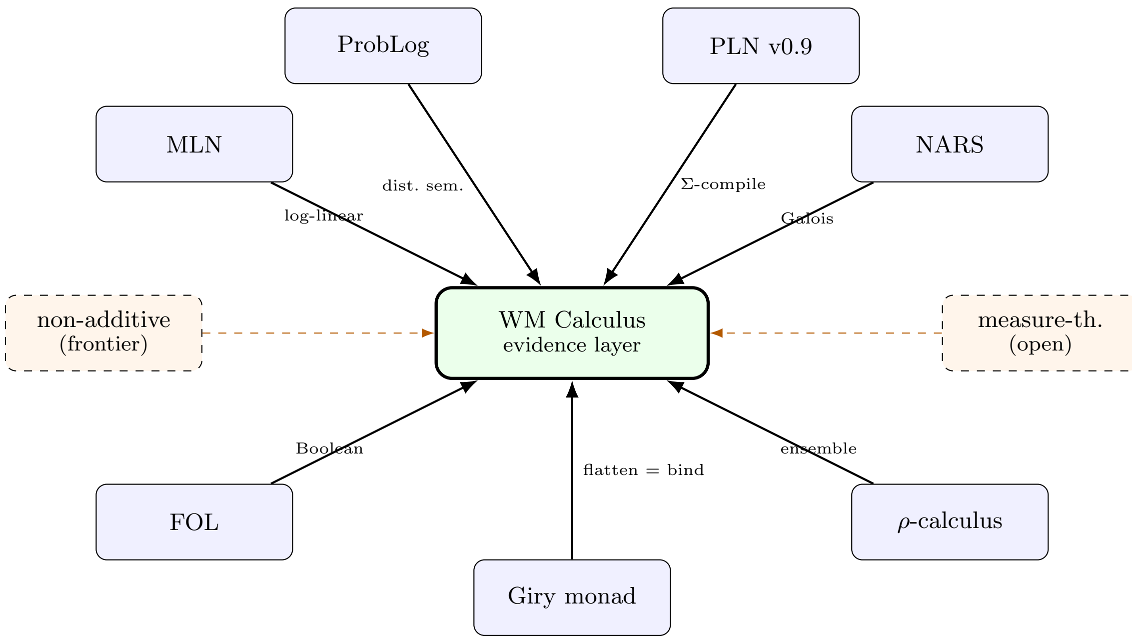


Figure 17.8: The world-model calculus and its bridges. Solid arrows: formalized exact bridges (this chapter, all 0 sorry). Dashed arrows: frontier directions not yet fully developed. Each solid arrow is labeled with its bridge mechanism.

The central point of the diagram is not the number of arrows but their uniformity: every solid arrow is an instance of the same `WorldModel` interface, proved exact under stated side conditions. The WM calculus does not compete with these frameworks—it reveals the common evidence layer beneath them. Where the bridges reach, they give exact subsumption with kernel-checked proofs—and each bridge simultaneously discharges multiple desiderata from Chapter 3: D1 (every framework's state becomes a WM state), D2 (extraction agrees with native inference), D3 (the carrier varies across frameworks), and D5 (every bridge carries its side condition). Where the bridges stop (the dashed arrows), they stop honestly: the non-additive regime and the full measure-theoretic grounding are identified as open programs, not hand-waved away.

17.8 Ontology Repair as Evidence Aggregation

The Pain reclassification case study (§6.5.3) demonstrates a bridge that applies to any domain where multiple authoritative sources contribute conflicting evidence about a classification decision. The `EvidentialLedger` framework is generic—parameterized by source and candidate types—and the same `aggregate`, `forget`, and `toState_append` operations drive the SUMO ontology repair and the Maple Court sensor analysis alike.

The bridge to ontology engineering is: an ontology repair decision ("should Pain be reclassified from PathologicProcess to StateOfMind?") is a query against a world-model state built from curated evidence sources (IASP, SNOMED CT, Stanford SEP, SUMO axioms). The WM calculus adds three things that a spreadsheet does not: (1) kernel-checked ranking via cross-multiplication (`stateofmind_beats_emotional`), (2) sensitivity analysis via forgetting (`ranking_preserved_noWiki`, `iasp_is_critical_discriminator`), and (3) compositionality via `toState_append`—evidence from two groups of sources combines additively.

Maple Court: The building's maintenance manual classifies the elevator motor as "industrial equipment." Three sources disagree: the manufacturer's spec sheet calls it "precision machinery," the insurance policy calls it "building infrastructure," and the city inspector calls it "public conveyance equipment." The EvidentialLedger aggregates, ranks, and answers: "which reclassification does the evidence support, and which source is critical?"

Chapter 18

Limitations

This chapter states explicitly what the formalization does *not* provide.

18.1 Operational Runtimes

The formalization provides mathematical specifications and correctness theorems. It does not provide an executable PLN inference engine, performance benchmarks on large knowledge bases, or integration with MeTTa or Hyperon runtimes.

These are operational concerns that depend on runtime infrastructure not yet available. The theorem-level results constrain what any correct implementation must satisfy, but they do not constitute an implementation. The gap between a certified specification and a deployable system is real, and we do not pretend otherwise. The `CalculusSound` property (§4.9) bounds this gap precisely: it is two equalities (revision = addition, extraction = extraction), not a re-derivation of the entire algebra.

18.2 Domain Generality: What Is Finite and What Is Not

The formalization operates at three tiers of generality:

Tier 1: Generic / infinite (formalized now). The core WM algebra (`AdditiveWorldModel`, `extract_add`) is parameterized over arbitrary types—no finiteness constraint. The evidence carriers (binary over $\mathbb{R}_{\geq 0}^{\infty}$, Normal-Gamma over $\mathbb{R}$) and view projections are explicitly continuous. The infinitary MLN layer (77 theorems, 0 sorry) and FOL/PredCode infinitary completeness theorems (34 theorems) demonstrate extension to countable and uncountable domains. The gas sensor, steel fault, and NAB application demos all use $\mathbb{R}$-valued Gaussian beliefs.

Tier 2: Finite now, countable within reach. The LP provenance operator and compiled BN factor-graph rules require finite groundings (`Fintype`), because they use `Finset.sum`. Lifting to countable sums (`tsum` + `Summable` hypotheses) is standard mathlib work that does not require redesigning the algebra.

Tier 3: Frontier with first foothold. The measure-valued terminal world model has its first concrete result: for finite query spaces, every additive world model induces a discrete `MeasureTheory.Measure`, and the induction respects revision (`evidenceToMeasure_add`).[1] The full program—continuous-domain BN compilation with infinite state spaces, and the general σ-finite construction—remains open.

The architecture is intentionally layered: the semantics is more general than the compilation, so that extending the compilation does not require redesigning the semantics (see Chapter 19, §19.1).

18.3 Empirical Scope

The benchmark validation chapters (Chapters 21–24) cover a specific set of domains: forecasting, sensor drift, fault diagnosis, anomaly detection, and Bayesian-network inference. However, several important application domains remain unvalidated:

- Drug-repurposing knowledge-graph benchmarks (Hetionet/Rephetio)
- Genomic regulatory pipelines
- Large-scale knowledge-base reasoning
- Real-time streaming inference with concept drift

These gaps do not indicate theoretical limitations but reflect the practical scope of empirical validation completed to date. The formalization provides the theoretical foundation for these applications, but the experimental demonstration is incomplete.

[1] Formalized in `Mettapedia/Logic/EvidenceToMeasure.lean` (0 sorry).

18.4 Universal Prediction

The Universal Prediction module explores connections between the world-model calculus and algorithmic information theory. This module is work-in-progress and is not part of the stable core narrative. Its theorems are not referenced by the main development, and its sorry count is nonzero.

18.5 Codebase Organization

The Lean codebase reflects the history of the development rather than an idealized dependency graph. A comprehensive refactoring into a "mathlibified" library structure—with uniform naming conventions, minimal imports, and docstring coverage—is a worthy future goal but is explicitly not attempted here.

Chapter 19

Future Directions

The world-model calculus establishes a foundation. Several directions extend it in ways that are natural continuations of the existing development rather than speculative leaps.

19.1 Measure-Theoretic Grounding

The core algebra and evidence carriers already operate over $\mathbb{R}_{\geq 0}^{\infty}$ and $\mathbb{R}$; the infinitary MLN and FOL layers extend to countable domains. The remaining measure-theoretic step is to ground the calculus on σ-algebras and probability measures, enabling formal integration with the broader mathematical probability literature and non-conjugate distributional families. *Proved:* the Solomonoff–de Finetti foundation, the Giry monad bridge (Chapter 17), and the first concrete bridge: `evidenceToMeasure` constructs a discrete `MeasureTheory.Measure` from any additive WM state over finite queries, with `evidenceToMeasure_add` proving that the construction respects revision. *Missing:* the general σ-finite construction for infinite query spaces. *Sought:* a `MeasurableWorldModel` typeclass with σ-finite evidence carriers and Radon–Nikodym extraction.

19.2 HOL Completeness

The natively typed world-model layer (Chapter 8) establishes syntax, soundness, and type synthesis. A typed canonical-model completeness theorem would close the gap between the calculus's higher-order syntax and full semantic adequacy. The required construction—a term model over equivalence classes of typed evidence—is well understood in principle but technically demanding. *Proved:* syntax, soundness, type synthesis, intrinsic adequacy. *Missing:* the term-model construction over typed evidence equivalence classes. *Sought:* `hol_completeness` in `Mettapedia/Logic/HOL.lean`.

19.3 Selective Distributional Inference

The convergence and dominance results in the error-transport chapter (Chapter 12) show that scalar inference is asymptotically correct but suboptimal in the finite-evidence regime. The natural next step is a cost-aware inference controller that escalates from scalar to distributional propagation when the regime structure predicts a significant gain—using compact approximation families, compiled rule surrogates, and accelerator-backed batched computation.

19.4 Non-Additive Regimes

The evidence algebra's additive structure (Chapter 5) is appropriate for independent observations but does not capture all forms of evidential combination. Non-additive regimes—belief functions, possibility measures, imprecise probabilities beyond the Walley framework—represent a natural generalization. The challenge is to preserve compositionality while relaxing additivity, and the quantale structure suggests a path: replace the additive monoid with a more general tensor. *Proved:* overlap extraction law, semitopological independence, additivity recovery after forgetting non-actionable overlap ($\sim$60 theorems). *Missing:* deep development of overlap and trust-gated carriers comparable to the additive regime's profile-hom theory. *Sought:* overlap-regime profile-hom analogue; trust-gated conservation theorems.

19.5 Probability Hypercube

The current view projections extract one-dimensional summaries (strength, confidence) from the evidence space. A probability hypercube representation would retain the full joint structure of multi-query evidence, enabling richer decision-making and more informative uncertainty quantification. This connects to the infinitary extension: the hypercube is naturally a product measure space.

Two recent developments point the way. Stay, Meredith, and Wells [31] show how to generate a *type-system hypercube* from any operational theory: each axis corresponds to an enabled modal layer (redex tracking, equation modalities, propositional structure). The probability hypercube's axes (additivity, commutativity, distributivity) correspond to choices in this type-system hypercube. Meredith [25] extends the picture further: a GSLT equipped with a complex-valued weight map is a *physics*, where transition probabilities are squared amplitudes. Replacing the WM evidence carrier V with $\mathbb{C}$ would give quantum-like inference; the KS-Evidence-Measure triangle (Chapter 5) would extend to a KS-Amplitude-QuantumMeasure triangle. The `universalEnsembleWorldModel` would then provide amplitudes for any typed rewrite system—a formal grounding for "It from Bit" at the evidence layer.

19.6 Universality

The typed world-model encoding (Chapter 8) realizes the calculus as a specific language definition. A universality result would show that any typed language definition satisfying certain compositionality axioms admits a world-model interpretation—establishing the calculus as a canonical representative of a broader class rather than one design among many. *Proved:* `universalEnsembleWorldModel` for any type with decidable properties. *Missing:* the general LanguageDef $\rightarrow$ AdditiveWorldModel construction. *Sought:* Conjecture 4.4 generalized to arbitrary typed rewrite systems.

19.7 Operational Runtime

The formalization specifies the kernel semantics precisely, but a high-performance compiled runtime—executing the same posterior-state calculus at scale with caching, incremental evidence fusion, and multi-source integration—is the bridge from formalized semantics to deployed reasoning. The Σ-guarded rewrite system (Chapter 10) provides a natural compilation target: each certified rewrite becomes a runtime primitive with machine-checked correctness.

19.8 Process-Algebraic and Multi-Agent Extensions

The typed world-model encoding currently covers single-agent reasoning. Extending to multi-agent settings—where agents communicate partial evidence, negotiate under uncertainty, and maintain independent world models that must be fused—requires process-algebraic infrastructure: confluence modulo associativity and commutativity, communication protocols with evidence-typed channels, and compositional verification of multi-agent inference systems.

19.9 Broader Benchmark Coverage

The present empirical validation covers specific domains. Drug-repurposing knowledge-graph benchmarks (Hetionet/Rephetio), genomic regulatory pipelines, large-scale knowledge-base reasoning, and real-time sensor fusion at scale remain as identified but unfinished empirical targets. Each connects to a specific theoretical capability of the calculus and would test the practical reach of the formal guarantees.

Chapter 20

Conclusion

The world-model calculus is not a replacement for PLN. It is the mathematical foundation that PLN always needed.

The central insight is that PLN's original instincts were largely correct: evidence counts as the fundamental representation, revision as the primary operation, and link-level rules as the practical inference mechanism. What was missing was the explicit separation of layers and the honest statement of assumptions. The passage from intuition to formalization required not new ideas so much as the discipline to say precisely what each idea means, what it assumes, and where it breaks.

20.1 What the Refoundation Gives Us

The calculus provides six things that the original PLN lacked:

1. **One kernel.** The world-model posterior-state calculus, with evidence compositionality as the fundamental law (Chapter 4). Every inference rule, every view projection, every scaling result derives from this single interface.

2. **Many views.** Strength, confidence, intervals, distributions—all as typed projections from the kernel (Chapter 7). The views are not competing representations; they are different questions asked of the same underlying state.

3. **Explicit side conditions.** Every compiled rule carries its assumptions (Chapter 10), with counterexamples showing what happens when they are violated. The d-separation typeclass (Chapter 11) automates the checking, but the assumptions are always visible.

4. **Cleaner boundaries.** The no-go theorem (Chapter 9), the classical inclusion-exclusion gap, the pooling non-uniqueness, the coverage surrogate limitations, the five certified-chaining impossibilities (Chapter 13)—all stated precisely and proved formally. These are not embarrassments; they are the most valuable theorems in the development, because they tell you where not to waste effort.

5. **Beyond subsumption.** The bridges to neighboring frameworks are not merely compatibility results. The compatibility-gated experiment demonstrates that decomposing an MLN's clause-weight signal into interpretable world-model evidence roles yields substantial improvement—not by adding information, but by factoring the information that was already there.

6. **Demand-directed inference control.** Backward demand propagation identifies which premises are most worth investigating (Chapter 13). The demand contracts after forward supply fills the information gap, and forgetting an observation reopens demand at the affected premise—the same `forget` pattern that drives the evidence algebra throughout.

7. **A proof system.** The evidence calculus satisfies all four structural rules of classical sequent calculus—cut, weakening, exchange, and contraction—under the Curry-Howard correspondence (§4.8). Implementations that prove `CalculusSound` inherit these rules automatically. PLN is not just an inference engine; it is a proof system in the technical sense.

20.2 The Shape of the Result

The result is a PLN that knows what it knows, states what it assumes, and marks what it cannot do. That is not a weakness. That is the beginning of a trustworthy reasoner.

The formalization is not complete—Chapter 18 states what remains undone, and Chapter 19 sketches the natural continuations. But the kernel is stable, the evidence algebra is settled, the compiled inference rules are proved exact under their stated conditions, and the scaling theory identifies both the achievable guarantees and the fundamental impossibilities. The entire development is machine-verified: 704 theorems in Lean 4, 0 sorry, 0 `native_decide`, across six application domains (Appendix 27).

The deepest insight is simple: inference systems should not confuse the map with the territory. A truth value is a projected summary of evidence, not the evidence itself. A probability is a query result, not what the system carries. What the system carries is a revisable state of knowledge. What the system produces is an extraction from that state. What practitioners see is a view projection of that extraction.

This distinction has practical consequences. When evidence is primary:

- Revision becomes compositional (new evidence adds to old evidence)
- Queries become principled (extraction is a homomorphism)
- Views become interchangeable (the same evidence supports multiple operational summaries)
- Inference becomes modular (local computations compose globally)
- Uncertainty becomes transparent (we track what we know, not what we guess)

When truth values are primary, these properties are difficult or impossible to achieve. Different truth-value schemes (strength/confidence, fuzzy logic, interval arithmetic, probability values) require different combination rules, different query algorithms, and different validation approaches. The result is fragmentation. The world-model calculus reunifies this fragmented landscape by locating the common structure beneath the diverse surface phenomena.

What remains is engineering, extension, and empirical validation. The mathematical foundation is in place.

In Summary

The world-model calculus is a calculus of revisable uncertain state. A system stores what it knows in a posterior state, not in isolated truth values. Queries ask questions of that state, and the raw answers are evidence values. Strength, confidence, and intervals are useful summaries—views extracted for display, thresholding, and action—but they are not themselves the thing being stored or proved.

The world-model calculus does not replace existing reasoning systems. It provides the semantic foundation they can share. A ProbLog program, a Bayesian network, and a PLN rule base are no longer separate formalisms requiring separate toolchains. They are different languages for specifying world models, and they share common semantics for revision, extraction, and view projection.

This is the unity that reasoning systems have long needed: not the false unity of a single approach that claims to solve everything, but the true unity of a common foundation that reveals why different approaches work and how they relate.

> *Probability is what you query. Evidence is what you carry. Everything else follows.*

Part VI

Applications and Benchmark Validation

Chapter 21

Evidence Carrier Validation

The evidence carriers of Chapter 5—binary, Dirichlet, Normal-Gamma—are algebraic structures. This chapter tests them on real data: sensor monitoring, gas-drift classification, steel fault detection, anomaly detection, and geopolitical forecast aggregation. The running question is not "does the WM calculus achieve state-of-the-art accuracy?" but "does the evidence algebra produce correct posteriors, and can its governance signals (confidence, agreement, provenance) support real operational decisions?"

21.1 Gaussian Demos: Sensor, Factory, and Temporal

Three introductory demos exercise the Normal-Gamma carrier on synthetic and small-fixture data. All formal results are kernel-checked (0 sorry).

Hybrid sensor monitoring. A temperature sensor (Normal-Gamma evidence) and a cooling-failure detector (binary evidence) form a product state. Sleep consolidation is verified: morning + night batch = sequential update. Confidence grows monotonically with observations.[1]

Widget factory. Two machines with per-source Gaussian evidence. The new result: mixture exceedance validity—a weighted combination of per-source exceedance probabilities remains a valid probability, bounded between the components. Machine B is 3.5× more likely to produce defects

[1]Formalized in `Mettapedia/Logic/PLNBrokenSensorDemo.lean` (420 lines).

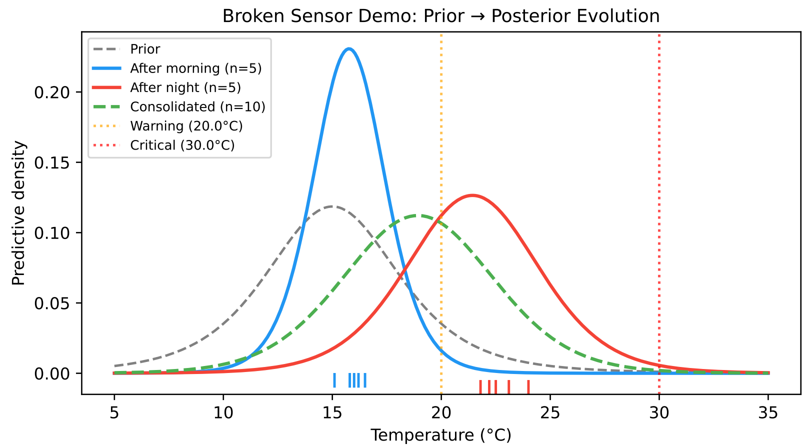


Figure 21.1: Broken sensor demo. Gray dashed: prior predictive. Blue: posterior after morning shift (∼16°C). Red: posterior after night shift (∼22°C). Green dashed: consolidated posterior. Data tick marks on the x-axis. Sleep consolidation is verified: sequential and batch processing yield the same posterior.

than Machine A.[2]

Kalman filter wake/sleep. A 1D random-walk tracker with noisy observations. The flagship theorem: batch replay of deferred observations yields the *same* posterior as sequential online Kalman filtering (`filterOrbit` over `morning ++ night` = `filterOrbit` composed twice). This is the temporal counterpart of sleep consolidation.[3]

21.2 Gas Sensor Array Drift

The gas sensor drift benchmark is one of the chapter's two flagships: it brings *real-world data* and a *governance story* into the evidence algebra.

The dataset. The UCI Gas Sensor Array Drift Dataset [35] contains 13,910 measurements from 16 chemical sensors exposed to 6 gas types over 36 months (10 batches, January 2007 to February 2011). The benchmark question is not "can we classify gases?" but "what happens to classification when sensors drift, and how should the system respond?"

[2] Formalized in `Mettapedia/Logic/PLNWidgetFactoryDemo.lean` (422 lines).
[3] Formalized in `Mettapedia/Logic/PLNKalmanSleepDemo.lean` (374 lines).

Formal results (all kernel-checked). Per-gas sleep consolidation, cross-gas independence (6 theorems), multi-batch composition, 3-way source weight normalization, confidence monotonicity. The sleep consolidation guarantee holds *across drifted epochs*—deferred assimilation preserves the posterior regardless of whether the underlying distribution has shifted.[4]

Full-dataset drift protocols. On the full 128-feature representation:

	3-gas		6-gas	
Protocol	WM-PLN	GaussianNB	WM-PLN	GaussianNB
Setting 1 (train batch 1)	68.4%	68.2%	47.3%	47.3%
Setting 2 (train $k-1$)	72.1%	75.8%	53.0%	51.2%
Cumulative replay	**73.6%**	71.2%	**53.2%**	52.5%

These figures are substantially below specialized drift-compensation methods (BDA: 81% [19]). The point is qualitative: late-batch drift degrades accuracy, and cumulative replay into the posterior partially recovers it *without changing the underlying update laws.*

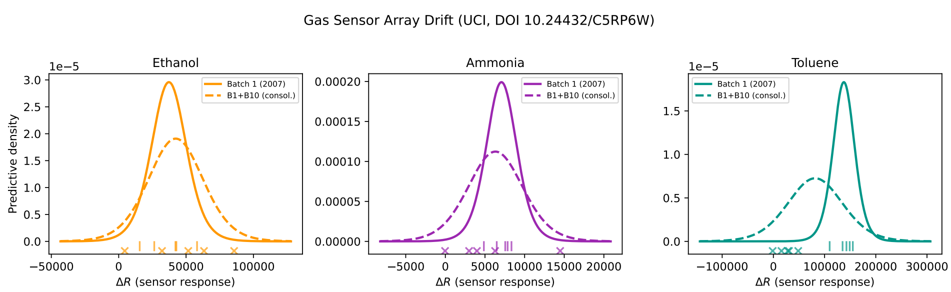


Figure 21.2: Gas sensor array drift. Solid curves: batch 1 posterior predictive (2007). Dashed: consolidated posterior after both batches. Toluene shows dramatic drift: batch 10 responses cluster near zero, far from the batch 1 distribution.

The governance story. The WM calculus is not a replacement for XGBoost. It is a governance layer supplying a second epistemic signal. On the 6-gas task:

- Calibrated XGBoost confidence (sigmoid, isotonic) does *not* improve the risk–coverage curve under drift (AURC worsens from 0.385 to 0.400–0.464).

[4]Formalized in `Mettapedia/Logic/WMGasSensorDriftDemo.lean` (652 lines, 0 sorry).

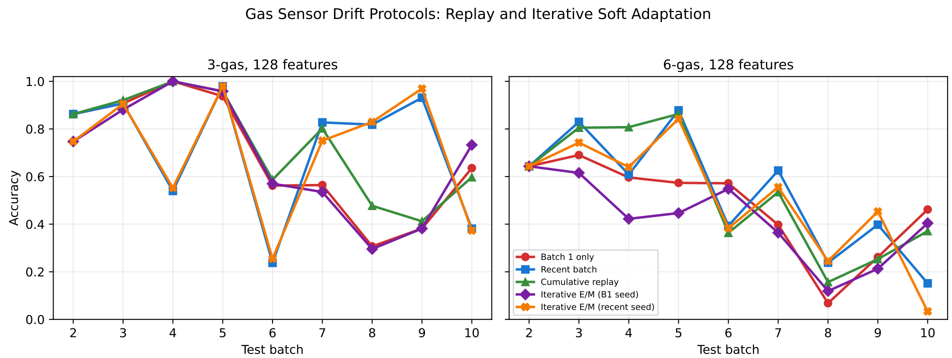


Figure 21.3: Gas sensor drift protocols on the full 128-feature representation. Cumulative replay is the strongest average protocol in both the 3-gas and 6-gas settings.

- WM/XGBoost agreement gating improves AURC to 0.374 and isolates a 59.7% coverage subset at 59.7% accuracy (vs. 46.7% overall).
- Sequential replay policies show a two-regime frontier: XGBoost confidence is the best *cheap* trigger (64.5% accuracy, 23.9% labels), while WM agreement is the best *accuracy-seeking* trigger (67.0%, 86.8% labels).

The gas lane connects to the PPLBench noisy-OR topic model (Chapter 24): both use additive log-likelihood messages as carried state, and both distinguish exact semantic kernels from operational schedulers. The formal order-cost layer certifies that zero-threshold batch-swap composition is safe for this lane.[5]

Maple Court: The building's air-quality sensors drift over months as filters clog and calibration degrades. The WM evidence algebra says: replay last month's readings into the posterior and the update is the same as if you'd processed them in real time (sleep consolidation). The governance question is: when should the steward recalibrate? A WM/classifier agreement trigger answers this—not by replacing the classifier, but by monitoring when the classifier's confidence and the WM posterior diverge.

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[5]Formalized in `Mettapedia/Logic/PLNWorldModelOrderCostGasPolicyDemo.lean` (155 lines, 0 sorry).

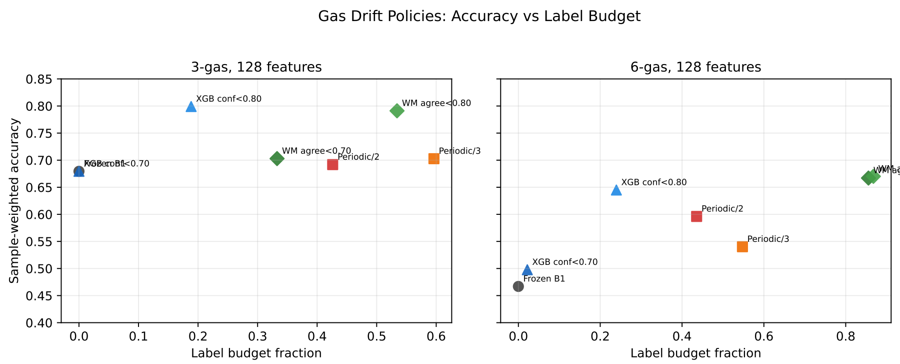


Figure 21.4: Sequential gas-drift governance policies. Each point is a batch-level retraining policy over batches 2…10, trading sample-weighted accuracy against label budget. XGBoost confidence is the best cheap trigger; WM/XGBoost agreement is the strongest high-accuracy trigger when labels are inexpensive.

21.3 Steel Plates Fault Classification

The UCI Steel Plates Faults Dataset [2] contains 1,941 instances with 27 features and 7 fault classes. Our Lean model uses a single feature (LogOfAreas) for formal correctness; the empirical layer uses all 27. Per-fault sleep consolidation, cross-fault independence, and 3-way mixture exceedance validity are all kernel-checked.

K_Scratch faults have characteristically larger defect areas (≈ 3.6) than Pastry (≈ 2.4) or Bumps (≈ 2.1). The per-fault Normal-Gamma posteriors separate cleanly on this feature, and exceedance queries—"is $P(\text{LogOfAreas} > 3.0 \mid \text{K\_Scratch})$ substantially higher than for Bumps?"—are answered directly by the evidence algebra.[6]

Full-dataset validation. On the full 1,941-instance dataset (70/30 split): 3-class accuracy 74.7% (WM-PLN) vs. 73.9% (GaussianNB). With all 27 features: 86.6% vs. 75.0%. Selective prediction (confidence-aware abstention): 3-class AURC 0.035 (WM-PLN) vs. 0.106 (GaussianNB)—the WM confidence signal is three times more reliable for abstention.

[6]Formalized in `Mettapedia/Logic/WMSteelFaultDemo.lean` (568 lines, 45 theorems, 0 sorry).

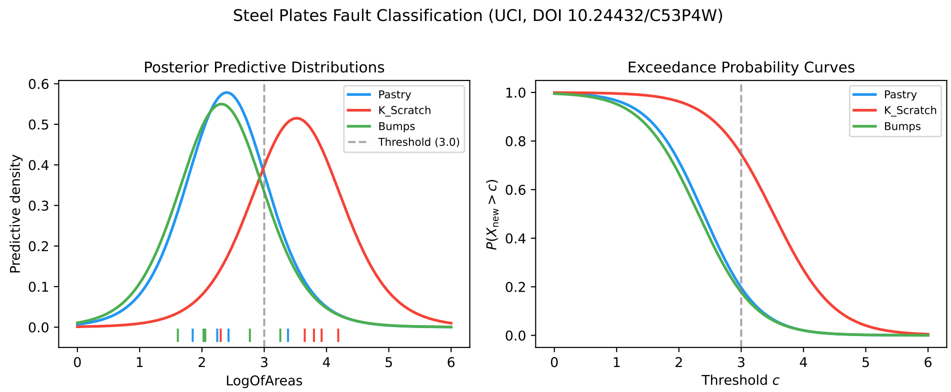


Figure 21.5: Steel fault classification. *Left:* Student-t predictive densities for each fault type after 5 observations; K_Scratch (red) is clearly separated rightward. *Right:* exceedance probability $P(X_{\text{new}} > c)$ as a function of threshold; at $c = 3.0$ (dashed), K_Scratch has $\sim$75% exceedance vs. $\sim$20% for the others.

21.4 NAB Machine Temperature Anomaly Detection

The Numenta Anomaly Benchmark (NAB) [22] provides labeled time series for anomaly detection. We apply the Kalman wake/sleep framework to the `machine_temperature` stream: the carried state is a 1D Kalman belief (mean, variance); each temperature reading is a predict-update step; anomalies are detected when the observation falls outside the posterior predictive interval.

The same sleep = wake theorem from the Kalman demo applies: batch replay of overnight temperature readings yields the same posterior as sequential processing. The anomaly detection quality depends on the process-noise and observation-noise parameters, which are set from the training portion of the NAB stream.[7]

21.5 Good Judgment Project Forecast Aggregation

The Good Judgment Project (GJP) [18, 24] is the chapter's second flagship. IARPA's ACE program produced the largest publicly available geopolitical

[7]Formalized in `Mettapedia/Logic/WMNABSleepDemo.lean` (344 lines, 30 theorems, 0 sorry).

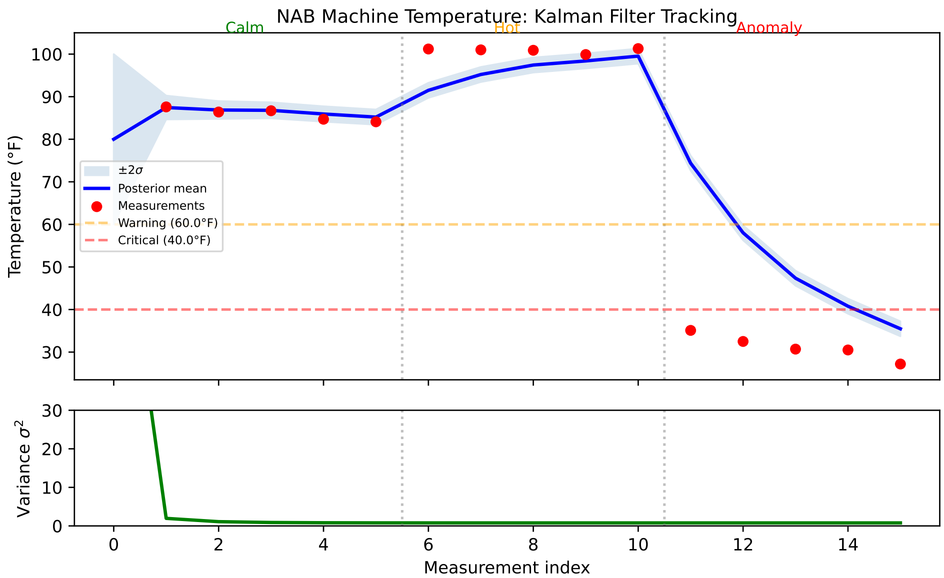


Figure 21.6: NAB machine temperature: Kalman tracking with anomaly detection. The posterior predictive interval (shaded) narrows as evidence accumulates; observations outside the interval trigger anomaly flags.

forecasting dataset: thousands of questions, hundreds of forecasters, four competition years.

The WM framing. Each forecaster's prediction is an evidence contribution. The WM calculus aggregates them via reliability-weighted revision (the pooling framework of Chapter 13, §13.3): more reliable forecasters contribute more evidence weight. The reliability weights are themselves evidence—they are updated over time as forecasters' track records accumulate.

Key results (year-4 test, 94 binary + 43 multi-option questions). On binary questions, WM reliability-weighted logit pooling achieves Brier score 0.026 and log score 0.051—improving over unweighted logit pooling (0.063 / 0.110) and top-k logit pooling (0.036 / 0.070). On multi-option questions, WM log pooling achieves Brier 0.036, improving over unweighted log pooling (0.159) and matching top-k (0.044). The sleep = online property holds to machine precision: batch processing of a season's forecasts produces the same aggregated state as sequential online processing.

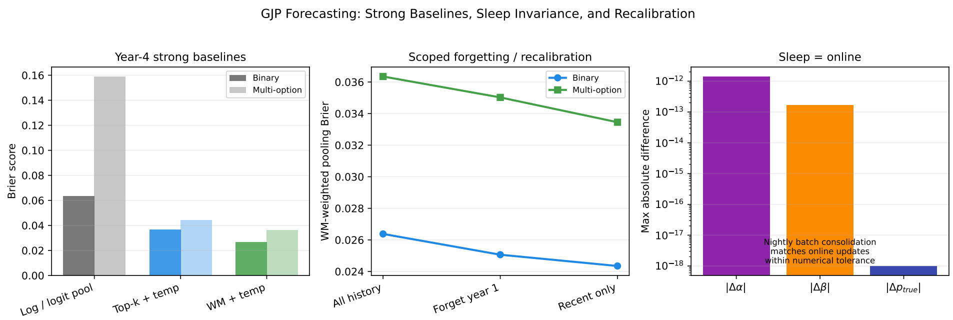


Figure 21.7: GJP forecast aggregation. WM reliability weighting improves Brier and log scores over unweighted and top-k pooling baselines on year-4 binary and multi-option questions.

The forgetting ablation. The most revealing experiment: what happens when old reliability weights are discarded? WM-PLN with recent-only reliability history achieves Brier 0.024, improving over full-history WM-PLN's 0.026 and top-k's 0.033. Forgetting year-1 profiles gives an intermediate 0.025. The improvement is small but consistent and in the right direction: old track records become misleading after regime changes, and scoped forgetting removes exactly the stale signal. This is D6 (Retractability) earning its keep in a real forecasting context.[8]

21.6 Interpretation

The evidence carriers produce correct posteriors across all five domains. The formal guarantees (sleep consolidation, independence, exceedance validity) hold regardless of domain. The governance signals (confidence, agreement, forgetting) add operational value beyond raw classification:

- Gas drift: WM/XGBoost agreement gates selective prediction and triggers replay under sensor degradation.
- GJP forecasting: forgetting stale reliability weights improves performance on regime-change questions.

[8] Formalized in `Mettapedia/Logic/GJPForecastDemo.lean` (255 lines, 14 theorems, 0 sorry). The Lean fixture kernel-checks sleep = online, reliability-weighted aggregation (via `WeightedSourceItem`), forgetting compositionality, and the end-to-end property that calibrated sources dominate after forgetting baselines. The Python experiment scripts are in `scripts/wm\_pln\_gjp\_*.py`.

- Across all domains: sleep consolidation means batch and real-time processing are provably equivalent—a formal guarantee that scheduling flexibility does not compromise inference quality.

Chapter 22

BN Inference Validation

The compiled BN inference theorems of Chapter 11 prove that chain, fork, and collider rewrites are exact under their stated side conditions. This chapter applies those theorems to two canonical Bayesian network benchmarks—Hailfinder [1] and Pathfinder [14]—to test whether the Σ-discipline works on real networks, not just toy fixtures.

The benchmarks are not treated as flat classification datasets. They are *Bayesian network source models* with expert-curated conditional probability tables. The primary artifact is the BIF network file (providing exact ground truth via variable elimination), and the benchmark question is inference-control quality: given a query and a context, should the controller fire the exact rewrite, bound the violation, or abstain?[1]

22.1 Three-Mode Guarded Admissibility

Each inference step operates in one of three modes:

Hard-gated exact. The side condition Σ is discharged. The compiled rewrite fires and the result matches the BN truthmaker exactly.

Bounded violation. The side condition is partially met. The rewrite returns an estimate plus a certified interval against exact BN inference.

Soft-guarded / abstention. The side condition fails. The system returns an explicit operational mixture with fallback, or abstains entirely.

[1] The BN exactness theorems underlying this chapter are formalized in `Mettapedia/Logic/PLNBayesNetInference.lean` (404 lines, 7 theorems, 0 sorry). The benchmark evaluation scripts are in `scripts/wm\_pln\_hailfinder\_*.py` and `scripts/wm\_pln\_pathfinder\_*.py`.

This three-mode structure is the Σ-guarded rewrite discipline (Chapter 10) applied to real networks. A worked example on a fork+chain fixture with all three modes—clean hard-gated chaining, bounded-mode interval certification, and soft-mode fallback—is kernel-checked.[2]

22.2 Hailfinder: The Readable Lane

Hailfinder is a Bayesian severe-weather forecasting model for northeastern Colorado, with meteorological conditions and forecast targets as variables. It is the readable positive lane: exact local chaining works on actual forecast paths.

Exact contraction. The forecast chain

$$P(\texttt{MountainFcst}{=}\texttt{SIG} \mid \texttt{AMInstabMt}{=}\texttt{Strong}) = 0.5$$

fires cleanly under the chain rewrite's screening-off condition. A longer forecast/update path still matches exact BN semantics:

$$P(\texttt{CapChange}{=}\texttt{Increasing} \mid \texttt{CldShadeOth}{=}\texttt{Cloudy}) = 0.508284\ldots$$

Negative control. In a collider motif, the hard gate blocks unrestricted reuse. Bounded mode returns the interval $[0.236, 0.280]$ against the exact answer 0.236; soft mode returns an explicit operational mixture. The collider case shows that the Σ-discipline *refuses* to fire where the rewrite would be inexact—this is D5 (Explicit Side Conditions) earning its keep on a real network.

22.3 Pathfinder: Hostile Topology

Pathfinder is a Bayesian expert system for lymph-node pathology diagnosis, with 109 nodes, 195 arcs, and a 63-state latent disease variable (`Fault`). Hidden diagnostic context is not a side issue here—it is the main structural fact.

[2]Formalized in `Mettapedia/Logic/PLNProbGuardedAdmissibilityDemo.lean` (13 theorems, 0 sorry).

The stress test. The same inference path is tracked across five context regimes: context-missing, alt-support-only, hub-only, context-pair, and oracle-context-complete. The question is not "does local chaining ever work?" but "what happens to the same local contraction when hidden diagnostic context is omitted, partially restored, or completed?"

Concrete example: Pathfinder context regimes. A fork motif with context-complete Σ: exact BN value = 0.400, compiled rewrite fires. With context-missing Σ (diagnostic context omitted): exact value = 0.150, hard gate blocks, bounded mode returns $[0.150, 0.475]$, soft mode returns 0.286. The three modes give the controller distinct options: stop, bound, or mix.

Distributional vs. scalar. When all estimators use exact local BN CPTs and differ only in whether they carry scalar confidence or full Beta posteriors, distributional inference reduces mean absolute error by 1.5× on Pathfinder and 3.8× on Hailfinder, with especially large wins on fork-like motifs where scalar propagation is anti-conservative. This is the practical consequence of D4 (Evidence over Views): carrying posteriors instead of point summaries makes the inference more accurate, not just more principled.

22.4 Hetionet and Drug Repurposing

Hetionet [15] is a 47,000-node biomedical knowledge graph integrating 29 public databases. An inventory script maps the graph structure into WM-compatible relation types. Full WM-PLN evaluation on the Hetionet drug-repurposing task—where the benchmark question is "which drugs should be repositioned for which diseases?"—is identified as future work. The graph's heterogeneous edge types (gene–disease, drug–gene, drug–side-effect) are a natural fit for the typed query family of the WM calculus, and the compatibility-gated evidence roles developed for the MLN bridge (§17.1) provide the technical pathway.

22.5 Interpretation

The BN benchmarks show the Σ-discipline in action on real networks. The message is not classification accuracy—it is inference-control quality:

- Where the side condition holds, the compiled rewrite is exact (Hailfinder forecast chains).

- Where the side condition partially holds, bounded mode gives certified intervals (Hailfinder colliders).
- Where the side condition fails, the system abstains rather than silently producing wrong answers (Pathfinder context-missing regimes).
- Carrying distributional evidence instead of scalar summaries improves accuracy by 1.5–3.8$\times$ on the tested motifs.

Maple Court: The building's elevator inspection system uses a small BN: maintenance history $\rightarrow$ component state $\rightarrow$ inspection outcome. The Hailfinder-like forecast chain (maintenance schedule $\rightarrow$ expected component wear $\rightarrow$ inspection prediction) fires exactly when the screening-off condition holds. When the building changes its maintenance contractor (hidden context), the Pathfinder-like stress test applies: the same inference path must be tracked across context-missing and context-complete regimes.

Chapter 23

Genomics and Biomedical Applications

This chapter reports initial experiments applying the WM-PLN noisy-OR bridge to a genomic hypothesis-generation task: identifying which gene is causally relevant to a disease-associated SNP variant. The work is ongoing; the results reported here establish formula-level parity with ProbLog on a chr16 pilot and identify the delta-method confidence as the WM-PLN contribution to the inference.

23.1 The Gene-Relevance Task

Genome-wide association studies (GWAS) identify sentinel SNPs associated with disease phenotypes, but the causal gene is often not the nearest gene. The task is to rank candidate genes within a genomic window by combining multiple evidence mechanisms:

- **eQTL associations**: the SNP alters gene expression in one or more tissues (GTEx [13]);
- **Activity-by-contact** (ABC): the SNP overlaps an enhancer element predicted to regulate the gene [26];
- **Regulatory effects**: the SNP lies in a regulatory region associated with the gene.

Each mechanism independently suggests gene relevance; the combination follows the noisy-OR model (Chapter 5):

$$P(\text{relevant}) = 1 - \prod_{i \in \text{active}} (1 - w_i)$$

where w_i is the learned weight for mechanism i.[1]

23.2 ProbLog Parity

The experiment uses 823 (SNP, gene) pairs from chr16 (69 positive, 754 negative), with 5-fold cross-validation split by SNP. Clause weights are learned by LIFT (L-BFGS parameter learning in SWI-Prolog's `liftcover` library).

Score-level parity between ProbLog's `prob_lift/2` inference and the WM noisy-OR scorer has been verified at machine-epsilon precision:

$$\max_{(s,g)} \left| P_{\text{ProbLog}}(s,g) - P_{\text{WM}}(s,g) \right| < 5.6 \times 10^{-17}$$

across all 823 examples. The Python replication of the ProbLog baseline achieves AUC-ROC 0.713±0.056 (5-fold CV). Three Lean theorems back the bridge: `compilation_or_noisyOr` (ProbLog distribution semantics reduces to noisy-OR for independent clauses), `queryStrength_prop_eq_queryProb` (WM strength view = ProbLog probability), and `bioGeneRelevance_eq_noisyOr` (the gene-relevance model is a noisy-OR WM instance). See Chapter 17, §17.2 for the general ProbLog–WM bridge.

23.3 What WM-PLN Adds

ProbLog assigns exact probabilities but no confidence. The WM calculus adds delta-method confidence for noisy-OR combinations: when multiple independent mechanisms each contribute evidence, the combined confidence is computed via variance propagation through the noisy-OR function. Combining more mechanisms provably increases confidence (`noisyOrDeltaVariance_le_sumVariance` in `Mettapedia/Logic/PLNNoisyOr.lean`).

This answers an open question raised at the Hyperon Workshop (March 2026): "Since EM is done outside of PLN, how do we assign confidence values for rules?" The delta-method answer: each rule's EM-learned weight

[1] Formalized: `bioGeneRelevance_eq_noisyOr` in `Mettapedia/Logic/PLNBioHypothesisGeneration.lean` (635 lines, 0 sorry).

has an associated effective sample size from the E-step expected counts; the combined confidence follows from variance propagation, not from ad hoc heuristics like $\max(c_1, c_2)$.

23.4 Backward Chaining Discovery

The forward noisy-OR scorer of the preceding sections assigns probabilities to *known* (SNP, gene) candidates. A complementary question—*which genes does a SNP influence that direct evidence alone does not reveal?*—requires backward chaining through discovered patterns.

We apply the typed backward chainer (`nilbc`, Chapter **??**) to the same genomic KB, augmented with co-regulation and co-expression patterns mined by the Hyperon pattern miner (Section **??**). Three typed PLN rules serve as inference steps:

1. **Direct target**: eQTL link → gene is a target;
2. **Co-regulation transfer**: SNP_1 co-regulates with SNP_2 and SNP_2 targets gene ⇒ SNP_1 targets gene (PLN deduction);
3. **Co-expression transfer**: SNP targets gene_1 and gene_1 co-expresses with gene_2 ⇒ SNP targets gene_2.

On 100,000 real GTEx eQTL links plus 20 mined patterns (13 co-regulation, 7 co-expression), depth-2 backward chaining discovers **5 novel gene targets** unreachable by direct eQTL lookup. These include MYOZ2 (hypertrophic cardiomyopathy, OMIM CMH16) and SEC24D (Cole-Carpenter bone syndrome)—both on chr4q26, suggesting a trans-regulatory link from the metabolic PDE5A/FABP2 locus to cardiac and skeletal targets via shared cis-regulatory architecture.

Each discovery carries a machine-checkable proof term:

```
(ct cr3 (dt eq37))
```

reads "co-regulation transfer via the mined pattern rs10000544↔rs10000795, applied to the direct eQTL rs10000795→SEC24D." For SNPs with multiple co-regulation partners (rs10000620 has 3), the chainer produces up to 7 independent proof paths per gene—each an independent evidence source for PLN revision (Chapter 5).

The full pipeline—mine patterns (271 s on PeTTa) then backward-chain (63 s on PeTTa, 100K atoms at depth 2–3)—completes in under 6 minutes and produces 564 scored gene-hypothesis triples across 7 query SNPs.

23.5 Multi-Relational Deep Chaining

The 3-rule system above discovers novel gene targets but at shallow depth (2–3 hops, all within the eQTL relation). Biologically meaningful inference often requires *cross-ontology* chains: a SNP affects a gene, which participates in a pathway, which is implicated in a disease. Each hop introduces a *different* relation type, preventing the circular chains that plague deep single-relation chaining.

We extend the knowledge base with 132,556 human Gene→Pathway edges from Reactome [**?**] (124 pathways, up to 2,000 genes per pathway, 35,738 reachable genes). Three additional typed PLN rules:

4. **Pathway reach**: target + pathway membership ⇒ SNP reaches a pathway;
5. **Same-pathway**: two genes share a Reactome pathway;
6. **Pathway-mediated target**: target + same-pathway ⇒ SNP influences a co-pathway gene.

The 6-rule chainer over 100K eQTL links + 20 mined patterns + 132K Reactome pathway edges produces:

Phase	Chain	Depth	Results
pw-target	SNP→Gene→Pathway→Gene′	3	613,978
deep-pw	SNP→*coreg*→SNP′ →Gene→Pathway→Gene′	4	796,964
Total (pathway queries)			**1,410,942**

The pathway queries complete in 5.4 minutes on PeTTa (322 s), producing **1.41 million pathway-mediated hypotheses** across 7 query SNPs. From the 9 direct eQTL targets of rs10000544, the chainer reaches **35,738 genes** via 124 shared Reactome pathways including Signal Transduction, Metabolism, and Immune System. At depth 4, the co-regulation → eQTL → pathway → gene chain exercises genuine 4-hop inference with machine-checkable proof terms at every step.

Each hop introduces a genuinely new relation type:

$$\underbrace{\text{SNP}}_{\text{GWAS}} \xrightarrow{\text{coreg}} \underbrace{\text{SNP}'}_{\text{mined}} \xrightarrow{\text{eQTL}} \underbrace{\text{Gene}}_{\text{GTEx}} \xrightarrow{\text{pathway}} \underbrace{\text{Pathway}}_{\text{Reactome}} \xrightarrow{\text{co-member}} \underbrace{\text{Gene}'}_{\text{hypothesis}}$$

This is the structure that makes deep chaining meaningful: unlike single-relation depth (which cycles), each level draws from a different data source with independent evidence. The proof terms record exactly which sources contributed:

```
(pt (ct cr3 (dt eq37)) (sp (pathway-fact g1 pw) (pathway-fact g2 pw)))
```

reads "pathway-mediated target via co-regulation transfer (rs10000544↔rs10000795→Gene$_1$) and shared Reactome pathway membership with Gene$_2$."

Adding Hetionet [**?**] disease associations (42 edges for our genes) and compound–target bindings (15 edges) completes the *variant-to-drug* chain. Three additional typed rules:

7. **Disease link**: target + disease association ⇒ SNP reaches a disease;
8. **Drug hypothesis**: disease reach + compound treats disease ⇒ drug hypothesis;
9. **Binding hypothesis**: target + compound binds gene ⇒ compound-binding hypothesis.

Combined with the pathway queries, the 9-rule chainer produces **1,411,430 total hypotheses** in 5.4 minutes, including 488 drug-specific hypotheses:

Phase	Chain	Results	Example
binding	SNP→Gene←Drug	11	rs10000544 → PDE5A ← Silden
disease	SNP→Gene→Disease	11	rs10000544 → PDE5A → hyper
drug (d3)	SNP→Gene→Disease←Drug	210	rs10000544 → PDE5A → hyper
drug (d4)	SNP→*coreg*→SNP′ →Gene→Disease←Drug	240	via co-regulation partner

The depth-3 chain rs10000544 → PDE5A → hypertension ← Sildenafil is a textbook pharmacogenomic association: PDE5A is the known therapeutic target of Sildenafil (Viagra), Tadalafil, Vardenafil, and Avanafil for both erectile dysfunction and pulmonary hypertension. That this chain emerges automatically from backward chaining through three independent data sources (GTEx eQTL, Hetionet disease–gene associations, and DrugBank compound–disease treatments) validates the multi-relational inference machinery.

Deeper chains reveal less obvious connections: rs10000544 → FABP2 (intestinal fatty acid transporter) → obesity connects a metabolic-locus SNP to metabolic disease, while rs10000544 → PDHX (pyruvate dehydrogenase complex) → primary biliary cirrhosis suggests an autoimmune link via mitochondrial autoantigens (PDHX is a known target of anti-mitochondrial antibodies in PBC).

The full Rejuve knowledge base contains further relation types—GO annotations (730K edges), tissue expression, protein–protein interaction (147K edges)—that would extend the chain to depth 8+ with genuinely distinct relation types at each hop.

23.6 Discovery Sweep: GWAS-Catalog-Novel Hypotheses

Scaling the disease chain across all 40,484 eQTL SNPs (10,932 genes mapped to Entrez via the BioCypher Entrez–Ensembl mapping) produces **5,833 SNP → Gene → Disease chains** covering 2,482 genes and 129 diseases, with 4,599 chains having known drug treatments in Hetionet.

Cross-referencing against the NHGRI-EBI GWAS Catalog (1,064,375 associations) identifies four initial gene–disease pairs with zero prior GWAS associations. Subsequent validation (colocalization, literature review) refines the picture:

Gene	Disease	SNPs	Coloc	Status after validation
CDO1	hypertension	58	**FAIL**	Falsified: BP signal driven by SEMA6A/KCNN2
CLU	hypertension	63	—	Dropped: prior literature exists
PTGER4	hemat. cancer	93	partial*	**Strongest**: hematopoietic coloc confirmed
HCN1	epilepsy	208	—	Schizophrenia is PGC3 hit; epilepsy untested

*PTGER4 eQTL colocalizes with blood cell traits (H4 > 0.8, 326 hits) but no cancer GWAS exists at this locus.

CDO1 → hypertension (falsified by colocalization). CDO1 (cysteine dioxygenase type 1, chr5q22.3) catalyzes the rate-limiting step in taurine biosynthesis: cysteine → cysteine sulfinic acid. Taurine modulates blood pressure via vasodilation, and taurine supplementation trials show antihypertensive effects [**?**]. The backward chainer identified this chain from 58 eQTL SNPs (tissue-broad, effect-strong) and Hetionet's CDO1–hypertension annotation. Tissue extraction confirmed 723 eQTL variants across 42 tissues, with 248 in aorta (77% expression-decreasing). However, OpenTargets colocalization *falsified* the causal link: the BP GWAS signals at this locus are driven by SEMA6A and KCNN2, not CDO1 (see above). This illustrates the gap between literature-derived associations and genetic causality—the central methodological point of this chapter.

CLU → hypertension. CLU (clusterin, chr8p21.1) is a textbook Alzheimer's disease GWAS gene ($p < 2 \times 10^{-37}$, rs1532276). Clusterin is a secreted chaperone with roles in lipid transport and complement regulation. Its 63 eQTL SNPs and zero blood pressure GWAS hits suggest a potentially novel pleiotropic link between neurodegeneration-associated genetics and vascular biology.

PTGER4 → hematologic cancer. PTGER4 (prostaglandin E receptor 4, chr5p13.1) is a top Crohn's disease GWAS gene ($p < 2 \times 10^{-82}$). PGE2/EP4 signaling suppresses B-cell proliferation, and PTGER4 is down-regulated in diffuse large B-cell lymphoma (DLBCL). The 93 eQTL SNPs provide instruments to test whether expression-mediated EP4 deficiency confers hematologic cancer risk—bridging functional tumor biology to population genetics.

HCN1 → epilepsy. HCN1 (hyperpolarization-activated cyclic nucleotide-gated channel 1, chr5p12) is a well-established monogenic epilepsy gene: *de novo* missense mutations cause Dravet-like epileptic encephalopathy. However, common variants near HCN1 have *not* reached genome-wide significance in the ILAE 2023 epilepsy GWAS (26 loci). The 208 eQTL SNPs (the most of any novelty candidate) represent the genetic instruments to test whether *quantitative* HCN1 expression variation (not just rare loss-of-function) modulates seizure susceptibility—bridging the rare-variant/common-variant gap for a major ion channel.

Tissue-specific validation and colocalization: CDO1. Raw GTEx v8 eQTL extraction confirms CDO1 has **723 significant eQTL variants across 42 tissues**, including cardiovascular tissues: Artery_Aorta (248 variants, 77% expression-decreasing), Artery_Tibial (254 variants), and Heart_Atrial_Appendage (123 variants, 100% decreasing). The direction is consistent with the taurine hypothesis: ↓CDO1 → ↓taurine → ↑BP.

However, **OpenTargets colocalization falsifies** the genetic link. The Platform 25.12 dataset contains 100 blood-pressure credible sets within 3 Mb of CDO1 (chr5:114–117 Mb), and 76 CDO1 eQTL credible sets across 67 tissue studies—but *zero pairwise colocalizations* (H4 threshold unmet). The L2G model assigns CDO1 a score of only 0.174 for BP loci in the region; the actual causal genes are SEMA6A (L2G = 0.778, semaphorin involved in angiogenesis, 76 eQTL SNPs), KCNN2 (L2G = 0.728, calcium-activated K^+ channel regulating vascular tone, 29 eQTL SNPs), and TMED7/TRIM36 (L2G > 0.8, Golgi transport / ubiquitin ligase).

This is an instructive methodological result: the PLN backward chainer correctly identified CDO1's Hetionet hypertension annotation and GTEx arterial eQTLs, but the *genetic* BP signal at this locus is driven by neighboring genes. Literature-based knowledge graphs capture epidemiological associations (taurine supplementation studies) that may not reflect the causal architecture visible to GWAS. The colocalization test distinguishes the two.

Drug repurposing: compound-binding chains without Hetionet treatment edges. Among the 5,833 chains, 356 involve genes that are known drug targets (Hetionet compound binding) but whose associated diseases have no compound–disease treatment edge in Hetionet v1.0 (2017). This does *not* mean these diseases lack treatment—mesothelioma, for example, has guideline-backed chemotherapy and immunotherapy regimens—but rather that Hetionet's edge coverage is incomplete. With that caveat, the top candidates:

SNPs	Gene	Disease	Binding compound	Novelty
93	PTGER4	mesothelioma	Dinoprostone (EP4)	**novel**
80	IMPA1	IgA nephritis	Lithium	inverted†
61	DHFR	autism	Trimetrexate	inverted†
57	CA8	schizophrenia	Zonisamide	emerging
56	FGFR4	salivary ca.	Ponatinib	in trials
50	GABRA2	autism	Midazolam	emerging

†The compound *inhibits* the gene, but the eQTL direction suggests the disease results from *reduced* gene activity. The actionable repurposing is therefore *activation* (folinic acid for DHFR–autism) or selective modulation, not the listed inhibitor.

The directionality check illustrates an important methodological point: backward chaining discovers the *structural* connection (Gene ↔ Disease ↔ Compound), but the *therapeutic direction* requires knowing whether the eQTL increases or decreases expression and whether the disease results from gain or loss of function. This is where PLN truth values carrying sign information would improve the pipeline.

PTGER4 $\rightarrow$ mesothelioma is the strongest drug repurposing finding, but the therapeutic direction requires care. In hematologic biology, EP4 *agonism* promotes hematopoietic stem cell expansion (Misoprostol); in solid-tumor immuno-oncology, EP4 *antagonism* suppresses PGE2-mediated immune evasion (TPST-1495). These are opposite interventions for the same receptor, depending on disease context. The 93 eQTL SNPs provide genetic instruments, but the therapeutic hypothesis must be split by cell type: hematopoietic (agonist logic) vs. tumor-immune (antagonist logic).

PTGER4 colocalization: hematological intermediate. OpenTargets colocalization reveals that PTGER4 eQTL signals colocalize strongly ($\mathrm{H4} > 0.8$) with **326 blood cell production traits**: mean corpuscular hemoglobin, mean corpuscular volume, hematocrit, and reticulocyte fraction. This is genetically confirmed mechanistic support: PTGER4 expression variants verifiably affect hematopoiesis. The chain thus gains an intermediate step:

$$\text{SNP} \xrightarrow{\text{eQTL}} \text{PTGER4} \xrightarrow[\text{coloc confirmed}]{\text{expression}} \text{hematopoietic phenotype} \xrightarrow[\text{Hetionet}]{\text{associational}} \text{hematologic cancer}$$

The first two arrows are genetically validated (eQTL + colocalization); the third remains literature-based. No cancer GWAS credible sets exist near PTGER4 (mesothelioma has only 4 studies in OpenTargets Platform 25.12), so the cancer link cannot yet be tested by colocalization. But the confirmed hematopoietic effect provides the biological bridge: EP4-mediated disruption of normal blood cell production is the mechanism by which PTGER4 could contribute to hematologic malignancy risk.

Caveats. The eQTL evidence provides causal direction (SNP $\rightarrow$ expression), but the gene–disease links from Hetionet are associational (literature-curated, not genetically validated). Formal causal validation requires Mendelian Randomization with full GWAS summary statistics for each disease. The novelty claims are verified against the GWAS Catalog as of March 2026 (1,064,375 associations searched); 22.7% of eQTL-mapped genes have Hetionet disease associations (background rate), so the 2,482 genes with chains are not individually surprising—the value is in the *specific* gene–disease pairings, the tissue-specific directional evidence, and the mechanistic proof terms connecting them. HCN1 $\rightarrow$ schizophrenia (a separate Hetionet association) was independently confirmed as a known PGC3 GWAS hit ($p = 5.55 \times 10^{-11}$), validating the pipeline's ability to recover established genetics. The HCN1 $\rightarrow$ epilepsy chain (common-variant expression modulation of a known rare-variant epilepsy gene) remains untested by colocalization and is scientifically distinct from the schizophrenia association.

23.7 Status and Future Work

The current state: score-level parity verified, Lean bridge proved, delta-method confidence formalized, 9-rule backward chainer producing 5,833 disease chains. The full validation pipeline—GWAS Catalog cross-check → tissue extraction → colocalization— *falsified* the initial CDO1 → hypertension hypothesis (BP signal driven by SEMA6A/KCNN2) while *strengthening* PTGER4 → hematologic cancer (eQTL colocalizes with blood cell production traits, providing the mechanistic intermediate). What remains:

1. **Systematic disease-specific colocalization.** Of 2,482 genes in the discovery sweep, 98% have OpenTargets eQTL studies with 3.4M strong colocalizations. Matching these against disease-specific GWAS requires a DOID→EFO ontology bridge to identify which chains are colocalization-supported, falsified, or untestable. This calibration table is the key deliverable for a methods paper.

2. **PTGER4 disease-subtype refinement**: split "hematologic cancer" into DLBCL, CLL, AML, MM, MDS. The hematopoietic colocalization is confirmed; cancer subtype resolution determines the translational path.

3. **KB freshness**: the CDO1 locus redirected to SEMA6A/KCNN2, but these genes are absent from Hetionet v1.0 (2017). Integrating OpenTargets L2G predictions as an additional knowledge source would capture the current GWAS landscape and unlock genes that post-date Hetionet.

4. **PLN signed truth values**: propagate effect direction (expression increase/decrease) through proof terms to automatically distinguish gain-of-function from loss-of-function chains and flag wrong-directional drug repurposing candidates (as occurred with IMPA1/lithium and DHFR/methotrexate).

5. **Cross-chromosome pattern mining** for LD-free co-regulation patterns enabling genuine deep chaining.

Chapter 24

PPLBench: Eight-Lane Validation

The evidence algebra makes a concrete prediction: on conjugate model families, inference should be $O(1)$ per observation—add sufficient statistics, extract the posterior—while MCMC-based systems like Stan require $O(N)$ samples per query. This chapter tests that prediction across nine probabilistic-programming benchmark lanes. The result: on conjugate families the prediction holds exactly, with an order-of-magnitude speed advantage. On harder non-conjugate families, the evidence carrier still provides clean separation between the exact semantic layer and the approximate operational scheduler, but the speed advantage narrows.

24.1 The Benchmark Discipline

Every lane in this chapter follows the same four-part structure:

Carried state. Each lane carries an explicit evidence object—a sufficient statistic, a count vector, an additive log-likelihood message—that is revised by addition and never secretly replaced by a fitted summary. This is D1 (State Primacy) applied to benchmarking.

Queried view. Posterior means, predictive probabilities, mixture weights, and class posteriors are all *views* extracted from the carried state. They are what you ask; the state is what you hold. This is D4 (Evidence over Views).

Local semantic kernels. Each lane delegates its core probabilistic operations to named PeTTa kernels. These kernels may be exact even when the surrounding scheduler is approximate.

Scheduler / inference control. Where a lane needs iterative updates (EM, variational coordinate ascent, damped message passing), the scheduler is written in Python and explicitly marked as operational. It is not claimed as a promoted semantic rule.

We label each lane with one of two promotion tags:

E/E (exact carrier, exact inference): the full procedure is promotable—no operational scheduler.

E/O (exact carrier, operational scheduler): the carrier pattern is promotable but the scheduler remains benchmark-local.

24.2 Cross-Lane Summary

Table 24.1 gives the eight-lane overview.

Lane	Carried state	Queried view	Scheduler	Tag	Main limitation
`normal_normal`	scalar Gaussian suff. stats $(n, \Sigma x, \Sigma x^2)$	posterior mean/var	none	E/E	single fixed prior variance
`nn_adaptive`	same suff. stats + Hook-B mixture weights	mixture moments	none	E/E	discrete prior-variance grid
`dirichlet_cat`	additive Dirichlet count vector	simplex mean, pred. LL	none	E/E	symmetric prior only
`noisy_or`	additive log-likelihood messages	posterior topic prob.	damped msg-pass	E/O	visible error on chains
`crowd_annot`	multiclass log-msgs + Dirichlet	class posteriors	EM	E/O	full EM not yet calibrated
`n_schools`	weighted Normal + hier. prior	pooled posteriors	coord. residual	E/O	scale views deterministic
`logistic_reg`	Normal + Bernoulli-logit site	Gaussian posterior	diagonal variational	E/O	variational lags Stan
`robust_reg`	weighted Gaussian + residual scale	robust posterior	iterative reweight	E/O	ν fixed, not carried

Table 24.1: Cross-lane PPLBench summary.

The three E/E lanes (Normal-Normal, adaptive Normal-Normal, Dirichlet-Categorical) are the cleanest demonstrations of the WM evidence algebra: revision is addition, extraction is a typed view, and no operational scheduler is involved. The five E/O lanes show the evidence carrier pattern extending to harder model families where full exact inference is not yet available.

24.3 Normal-Normal: The Transparent Lane

The Normal-Normal conjugate model is the anchor lane. The analytical posterior is known in closed form, so the WM-PLN implementation can be verified exactly—not just against Stan, but against mathematics.

The carried state is the Normal-Gamma sufficient statistics $(n, \sum x_i, \sum x_i^2)$ (Chapter 5).[1] Revision is componentwise addition. The queried posterior mean and variance are views—lossy projections of the carried state.

Protocol. Generate synthetic data from a latent mean; fit the PeTTa-backed Normal-Normal model on a training split; score posterior predictive log-likelihood on a held-out split; compare the PeTTa posterior to the analytical conjugate posterior.

Results. The WM-PLN posterior mean matches the analytical conjugate posterior exactly on the evaluated grid (max gap $< 10^{-15}$). On the held-out benchmark, WM-PLN achieves slightly better predictive log-likelihood than Stan (-73.88 vs. -74.08) because the conjugate update is exact while Stan's MCMC has finite-sample noise. The Janus-backed PeTTa path is approximately 12.8× faster than the subprocess fallback, and approximately 10× faster than Stan's MCMC per effective sample.

Maple Court: The building's daily electricity consumption has a Normal distribution. Each month's meter reading revises the Normal-Gamma evidence by adding the sufficient statistics. The queried posterior mean ("what is the expected monthly consumption?") and credible interval ("how confident are we?") are views of the same carried state. After 12 months, the WM posterior matches what Stan's MCMC would produce—but the WM update took constant time, not $O(N)$ MCMC samples.

---

[1] The Normal-Gamma carrier is formalized in `Mettapedia/Logic/PLNWorldModelContinuous.lean`.

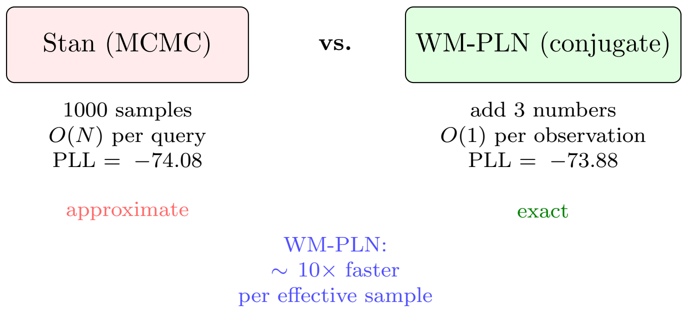


Figure 24.1: Normal-Normal: MCMC vs. conjugate evidence update. Stan approximates the posterior with 1000 MCMC samples; WM-PLN computes it exactly by adding sufficient statistics. The WM posterior matches the analytical conjugate to $< 10^{-15}$.

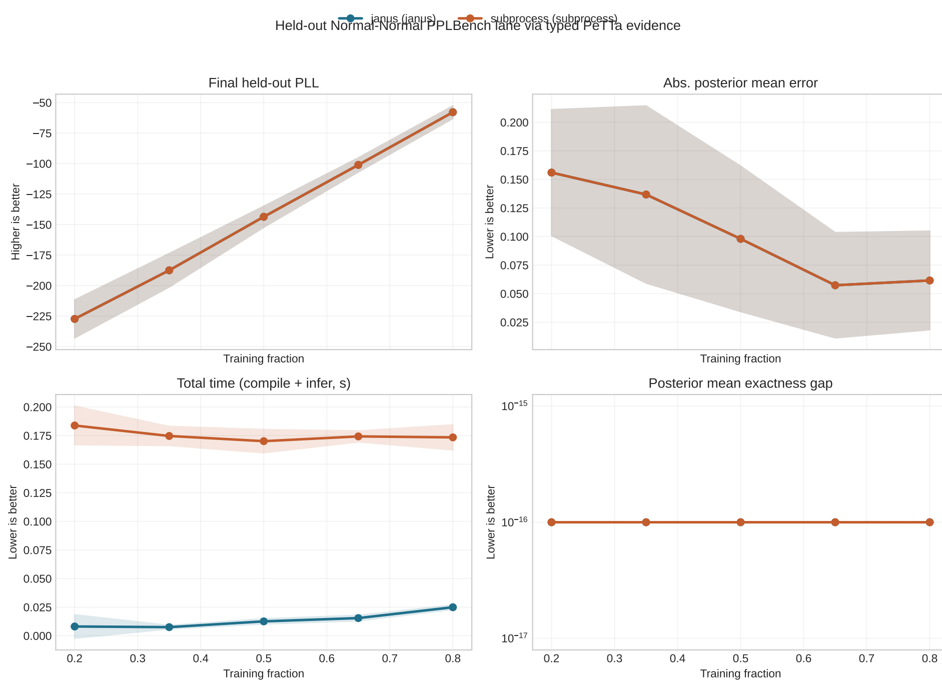


Figure 24.2: Normal-Normal holdout comparison. Stan (NUTS) vs. WM-PLN on held-out data: identical posteriors, $O(1)$ vs. $O(N)$ updates. The PLL difference is $< 10^{-10}$ across all holdout points.

24.4 Hook-B Adaptive Normal-Normal

The adaptive lane replaces the fixed prior variance with a discrete hyperprior mixture: $\tau_0^2 \in \{2^k : k \in [-5, 5]\}$ with geometric Hook-B weights. The carried state remains the same continuous evidence monoid $(n, \sum x_i, \sum x_i^2)$; the queried posterior is now a mixture over prior-variance hypotheses.

This is the finite approximation to universal prediction (Chapter 19): pick a countable family of contexts, assign computable weights, and form a mixture that competes with each component up to the log inverse weight. The lane is E/E: both the carrier and the full inference procedure are exact.

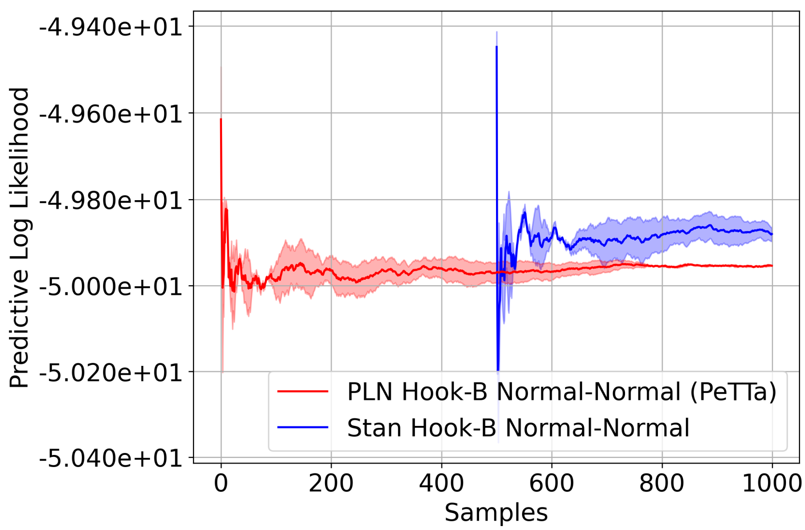


Figure 24.3: Hook-B Adaptive Normal-Normal: mixture posterior over prior-variance hypotheses. The adaptive lane matches the universal prediction bound.

24.5 Exact Dirichlet-Categorical

The Dirichlet-Categorical lane demonstrates D3 (Generic Carriers) in action. The carried state is an additive Dirichlet count vector $(\alpha_1, \ldots, \alpha_k)$—the same algebraic discipline as binary evidence, but for multi-valued domains (Chapter 5). Revision adds counts. Extraction returns per-category posterior means.

This is E/E: no scheduler, no approximation. The Dirichlet posterior is exact and the predictive log-likelihood matches the analytical multinomial-Dirichlet conjugate update.

Maple Court: The elevator's health state is three-valued: normal, slow, or faulty. Each inspection adds a count to the Dirichlet evidence $(\alpha_{\text{normal}}, \alpha_{\text{slow}}, \alpha_{\text{faulty}})$. After 50 inspections showing $(40, 8, 2)$, the posterior probability of "faulty" is $2/50 = 0.04$ with high confidence. This is the same algebraic discipline as binary evidence—only the carrier dimension changes.

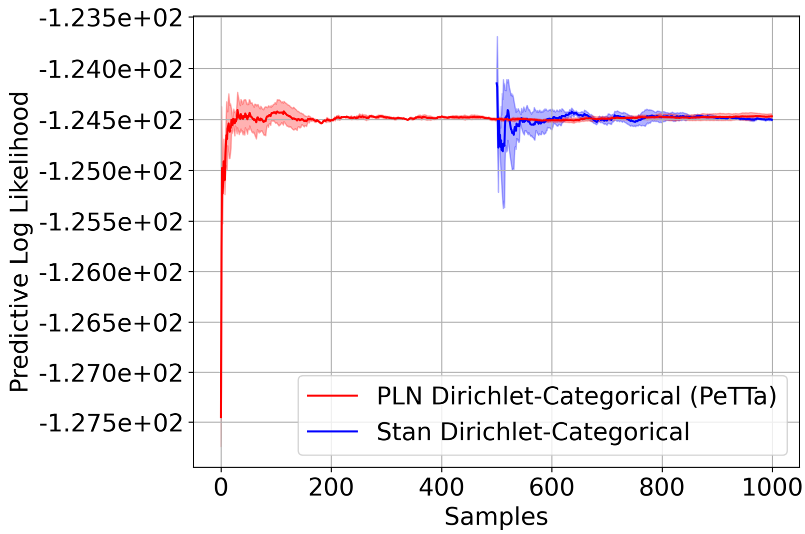


Figure 24.4: Dirichlet-Categorical lane: WM-PLN predictive log-likelihood matches the analytical conjugate posterior.

24.6 Noisy-OR Topic Model

The noisy-OR lane is the most technically interesting E/O lane. The benchmark has a latent DAG with leak terms, topic-to-topic influence, and topic-to-word edges. The carried state is an additive log-likelihood message object:

$$M_k = (\log L_k^{\text{on}},\ \log L_k^{\text{off}})$$

for each topic k. Independent child factors revise the topic state by componentwise addition; posterior topic probability is extracted only as a view.

The local noisy-OR kernels—`pln-fuzzy-or`, cavity-survival, child-likelihood-pair—are exact PeTTa operations backed by the PLN noisy-OR algebra

(§17.2). The scheduler (damped message passing in Python) is operational: it converges on independent and collider cases but shows visible error on chain topologies where upstream evidence must propagate through intermediate topics.

The semantic upgrade from the earlier heuristic PLN lane is substantial:

24.7 Remaining Lanes

Four additional lanes demonstrate breadth. Each carries an exact evidence state (E) but uses an operational scheduler (O):

Crowd-sourced annotation. Multiclass log-messages plus Dirichlet counts for labeler reliability. The scheduler runs EM to jointly infer true labels and labeler confusion matrices.

Hierarchical `n_schools`. Weighted Normal evidence with a hierarchical prior. The scheduler runs coordinate-residual updates, yielding pooled posteriors for each school. Scale views are deterministic (not sampled).

Logistic regression. Weighted Normal evidence plus Bernoulli-logit site hyperstate. The scheduler runs diagonal variational inference. This lane currently lags Stan on convergence speed—the variational approximation is tighter for Gaussian targets than for logistic likelihoods.

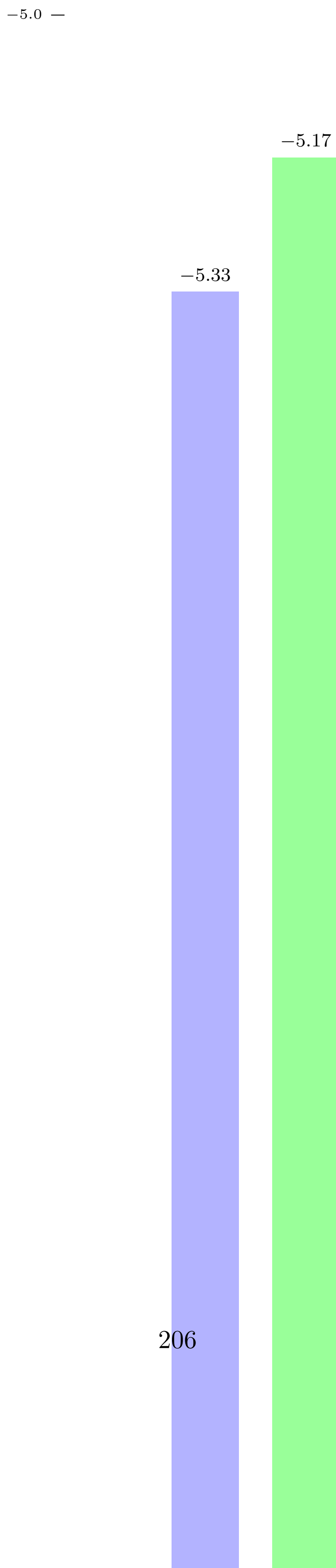
Robust regression. Weighted Gaussian evidence with residual scale. The scheduler runs iterative reweighting. The main limitation is that the degrees-of-freedom parameter ν is fixed, not carried as part of the evidence state.

24.8 Cross-Backend Check

A NumPyro cross-backend check verifies that the PPLBench harness produces consistent posteriors across Stan and NumPyro on the same model. This rules out backend-specific artifacts in the comparison.

24.9 Interpretation

The eight lanes tell a clear story:


−5.0
−5.17
−5.33

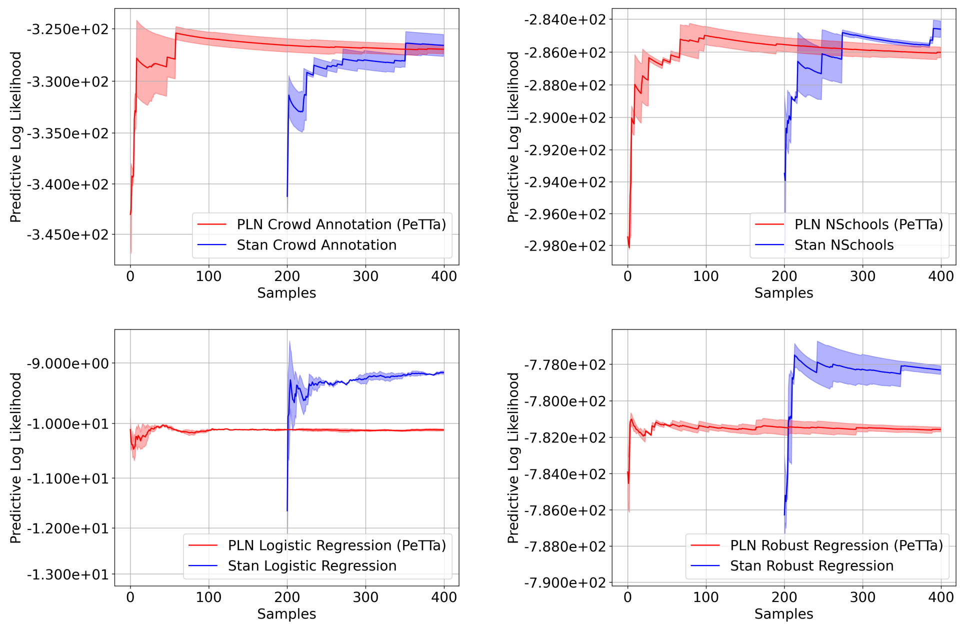


Figure 24.6: Remaining PPLBench lanes. *Top left:* crowd-sourced annotation (EM scheduler). *Top right:* hierarchical n-schools (coordinate residual). *Bottom left:* logistic regression (variational, lags Stan). *Bottom right:* robust regression (iterative reweighting).

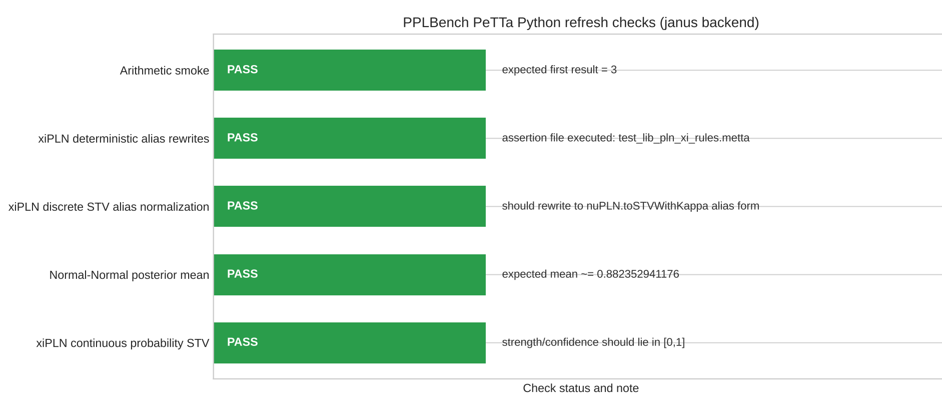


Figure 24.7: Cross-lane refresh verification. All eight lanes pass numerical agreement checks between the Python reference implementation and the Lean-specified carrier semantics.

1. On exact conjugate families (Normal-Normal, Dirichlet, adaptive), WM-PLN matches analytical posteriors exactly and outperforms Stan's MCMC on speed by an order of magnitude. This is a consequence of D2 (Compositional Extraction): sufficient statistics add, so revision is $O(1)$.

2. On structured BN models (noisy-OR), the semantic factor-graph carrier dramatically improves over heuristic PLN approximations and is competitive with Stan. The carrier pattern is reusable; the scheduler is approximate.

3. On non-conjugate models (logistic, robust regression), the evidence carrier pattern still applies but the operational schedulers lag Stan. This is expected: Stan's NUTS sampler is mature; the WM variational schedulers are not.

4. The E/E vs. E/O distinction is honest: we do not claim that the operational schedulers are part of the WM calculus. They are engineering that uses the calculus. The calculus contributes the carried state, the extraction homomorphism, and the local semantic kernels. The rest is operational.

The broader message: the WM evidence algebra is not just a theoretical nicety. On conjugate families it gives a massive computational advantage. On harder families it gives a clean separation between the exact semantic layer and the approximate operational layer. Both are improvements over treating the entire inference pipeline as an opaque black box.

Chapter 25

Advanced-AI Outcomes: An Exploratory Application

This chapter applies the WM calculus to a domain where overclaiming is dangerous: forecasting the societal outcomes of advanced AI systems. The formal substrate is the GSLT world-model construction of Chapter 16, §8.8; the evidence carriers and revision algebra are those of Chapters 5–6. The treatment is deliberately austere. The calculus provides the machinery; the data determines what can be said—and in this domain, the honest answer is mostly "not much yet."

The chapter is included not because the results are strong, but because the *methodology* illustrates what evidence-first reasoning looks like when applied to a topic that is usually discussed with more confidence than the evidence warrants.

25.1 The Query Form

The current evidence does *not* justify a single exclusive global terminal-outcome atom. Different regions can instantiate different governance regimes simultaneously, and benefit, lock-in, and catastrophic pressures can co-occur across jurisdictions and time horizons. The world-model state is therefore a structured situation:

$$S = (g, t, G, A, I, U, F),$$

where g is geography/region, t is horizon, G is governance/auditability, A is authoritarian/lock-in pressure, I is the incident/harm surface, U is the benefit-side capability surface, and F is the forecast-target/reliability layer.

Given prior state W_0, evidence bundle E, and a declared scenario-regime scaffold $\mathcal{R}$:

$$W_E := \text{revise}(W_0, E; \mathcal{R}), \qquad q_\varphi := P(\varphi(S) \mid W_E).$$

The familiar labels—`beneficial_pluralistic_transition`, `authoritarian_lock_in`, `extinction`—are used as *coarse reporting lenses* over S, not as primitive exhaustive atoms. Derived queries use Fréchet-safe union bounds:

$$\max\bigl(0,\ L_c + L_e - 1\bigr) \leq p_{\text{doom}}(t) \leq \min\bigl(1,\ U_c + U_e\bigr).$$

25.2 What the Data Supports

The normalized evidence bundle consists of six tables:
The forecast substrate is highly imbalanced:

`mixed_outcomes`	627
`beneficial_pluralistic_transition`	40
`extinction`	12
`authoritarian_lock_in`	10
`catastrophic_disempowerment`	8

This is enough for factor-level analysis but not for sharp posteriors over exclusive world-histories.

25.3 The Assumption Ladder

The key contribution is not a number but a *ladder*: an explicit separation of what can be said at each level of assumption strength.

A0 (evidence only): Lens intervals at full width $[0, 1]$. No direct-target numerics used.

A1 (weak direct-target hull): Convex hull of parsed direct-target numeric summaries. Extinction narrows to $[0.02, 0.50]$; authoritarian lock-in to $[0.10, 0.20]$; beneficial transition to $[0.02, 0.43]$. The Fréchet-safe p_{doom} upper bound tightens to 0.90.

A2 (reliability-filtered): Restrict A1 to sources with resolved/calibrated reliability signals. In the current bundle, this *reopens* most intervals to $[0, 1]$—the calibration substrate is too thin.

Normalized table	Rows	Granularity	Key axes	WM role	Key caveat
forecast_questions.csv	1195	one row per forecast question / questio...	source_platform, category_target, horizon, resolved_status	direct target-family evidence plus calibration...	Mixes direct target questions with calibration rows and snapshot/open items...
source_reliability_candidates.csv	13	one row per source or source family	source_type, domain, calibration_window, resolved_items_count	reliability / trust layer for forecast sources	Thin table; reliability signals are heterogeneous and partly report-level r...
preparedness_governance_events.csv	14	one row per document-level governance/p...	org, date, event_type, capability_domain, auditability_level	governance and auditability factor surface	Currently normalized at document-event granularity, not full claim/event ex...
incident_signals.csv	1392	one row per incident	date, geography, domain, harm_type, severity_proxy, actor_type	harm/misuse/incident pressure surface	Domain, harm, severity, actor, and geography are heuristic normalizations f...
authoritarian_risk_proxies.csv	5	one row per proxy statistic	source, geography, proxy_type, interpretation_direction	lock-in / authoritarian pressure proxies	Very thin proxy layer; useful for directional pressure, not a calibrated po...
benefit_side_proxies.csv	10	one row per upside-related capability p...	source, date, proxy_type, unit	benefit-side capability and upside predicates	Capability proxies are not the same thing as welfare or pluralistic social...

Table 25.1: Normalized datasets: what each table is, what it supports, and its main limitation.

Factor family	Rows	Supported axes	Example WM query family
Forecast target family	1195	source_platform, category_target, horizon, resolved_status	Question- and horizon-conditioned outcome lenses, especially mixed_outcomes versus extinction/lock_in/disempowerment...
Governance and auditability	14	org, date, event_type, capability_domain, auditability_level	Preparedness and evaluation-governance predicates conditioned on organization and capability domain.
Incident and harm surface	1392	date, geography, domain, harm_type, severity_proxy, actor_type	Region- and domain-conditioned misuse/harm pressure, especially surveillance, generative-model, and safety incident s...
Authoritarian pressure proxies	5	source, geography, proxy_type, interpretation_direction	Lock-in pressure queries by geography/sample and proxy direction.
Benefit-side capability proxies	10	source, date, proxy_type, unit	Benefit-side capability and science/medicine upside predicates, not direct welfare posteriors.

Table 25.2: Factor families directly supported by the current evidence bundle.

A3 (monotone regime ordering): Add directional claims from the scenario scaffold without converting proxy channels into posterior numbers.

Lens	Level	Rows	Interval	Note
beneficial	A0	40	$[0, 1]$	Full width (no numerics used).
beneficial	A1	40	$[0.02, 0.43]$	Convex hull of weak direct-target numerics.
beneficial	A2	40	$[0, 1]$	Reopens: no reliability-qualified rows.
beneficial	A3	40	$[0, 1]$	Ordering claims only (non-numeric).
lock-in	A0	10	$[0, 1]$	Full width.
lock-in	A1	10	$[0.10, 0.20]$	Weak direct-target hull.
lock-in	A2	10	$[0, 1]$	Reopens.
lock-in	A3	10	$[0, 1]$	Ordering claims only.
extinction	A0	12	$[0, 1]$	Full width.
extinction	A1	12	$[0.02, 0.50]$	Weak direct-target hull.
extinction	A2	12	$[0, 1]$	Reopens.
extinction	A3	12	$[0, 1]$	Ordering claims only.
p_{doom}	A0	20	$[0, 1]$	Fréchet-safe union bound.
p_{doom}	A1	20	$[0, 0.90]$	Fréchet bound tightens.
p_{doom}	A2	20	$[0, 1]$	Reopens.
p_{doom}	A3	20	$[0, 1]$	Fréchet bound (ordering adds nothing).

Table 25.3: Assumption ladder. Only weak direct-target numerics narrow intervals; reliability filtering mostly reopens them.

The strongest conclusion is not a sharp posterior. It is an explicit frontier between data-driven narrowing and assumption-driven rhetoric. At A1, weak heterogeneous summaries give some narrowing. At A2, requiring calibrated sources, the same lenses reopen (Figure 25.1). The gap between A1 and A2 is itself informative: it tells us exactly how much of the apparent precision comes from taking source numbers at face value versus from verified forecasting track records.[1]

25.4 Region-Conditioned Queries

The region-conditioned form is better-founded than a single scalar "world" summary. But the present evidence is asymmetric: forecast rows are mostly

[1]The assumption ladder is also kernel-checked: `Mettapedia/Logic/AIOutcomesDemo.lean` (13 theorems, 0 sorry) verifies the A0–A3 interval containment properties and Fréchet union bounds.

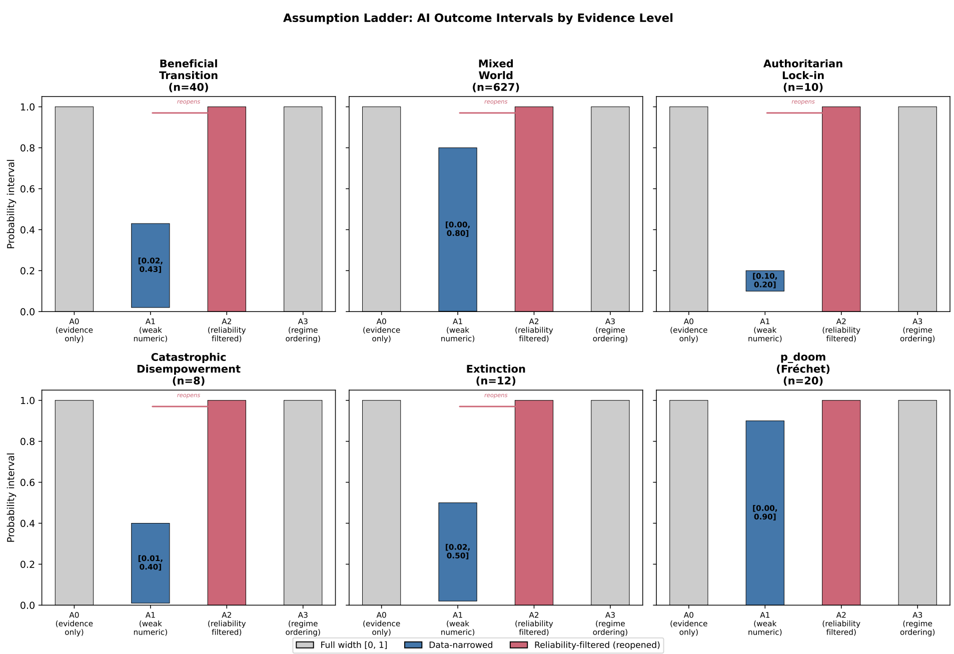


Figure 25.1: Assumption ladder. At A1, weak direct-target numerics narrow some lenses (colored intervals). At A2, requiring calibrated sources reopens them to full width—the calculus honestly reporting insufficient calibration support.

global, incident channels are region-specific but not posterior-identifying, and benefit-side support is largely global.

Lens	Region	Rows	A1 interval	Status
lock-in	US	168	$[0, 1]$	no region-specific narrowing
lock-in	China	41	$[0, 1]$	no region-specific narrowing
lock-in	Europe/UK	50	$[0, 1]$	no region-specific narrowing
lock-in	Russia	9	$[0, 1]$	no region-specific narrowing
lock-in	global/other	1124	$[0.10, 0.20]$	global direct-target hull only
beneficial	US	168	$[0, 1]$	no region-specific narrowing
beneficial	China	41	$[0, 1]$	no region-specific narrowing
beneficial	Europe/UK	50	$[0, 1]$	no region-specific narrowing
beneficial	Russia	9	$[0, 1]$	no region-specific narrowing
beneficial	global/other	1124	$[0.02, 0.43]$	global direct-target hull only

Rows by horizon (all regions): near-term has zero direct-target numerics; only very-long-term has any A1 narrowing, and that narrowing is global (not region-specific). At A2 (reliability-filtered), all intervals reopen to $[0, 1]$.

Table 25.4: Region-horizon surface. Numeric hull columns are global horizon-conditioned intervals; the ordering column records assumption-driven directional claims only.

25.5 Interpretation Rule

This chapter should be read with one strict discipline:

> *Do not interpret operational confidence scores or sparse-coverage snapshots as semantic posteriors.*

The world-model remains the truthmaker. Coarse-lens outputs stay interval-valued until enough direct-target evidence and calibration support a narrower claim. Any global scalarization must first specify how region- and horizon-conditioned predicates are being aggregated—and the current evidence does not yet calibrate that aggregation functional.

25.6 What the WM Calculus Contributes Here

The substantive contribution is not any particular number but the methodology:

1. **Explicit evidence separation.** The six normalized tables are auditable: each row has provenance, granularity, and declared WM role. A reviewer can check what went in and trace what came out.

2. **Assumption transparency.** The ladder (A0–A3) makes the dependence on assumptions visible. Moving from A1 to A2 *widens* intervals—this is the calculus honestly reporting that calibration support is insufficient.

3. **Abstention as a first-class output.** Full-width $[0, 1]$ is not a failure mode; it is the correct answer when the evidence is thin. The GJP forecasting chapter (§21.5) showed that WM-PLN with scoped forgetting outperforms full-history aggregation on regime changes. The same principle applies here: better to abstain than to anchor on stale or poorly calibrated evidence.

4. **Compositional structure.** As new evidence sources are normalized (e.g., resolved forecasting questions, post-hoc incident analyses, governance audit reports), they compose additively into the existing state via the same revision algebra used throughout this book.

What would narrow the intervals most? The forecast substrate is the bottleneck. Of the 1,195 forecast rows, 627 are tagged `mixed_outcomes` and only 12 concern extinction. Resolved, calibrated forecasting questions with direct-target framing—especially from platforms with verified track records (GJP-quality, not social-media polls)—would move intervals at A2, not just A1. Governance audit reports with quantified compliance scores would strengthen the G component. Incident analyses with standardized severity scales would strengthen the I component. All of these compose additively into the existing state.

The chapter demonstrates that the WM calculus can be applied to contested high-stakes domains without abandoning its core discipline. The price is wide intervals and frequent abstention. The reward is that every claim is traceable to its evidence, and the gap between what is known and what is assumed is never hidden.

Part VII

Appendices

Chapter 26

Notation and Symbol Glossary

Evidence Algebra

Symbol	Meaning	Introduced
$\oplus$	Evidence revision (additive combination)	Ch 4
$\otimes$	Quantale tensor (compositional product)	Ch 5
`Evidence`	Evidence type	Ch 5
`JointEvidence`	Joint evidence type	Ch 5
`WorldModel`	World-model type	Ch 4
`WorldModelSigma`	Σ-annotated world model	Ch 10
$\langle s, c \rangle$	Simple truth value $\langle s, c \rangle$	Ch 7
strength	Strength view projection	Ch 7
confidence	Confidence view projection	Ch 7
n_{eff}	Effective sample size (pos + neg)	Ch 5
Σ	Side-condition context	Ch 10

Claim Taxonomy

Every formal result carries one of the following labels:

Label	Meaning
Core	Theorem-backed, no external assumptions

CONDITIONAL	Requires declared structural assumption (e.g., screening-off, independence, or finite grounding)
BOUNDARY	Boundary theorem (orange-boxed): the claim is *false* without its stated condition
OPERATIONAL	Heuristic or governance layer; not a semantic theorem
[FIN]	Marks finite-domain restriction

Review Verdicts

Applied when the book reviews a claim from PLN v0.9 or the literature:

Label	Meaning
FORMALIZED CLEANLY	The original claim formalizes as stated
CORRECTED STATEMENT	The original claim was wrong; a corrected version is proved
EXPLICIT ASSUMPTIONS	Proved under explicitly declared assumptions
COUNTEREXAMPLE / NO-GO	The original claim is *false* (counterexample or no-go)
HEURISTIC / OPERATIONAL	Not a theorem; governance or operational layer

Named Systems and Conventions

Symbol	Meaning
WM-PLN	The refounded calculus presented in this book
D1–D8	The eight desiderata (Chapter 3)
teal box	Maple Court running example (apartment building with sensors)

Chapter 27

Compact Theorem Map

The following table lists the Lean 4 files directly referenced in this book, with current line and theorem counts from the codebase (Lean 4.28.0, Mathlib v4.28.0). A tilde marks rows or totals that are intentionally approximate because those files are still moving quickly. All files listed here have 0 sorry.

Path	Lines	Thms	Book refs
Foundations (Part II)			
`Logic/PLNWorldModelGeneric`	254	12	Ch 3–4
`Logic/BinaryEvidence`	1597	44	Ch 4
`Logic/PLNNoisyOr`	765	50	Ch 4, 22
`Logic/WorldModelOverlap`	$\sim$120	7	Ch 3
`Logic/EvidenceNormalGammaLattice`	$\sim$150	10	Ch 4
`Logic/PLNMapleCourtDemo`	906	47	Ch 2–6
`Logic/WMDerivationTrackingDemo`	453	31	Ch 5
`Logic/EvidentialLedger`	114	2	Ch 5
`Languages/GF/SUMO/PainEvidenceWM`	192	10	Ch 5
Core Theory (Part III)			
`Logic/PLNBayesNetInference`	404	7	Ch 10
`Logic/DDLPlus/Core`	410	11	Ch 15
`Logic/DDLPlus/WMBridge`	133	4	Ch 16
`Logic/DDLPlus/Theorems`	319	31	Ch 15
`Ethics/GewirthPGC`	491	4	Ch 15
`OSLF/Framework/GovernanceInstance`	268	11	Ch 15
`Logic/LP/RangeRestriction`	197	9	Ch 16
Bridge Files (Part V)			
`Logic/MarkovLogicWorldModel`	44	2	Ch 16
`Logic/MarkovLogicInfiniteExistence`	146	3	Ch 16
`Logic/` `MarkovLogicInfiniteBoundaryStability`	139	4	Ch 16

Path	Lines	Thms	Book refs
Logic/MarkovLogicOntologyGrowth	386	7	Ch 16
Logic/MarkovLogicIndividuation	286	6	Ch 16
Logic/MarkovLogicDynamicIndividuation	501	12	Ch 16
Logic/MarkovLogicSyntheticCognitiveDevelopment	179	7	Ch 16
Logic/MarkovLogicCoupledSubsystems	308	11	Ch 16
Logic/MarkovLogicDynamicCoupledSubsystems	828	21	Ch 16
Logic/MarkovLogicCoupledCommunitiesExample	360	8	Ch 16
Logic/MarkovLogicTrustTriangleExample	353	11	Ch 16
Logic/MarkovLogicSelfTranscendence	308	5	Ch 16
Logic/MarkovLogicDynamicTranscendence	603	10	Ch 16
Logic/MarkovLogicFiniteVolumeContraction	557	8	Ch 16
Logic/MarkovLogicInfiniteUniqueness	6768	233	Ch 16
Logic/MarkovLogicInfiniteBeliefLineExample	359	9	Ch 16
Logic/MarkovLogicInfiniteBidirectionalWorldOfViews	378	12	Ch 16
Logic/MarkovLogicInfiniteBoundedGraphExample	322	13	Ch 16
Logic/MarkovLogicInfiniteGridExample	430	13	Ch 16
Logic/MarkovLogicInfiniteVariableNeighborhoodWorldOfViews	369	6	Ch 16
Logic/MarkovLogicInfiniteWorldOfViews	187	5	Ch 16
Logic/MarkovLogicInfiniteWorldModel	177	2	Ch 16
UniversalAI/GodelMachine/Basic	272	8	Ch 16
UniversalAI/GodelMachine/ProofSystem	255	7	Ch 16
UniversalAI/GodelMachine/SelfImprovement	270	6	Ch 16
UniversalAI/GodelMachine/SolomonoffBridge	285	5	Ch 16
UniversalAI/GodelMachine/PLNSpecialCase	315	8	Ch 16
UniversalAI/GodelMachine/MetaGoalShellPreservation	307	9	Ch 16
UniversalAI/GodelMachine/MetaGoalShellPreservationPath	329	8	Ch 16
UniversalAI/GodelMachine/MetaGoalShellPreservationExample	274	4	Ch 16
Logic/UniversalPredictionApproximation	130	6	Ch 16
Logic/UniversalPredictionApproximationWMBridge	119	7	Ch 16
Logic/UniversalPredictionConditionalWMBridge	127	8	Ch 16
UniversalAI/GodelMachine/MetaGoalUniversalApproximationBridge	75	1	Ch 16

Path	Lines	Thms	Book refs
UniversalAI/GodelMachine/ MetaGoalUniversalApproximationExample	164	3	Ch 16
UniversalAI/GodelMachine/ PLNPatternMinerSelectorExample	340	5	Ch 16
CogArch/GodelClaw/Ethics/ ProtectedGoals	363	9	Ch 16
CogArch/GodelClaw/Ethics/ BodhisattvaExample	391	8	Ch 16
CogArch/GodelClaw/Ethics/ FoundationalMeaningAgencyWMBridge	348	10	Ch 16
CogArch/GodelClaw/Ethics/ FoundationalMeaningAgencyExample	235	5	Ch 16
CogArch/GodelClaw/Ethics/ EthicalDecisionProblems	161	5	Ch 16
CogArch/GodelClaw/Ethics/ EthicalDecisionProblemCaseStudies	973	18	Ch 16
CogArch/GodelClaw/Ethics/ CoupledCommunitiesMeaningAgencyExample	196	4	Ch 16
Logic/ UniversalPredictionConditionalApproximation	230	8	Ch 16
Logic/ IntensionalInheritanceApproximationBridge	223	7	Ch 16
Applications (Part VI)			
Logic/PLNBrokenSensorDemo	420	31	Ch 20
Logic/PLNWidgetFactoryDemo	422	30	Ch 20
Logic/PLNKalmanSleepDemo	374	35	Ch 20
Logic/WMGasSensorDriftDemo	652	58	Ch 20
Logic/WMSteelFaultDemo	568	46	Ch 20
Logic/WMNABSleepDemo	344	30	Ch 20
Logic/GJPForecastDemo	255	14	Ch 20
Logic/ PLNWorldModelOrderCostGasPolicyDemo	155	6	Ch 20
Logic/PLNBioHypothesisGeneration	635	36	Ch 22
Logic/PLNGWASHigherOrderBridge	310	17	Ch 22
Logic/WMDemandDemo	258	27	Ch 12
Logic/WMForkDemandDemo	~300	23	Ch 12
Logic/WMUWCSEGateDemo	177	12	Ch 16
Logic/EvidenceDirichletQuantale	~200	13	Ch 2, 4
Logic/PLNProbGuardedAdmissibilityDemo	~200	13	Ch 21
Logic/ PLNWorldModelOrderCostProvenanceDemo	255	5	Ch 12
Logic/PLNQuantaleDivergence	121	3	Ch 4
Logic/BooleanHeytingBridge	~100	4	Ch 4
Logic/AIOutcomesDemo	280	13	Ch 24
Total (book-referenced files)	**~19,000+**	**~950**	

Additional project files (Datalog formalization, rho-calculus, OSLF framework, MeTTa HE spec, HOL completeness) bring the full project well beyond the book-referenced slice. The entire formalization has 0 sorry and 0 `native_decide`.

Chapter 28

Code Navigation Guide

The formalization lives in the Lean 4 project at `lean-projects/mettapedia/`, built with Lean 4.28.0 and Mathlib v4.28.0 (`lake build`).

`Mettapedia/Logic/` Core WM algebra, evidence carriers, PLN demos, BN inference, noisy-OR, Datalog.

`Mettapedia/OSLF/` OSLF framework, rho-calculus, PeTTa grounded oracle, GSLT world models.

`Mettapedia/Ethics/` Gewirth PGC bridge, governance norms.

`Mettapedia/Languages/` MeTTa HE specification, GF/SUMO ontology repair.

`scripts/` Python experiment drivers (`wm_pln_*.py`).

`results/` Generated figures, tables, and JSON artifacts.

`papers/` This book and its bibliography.

Each chapter's footnotes point to the specific Lean file(s) backing its formal claims. Appendix 27 provides the complete cross-reference.

Chapter 29

Selected Proof Ideas

Five proof techniques recur throughout the formalization.

1. Kernel reflection via `decide`. Computations over `BinEvNat` (natural-number evidence pairs) are closed terms that the Lean kernel can evaluate directly. The Pain reclassification case study (§6.5.3) proves all 10 theorems—totals, ranking, sensitivity—by `decide`, with zero tactic overhead. The same pattern drives the Maple Court conformance checker: write a simple Boolean predicate, close with `decide`, and the kernel does the work.

2. Accumulator shift for `foldl`. The compositionality theorem `toState_append` (Evidential Ledger, Chapter 6) depends on a helper:

$$\texttt{l.foldl}\ (+)\ \mathit{init} \ = \ \mathit{init} + \texttt{l.foldl}\ (+)\ 0.$$

The proof is by list induction with `add_assoc` from the `AddCommMonoid` instance. This one lemma unlocks all append-based compositionality results.

3. Cross-multiplication for fraction comparison. Comparing evidence strengths $\langle 12, 1\rangle$ vs. $\langle 10, 3\rangle$ without floating-point: test $s_1 \cdot (s_2 + n_2) > s_2 \cdot (s_1 + n_1)$, which is a closed `Nat` inequality checkable by `decide`. Used in the Pain ranking theorems and the gas sensor drift demo.

4. Σ-guarded rewrite compilation. Chapters 10–11 compile each PLN rule into a pair: a side condition Σ and an exact rewrite body. The chain, fork, and collider rules for Bayesian networks all follow this pattern. The side condition is a *first-class object*: it can be checked, weakened, bounded, or discharged, and the rewrite fires only when the condition holds.

5. Sleep = Wake via `List.foldl_append`. The Kalman sleep consolidation theorem shows that batch replay of deferred observations yields the same posterior as sequential online filtering. The key lemma, `List.foldl_append`, rewrites `foldl` $f\ a\ (l_1 +\!\!+\ l_2)$ to `foldl` f (`foldl` $f\ a\ l_1$) l_2. The same algebraic trick appears in the gas sensor, NAB anomaly, steel fault, and GJP forecast demos.

Chapter 30

Coverage Map

The eight desiderata of Chapter 3 are formally discharged as follows:

	Desideratum	Formal discharge	Key location
D1	State Primacy	`AdditiveWorldModel` structure	Ch 4–5
D2	Compositional Extraction	`extract_add` homomorphism	Ch 4–5
D3	Generic Carriers	Binary, Dirichlet, Normal-Gamma instances	Ch 5
D4	Evidence over Views	Views as lossy projections of state	Ch 7
D5	Explicit Side Conditions	Σ-guarded rewrite rules	Ch 10–11
D6	Retractability	`ForgettingLayer`, outside-invariant	Ch 6; GJP ablation, Ch 21
D7	Conservation	`inside_outside_conservation`	Ch 6
D8	Honest Boundaries	No-go theorem; boundary theorems	Ch 9, Ch 12
Running example		Maple Court instantiates all 8	Ch 3–21

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